# Risk Management and Financial Institutions

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# Preface

## Achieve Mastery Over the Art of Financial Risk Management

Ever wondered how financial institutions remain stable even in the stormiest of economic climates? The answer lies in effective risk management. Our course, "Risk Management and Financial Institutions", is a detailed, practical, and comprehensive exploration of this critical field.

Through this course, you will acquire a deep understanding of the sources of risk, and how financial institutions identify, measure, and manage these risks. It provides a clear and complete overview of risk management's significant aspects, spanning interest rate risk, market risk, credit risk, operational risk, and climate risk, among others.

## Here's what we'll explore in our journey:

1. **Introduction to Risk Management:** We'll delve into various sources of risk, setting a strong foundation for the rest of the course.
2. **Revision of the Basic Concepts:** Understand the necessary background from financial derivatibes, banking, and financial markets.
3. **Forwards and Futures Strategies:** We'll dissect forward and futures contracts and their use in protecting against potential losses.
4. **Options Strategies:** Understand the mechanism behind options and how they are used as effective hedging tools in risk management.
5. **Volatility, Correlations, and Copulas:** We will explore how volatility, correlations, and copulas are used in risk managment.
6. **Market Risk I:** Delve deep into the complexities of market risk, its components, measurement techniques, and management strategies.
7. **Reading Week:** Take some rest and revise the first six topics.
8. **Market Risk II:** Delve even deeper into the complexities of market risk, its components, measurement techniques, and management strategies.
9. **Credit Risk:** Learn to evaluate and manage credit risk, an essential aspect of risk management.
10. **Operational Risk, Liquidity Risk, and Model Risk:** Understand the different operational risks financial institutions face, liquidity risk management, and the potential pitfalls of financial modeling.

11. **Climate Risk and Enterprise Risk Management:** Explore the emerging field of climate risk management and understand the holistic approach of enterprise risk management.
12. **Regulation, Economic Capital, and RAROC:** Get an in-depth understanding of the regulatory environment, how economic capital is calculated, and the importance of Risk-Adjusted Return on Capital.
13. **Financial Innovation and Risk Management Mistakes:** Learn about the role of innovation in risk management and explore common risk management mistakes.

Finally, we will review the entire course and ensure you're prepared to apply your new skills and knowledge in the real world.

## Why Enroll in this Course?

- Learn from the best: Our curriculum is designed based on Hull (2023) and Pirie and Kritzman (2017), offering you top-notch content.
- Practical focus: We emphasize real-world applications of theoretical concepts, arming you with practical skills to handle real-life scenarios.
- Dynamic and engaging: Our course is far from dry textbook material. We make learning enjoyable and engaging through interactive exercises and relatable examples.
- End-to-end understanding: From the basics of risk management to advanced strategies, our course covers everything you need to master risk management in financial institutions.
- Empowering knowledge: Upon completion, you'll be equipped to make sound financial decisions, manage various risks effectively, and drive success in your or any financial institution.

**Take the step and transform your understanding of risk management!**

## References

The whole course is based on the following textbooks:


- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.


The course was created using a support of the ChatGPT.

# 1 Introduction to Risk Management

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management (Subchapters 1-4)
- CFA Program Curriculum, Level I, Volume 6 - Portfolio Management and Ethical and Professional Standards
  - Reading 53 - Introduction to Risk Management

**Learning Outcomes:**

- Define risk and risk management.
- Describe features of a risk management framework.
- Identify financial and non-financial sources of risk and describe how they may interact.
- Describe methods for measuring and modifying risk exposures and factors to consider in choosing among the methods.

## 1.1 Introduction

In financial institutions, risk is an inherent aspect of all investment decisions, making effective risk management vital for long-term success. The process of risk management involves identifying, measuring, and controlling risks to ensure alignment with the institution's financial objectives and regulatory requirements. Portfolio managers must be proficient in handling risks at both the portfolio level—focusing on specific assets and their correlations—and the enterprise level, where systemic and institutional risks are addressed. Continuous assessment is essential to classify risks into those that offer potential rewards, those that require mitigation, and those that should be avoided to preserve institutional stability and performance.

## 1.2 The Risk Management Process

Definition of risk

***Risk*** → exposure to uncertainty.
Risk is the chance of a loss or adverse outcome as a result of an action, inaction, or external event.

- Managing risk doesn't mean avoiding it but controlling it effectively.
- **Risk and return**: These are interrelated forces in finance; only risk can be controlled, not return.
- **Risk exposure**: The state of being vulnerable to potential risks.

Example

Consider an announcement in Japan that could lead to a 1% appreciation or depreciation of the yen. The potential risk reflects a range of outcomes, typically described by a probability distribution of future returns. For an underlying amount of ¥1,000,000, the risk exposure is ¥10,000 (1% of ¥1,000,000). The risk management process helps decide whether to invest in this risky currency.

- **Risk driver**: The underlying uncertainty (e.g., +1% or -1% yen movement).
- **Risk position**: The action or quantity at risk (e.g., ¥1,000,000 worth of yen).
- **Risk exposure**: The potential valuation change (e.g., ±¥10,000).

Definition of risk management

Risk management is the process of defining, measuring, and adjusting the level of risk to optimize a portfolio's or organization's value. It involves balancing risk to achieve goals with an acceptable chance of failure.
*Alternative definition*: Risk management involves decisions and actions that maximize personal or organizational objectives within tolerable risk levels.

- Effective risk management quantifies, monitors, and adjusts risks to balance goals with failure chances.
- It provides a framework for decision-making before, during, and after risk events, rather than simply preventing losses.
- **Dynamic nature**: Risks evolve, making risk management a continuous process.

## 1.3 The Risk Management Framework

A risk management framework provides the infrastructure, processes, and analytics necessary for effective risk management. It integrates risk and return into decision-making to achieve the organization's goals within its risk tolerance. While organizations face numerous risks, they are typically driven by a small number of key risk factors.

### 1.3.1 Key Components of the Risk Management Framework

1. **Risk Governance**
   This top-down process guides risk management activities. Enterprise risk management (ERM) ensures alignment between risk practices and the organization's overall objectives, health, and value.

2. **Risk Identification and Measurement**
   Identifying relevant risk drivers and tracking exposures within the business or portfolio. This includes calculating risk metrics under various scenarios to size potential risks.

3. **Risk Infrastructure**
   Consists of the personnel and systems required for risk tracking and quantitative analysis, such as databases and analytic models.

4. **Defined Policies and Processes**
   Operationalizes governance by setting daily guidelines, limits, and processes that align with the organization's risk tolerance and regulatory requirements.

5. **Risk Monitoring, Mitigation, and Management**
   This involves ongoing reviews and adjustments to align risk exposure with the organization's tolerance levels. It is a dynamic process that integrates all aspects of the framework.

6. **Communication**
   Ensures clarity in risk governance, metrics, and exposure changes throughout the organization. Effective communication enables timely and appropriate responses to risks.

7. **Strategic Integration**
   Proactive use of risk management to enhance performance, optimize decision-making, and increase overall organizational value.

## 1.4 Risk Tolerance and Risk Budgeting

**Risk tolerance** defines the level of risk an organization is willing to accept, *including potential losses, opportunity costs, and the risk of failing to meet* objectives. It helps identify critical shortfalls in goal achievement and exposure to *uncertain factors. Risk tolerance must be formally established and communicated in* advance, guiding strategic risk selection.

**Risk budgeting** is the implementation of risk tolerance decisions approved at the *strategic or governance level. It quantifies risk using specific metrics and allocates* it across the organization. Common single-dimensional measures include standard *deviation, beta, value at risk (VaR), and scenario analysis. Multi-dimensional* approaches categorize risks by underlying classes (e.g., equity, fixed income, *commodities) and allocate investments accordingly through factor analysis.

Risk budgeting fosters intentional risk trade-offs and encourages a decision-making culture where risk is integral. Its goal is to invest where the return per unit of risk is highest, influencing decisions such as hedging or choosing lower-risk investments.

## 1.5 Identification of Risk

### 1.5.1 Financial Risks

Definition

**Financial risks** stem from financial markets, such as fluctuations in prices or interest rates.

- **Market risk**: Risk due to movements in interest rates, stock prices, exchange rates, and commodity prices.
- **Credit risk** (also known as default or counterparty risk): The risk of loss if one party fails to meet its obligations on bonds, loans, or derivatives.
- **Liquidity risk**: The risk of a significant downward price adjustment when selling a financial asset, often influenced by market transaction costs (bid-ask spread), asset liquidity, and position size.

### 1.5.2 Non-Financial Risks

> Definition
>
> **Non-financial risks** originate from sources outside financial markets, including internal organizational actions or external factors like relationships with regulators, governments, suppliers, or customers.

- **Settlement risk**: Risk of payment failure before default (e.g., sending funds to a counterparty that defaults).
- **Legal risk**: Risk of litigation or the failure of contracts to be upheld.
- **Compliance risk**: Encompasses regulatory, accounting, and tax risks.
- **Model risk**: Risk of errors from using incorrect models or misapplying correct ones.
- **Tail risk**: Risk of extreme events that occur more frequently than anticipated.
- **Operational risk**: Risk arising from internal failures (e.g., inadequate personnel, systems, or policies) or external events (e.g., fraud, natural disasters, cyber-attacks).
- **Solvency risk**: Risk of running out of cash, despite being otherwise solvent (distinct from liquidity risk in financial markets).

## 1.6 Interactions Between Risks

Risks often overlap across categories. For instance, market risk can lead to credit risk, which may, in turn, cause operational risk. Similarly, credit risk is linked to settlement risk, and legal risks can arise from market or credit risk when parties seek contract loopholes.

The interaction of risks can compound and amplify costs and reduce expected returns, particularly under stressed conditions, as seen during the 2008 financial crisis. **Combined risks are usually worse than the sum of individual risks**, following a **non-linear pattern**. Scenario planning helps identify these interactions early. Risk models often fail to capture these interactions, worsening the consequences. Governance bodies, management, and analysts should avoid treating risks as isolated and unrelated.

## 1.7 Measuring Risk

All risks stem from the uncertainty of the future. Financial markets reflect a blend of global and domestic economic conditions, industry dynamics, and company-specific factors. Government policies, regulations, and broader economic factors influence this environment, impacting company activities.

- **Systematic risk** affects entire economies or industries.

- **Idiosyncratic (unsystematic) risk** is specific to individual companies but can sometimes evolve into systematic risk, as with poor credit risk management triggering a global crisis.
- Diversified portfolios aim to eliminate unsystematic risks, though these risks still matter to company managers, poorly diversified investors, and financial analysts.

### 1.7.1 Metrics for Measuring Risk

Risk metrics are used to quantify exposure. Common measures include:

- **Probability**: Frequency of outcomes.
- **Standard deviation**: Measures the range of outcomes but has limitations for non-normal distributions.
- **Beta**: Sensitivity to market returns.
- **Greeks**: Measures specific to derivatives (e.g., delta, gamma).
- **Duration**: Applies to fixed-income assets.
- **Value at Risk (VaR)**: Estimates extreme loss potential but is subject to model limitations. Variants include Conditional VaR, extreme value theory, scenario analysis, and stress testing.

For credit risk, assessments include factors like **credit ratings**, liquidity, solvency, profitability, and leverage, along with macroeconomic indicators. Credit risk is difficult to measure due to the rarity of credit events, but default probabilities can be estimated using similar company characteristics or analyzing derivative instruments like credit default swaps. Operational risk, such as from cyberattacks, is also challenging to quantify, requiring assessment of rare but impactful events. Subjective evaluation may be needed for risks related to changes in accounting or regulation.

## 1.8 Methods of Risk Modification

Risk modification aims to align actual risk with acceptable levels and strategic objectives. It involves reducing or hedging risks and varies based on organizational goals. The primary methods include prevention and avoidance, acceptance, transfer, and shifting.

### 1.8.1 Risk Prevention and Avoidance

Risk avoidance is not always possible or desirable due to trade-offs between cost and benefits. Strategic decisions at the governance level help fine-tune exposure to desired risk levels. If risk exceeds acceptable thresholds, mitigation options include self-insuring, transferring, or shifting the risk.

### 1.8.2 Risk Acceptance: Self-Insurance and Diversification

Self-insurance involves accepting a risk that is too costly to eliminate through external means. This can either mean bearing the risk as is or setting aside reserves to cover potential losses. It must align with the enterprise's risk tolerance. Diversification, a key risk-mitigation technique, spreads investments across assets or sectors, and is central to modern portfolio theory.

### 1.8.3 Risk Transfer

Risk transfer shifts risk to another party, often through insurance. Insurers manage diversified risks by charging premiums and offering specialized coverage for unique risks.

### 1.8.4 Risk Shifting

Risk shifting alters the distribution of risk outcomes, typically using derivatives. Forward commitments and contingent claims efficiently shift risk across probability distributions and between parties. Derivatives can be traded on exchanges or created privately.

### 1.8.5 Choosing a Method for Modifying Risk

Organizations often **combine multiple risk-modification strategies** based on costs, benefits, and risk tolerance. For example, companies with foreign operations commonly hedge exchange rate risks using derivatives. Risk prevention and avoidance are preferred for low-benefit risks, while self-insurance may suit firms with strong cash flows. However, insurance may not cover risks affecting multiple parties simultaneously. Derivatives are often used to shift financial risks that exceed an organization's risk appetite. Each method affects risk profiles differently and requires careful evaluation.

# 2 Revision of the Basic Concepts

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 1 - Introduction: Risk-Return Trade-offs
  - Chapter 2 - Banks
  - Chapter 3 - Insurance Companies and Pension Plans
  - Chapter 4 - Fund Managers
  - Chapter 5 - Financial Instruments
  - Chapter 6 - The OTC Derivatives Market
  - Chapter 7 - Securitization and the Global Financial Crisis

**Learning Outcomes:**

- Quantify risk and return, and understand diversification and the efficient frontier.
- Analyze CAPM, systematic/nonsystematic risk, and $\alpha$ and $\beta$.
- Evaluate leverage, capital structure, and Modigliani-Miller propositions.
- Examine derivatives and the impact of the 2007-2008 crisis on OTC markets.
- Assess financial institutions' risks and differences between funds.
- Explore systemic risk, the 2007-2008 crisis, and regulatory frameworks.

## 2.1 Risk vs. Return for Investors

- How do investors quantify risk and return, both individually and in portfolios?
- What is the relationship between risk and expected return, and how does it influence investment decisions?
- How does diversification impact portfolio risk, and what is the efficient frontier's role in portfolio construction?
- How does adding a risk-free asset alter the efficient frontier, and how is the concept of the "market portfolio" derived?
- How do changes in market conditions, such as interest rate fluctuations or economic downturns, impact the risk-return trade-off in an investment portfolio?

## 2.2 The Capital Asset Pricing Model

- What is the CAPM model, and how is it defined?
- What is the systematic and nonsystematic risk, and where can you find it in the CAPM model?
- What is represented by $\alpha$ and $\beta$ in the CAPM model?
- What are the theoretical assumptions of the CAPM model? Is this model still useful in the real world?
- How is it possible to achieve positive $\alpha$, and how much is the weighted average $\alpha$ of all investors?

## 2.3 Capital Structure and Leverage

- How does leverage impact a firm's risk and return profile?
- What is the optimal capital structure for a company in terms of risk management?
- How does debt affect the company's risk of financial distress or bankruptcy?
- How do Modigliani-Miller propositions relate to risk management in firms with different capital structures?
- What role does equity financing play in reducing risk, and how does it compare to debt in terms of cost and financial flexibility?

## 2.4 Risk Management Tools and Derivatives

- How do exchange-traded and OTC derivatives differ in terms of risk exposure and regulation?
- What are the main challenges of using derivatives for risk management?
- What is meant by netting, and how is it related to events of default?
- How does financial crisis 2007-2008 changed OTC derivatives market?
- What is "rehypothecation", and what issues it causes on the OTC market?

## 2.5 Financial Institutions

- What are the roles of credit rating agencies in assessing the creditworthiness of institutions and securities?
- What are the potential conflicts of interest between commercial banking, securities services, and investment banking when they are all conducted under the same corporate umbrella?
- What is the definition of moral hazard and adverse selection?

- What are the main risks for insurance companies?
- What are the similarities and differences between mutual funds, ETFs, and hedge funds?

## 2.6 Regulatory and Systemic Risk

- What is systemic risk, and why is it crucial in the context of financial institutions?
- What were the origins and causes of the 2007 - 2008 financial crisis?
- What role do regulatory frameworks play in managing systemic risk?
- What went wrong during the crisis?
- What lessons can be learned from the crisis?

# 3 Forwards and Futures Strategies

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 7 - Risk Management Applications of Forward and Futures Strategies
- HULL, John. Options, futures, and other derivatives. Ninth edition. Harlow: Pearson, 2018. ISBN 978-1-292-21289-0.
  - Chapter 6 - Interest Rate Futures

**Learning outcomes:**

- Demonstrate how to use equity futures to achieve a target portfolio beta and calculate the required number of contracts.
- Create a synthetic stock index fund using cash and stock index futures.
- Convert a long stock position into synthetic cash with stock index futures.
- Rebalance a portfolio between equity and debt using equity and bond futures.
- Use futures to shift portfolio allocations across equity sectors.
- Define exchange rate risk and show how forward contracts mitigate risk for future foreign currency transactions.

## 3.1 Introduction

Definition

- **Forward and futures contracts** are agreements where one party commits to buy, and the other to sell, an underlying asset at a predetermined price on a specified future date.
- Forward contracts are customized between parties, while futures contracts are standardized, traded on exchanges, and regulated by federal authorities.

This material explores scenarios where these contracts are used in risk management. Businesses often hedge to mitigate various risks, using market positions to protect against adverse price movements. Here, we emphasize a broader view of risk management beyond simple hedging.

For further review, see the topics from the Financial Derivatives course on Forwards and Futures and the Determination of Forward and Futures Prices.

## 3.2 Managing Interest Rate Risk

Definition

**Interest rate risk** is the potential for financial loss due to fluctuations in interest rates. It can affect both the value of financial assets and the cost of liabilities.

- Financial institutions are particularly vulnerable to interest rate risk, as changes in rates can directly impact their profitability, net income, and balance sheet stability due to their reliance on lending and borrowing activities.
- This risk also affects non-financial companies, especially when planning long-term borrowing or managing fixed-rate loans, as unexpected changes in interest rates can lead to higher costs or cash flow disruptions.
- Effective management of interest rate risk is essential to maintaining financial health. To mitigate this risk, institutions and corporations often use forward rate agreements (FRAs) and interest rate futures, allowing them to hedge against unfavorable interest rate movements.

While both interest rate futures and FRAs serve as tools for managing interest rate risk, FRAs offer greater flexibility due to their customizable nature. Since financial institutions often require bespoke solutions to match their unique interest rate exposures, FRAs are more commonly used in practice than standardized futures contracts. Therefore, the following text will focus on FRAs as the preferred tool for managing interest rate risk.

### 3.2.1 Interest Rate Futures

Definition

Interest rate futures are standardized contracts traded on exchanges that allow market participants to buy or sell a notional amount of a debt instrument or interest rate index at a specified future date and price.

These contracts are typically used to hedge against short-term interest rate fluctuations. Since they are exchange-traded, they offer high liquidity and are subject to daily mark-to-market adjustments, meaning gains and losses are settled at the end of each trading day, which can lead to margin calls.

Examples of commonly traded interest rate futures include futures on government bonds (e.g., U.S. Treasury bonds) or short-term interest rates (e.g., Eurodollar futures). They are often used by institutional investors, banks, and corporations to protect themselves from unexpected changes in interest rates that can affect the cost of borrowing or the value of their bond portfolios.

**Duration-based hedging** is similar to managing beta in equity portfolios, but instead of using beta, we use duration to manage interest rate risk. Duration measures the sensitivity of a bond's price to changes in interest rates, much like beta measures a stock's sensitivity to market movements. We'll explore the concept of managing beta in more detail when we discuss market risk in a later section.

**Duration matching** is a risk management technique used by financial institutions, pension funds, and insurance companies to protect against interest rate risk. The idea is to match the duration of assets with the duration of liabilities so that changes in interest rates have an equal but offsetting effect on both. This minimizes the institution's exposure to fluctuations in interest rates.

Interest rate futures are more complex than simple buy/sell agreements, and understanding their specifics is crucial for effective risk management.

#### 3.2.1.1 Price Quotations

Interest rate futures, like Eurodollar futures, are often quoted as 100 minus the interest rate. For example, if the quoted price is 98.50, it implies a 1.5% interest rate (100 - 98.50 = 1.50%). This inverted quotation system can be confusing but is designed to reflect rising interest rates causing a fall in futures prices and vice versa. Understanding these price quotations is key to analyzing market trends and deciding on hedge positions.

#### 3.2.1.2 Day Count Conventions

Interest rate futures follow specific day count conventions to calculate interest and contract values. For example, Eurodollar futures use an actual/360 convention, meaning the interest is based on the actual number of days in the contract period, divided by 360. U.S. Treasury futures use a 30/360 convention, assuming each month has 30 days and the year has 360 days. These conventions affect the pricing and calculation of settlement values.

#### 3.2.1.3 Conversion Factors

For bond futures, conversion factors are used to adjust the price of various bonds so they can be compared to the futures contract. The conversion factor represents the price at which a bond with a 6% coupon would be delivered. Bonds with different coupons or maturities are

adjusted through this conversion factor to account for discrepancies, ensuring that different bonds are appropriately valued for delivery. This mechanism allows flexibility in delivery but can complicate the pricing and hedging process.

#### 3.2.1.4 Cheapest-to-Deliver Bonds

In futures contracts involving bonds, such as U.S. Treasury futures, the seller can deliver any bond that meets the contract's maturity criteria. However, sellers will choose the bond with the lowest cost of delivery, known as the "cheapest to deliver." The selection of this bond impacts the actual delivery value and creates opportunities for arbitrage. Understanding how to calculate the cheapest-to-deliver bond is essential for participants hedging fixed-income portfolios.

#### 3.2.1.5 Delivery Options in Treasury Bond Futures

In Treasury bond futures, the short position holder has key delivery options:

1. Can deliver on any day during the delivery month.
2. Can choose from several eligible bonds, including the cheapest-to-deliver.
3. Can notify intention to deliver based on the 2:00 p.m. settlement price but decide later in the day.

These options add flexibility for the short position and generally reduce the futures price.

### 3.2.2 Forward Rate Agreements

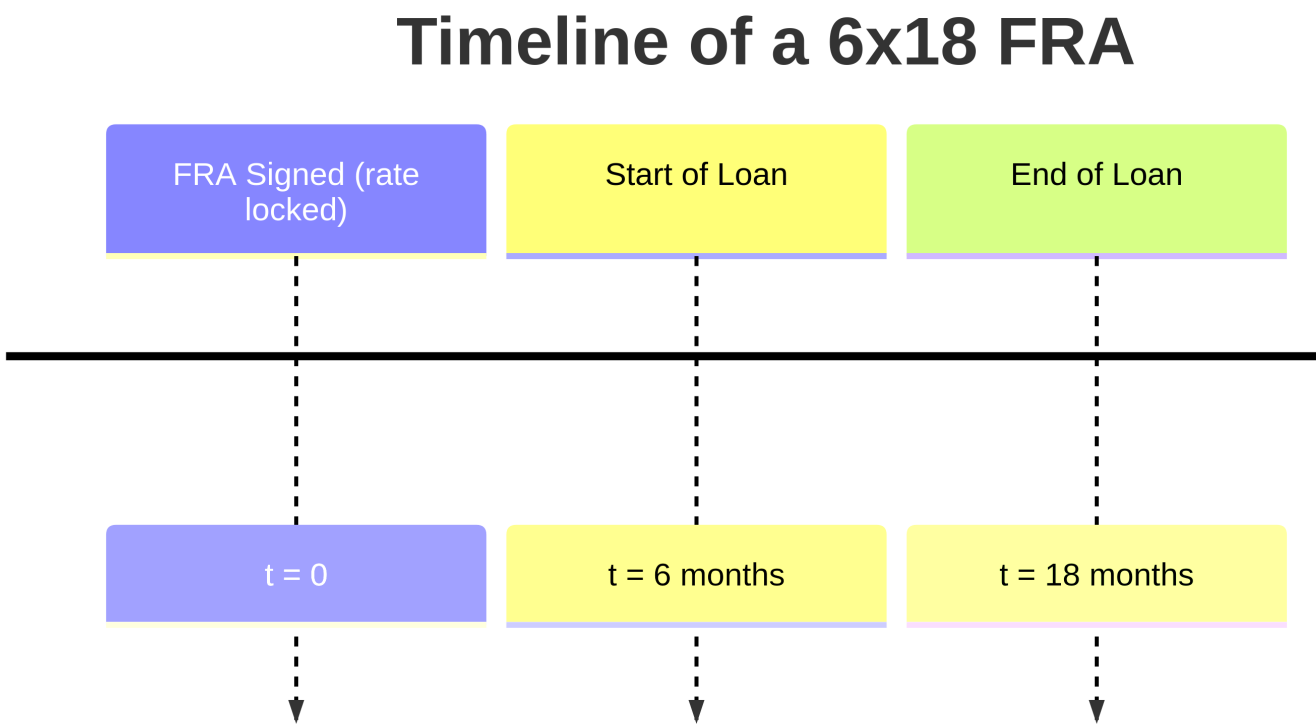
Definition

A Forward Rate Agreement (FRA) is an over-the-counter (OTC) derivative contract between two parties to exchange interest payments based on a notional principal amount at a specified future date.

The buyer of the FRA agrees to pay a fixed interest rate, while the seller agrees to pay a floating rate (usually based on LIBOR or another reference rate). The actual notional principal is never exchanged; instead, the parties settle based on the difference between the agreed-upon rate and the prevailing market rate. FRAs are used to hedge interest rate exposure. Borrowers can lock in future borrowing rates by buying a FRA, protecting against rising rates, while lenders can short a FRA to lock in future lending rates, guarding against falling rates.

In a **6x18 FRA**, a company locks in the interest rate for a loan that begins 6 months from now and lasts for one year. If market rates rise by then, the borrower pays the agreed lower rate; if they fall, the borrower pays the difference.

Banks use FRAs to hedge the risk of rising interest rates on floating-rate loans. Some prefer options instead, limiting losses to the option premium if the loan isn't needed.




The fair price for a FRA that prevents arbitrage is called **forward rate**. Forward rates are the rates implied by current zero rates for future periods. **Zero rates** (or spot rates) are the interest rates on zero-coupon bonds, which pay no interim interest but a single payment at maturity.

Forward rates are calculated using the difference between zero rates for different maturities. They represent the interest rate for a future period that makes the total return over multiple periods consistent with the zero rates.

$$R_F = \frac{R_2 T_2 - R_1 T_1}{T_2 - T_1}$$

- $R_F$ is forward rate.
- $R_1$ and $R_2$ are zero rates for maturities $T_1$ and $T_2$, respectively.
- $T_1$ and $T_2$ are maturities in years.

Example

| Maturity (Years) | Zero Rates (%) | Forward Rates (%) |
|---|---|---|
| 0.25 | 3.00 | N/A |
| 0.50 | 3.20 | 3.40 |
| 0.75 | 3.40 | 3.80 |
| 1.00 | 3.50 | 3.80 |

The forward rate for the period between 0.5 and 0.75 years is calculated as:

$$R_F = \frac{3.40 \times 0.75 - 3.20 \times 0.50}{0.75 - 0.5} = 3.8$$

This means the implied forward rate between 6 months and 9 months is 3.8%, based on the zero rates for those periods.

#### 3.2.2.1 Single-Payment Loan

XYZ Corporation plans to borrow \$10 million in six months for a year to finance business expansion and is concerned about rising interest rates. To hedge this risk, they enter into a 6x18 FRA with a bank at a fixed rate of 5.5%.

Two possible outcomes:

1. If interest rates rise to 7%, XYZ Corporation borrows at 7%, but the bank compensates them for the difference, effectively reducing the borrowing cost and making the hedge beneficial.
2. If interest rates fall to 4%, XYZ pays the bank the difference, resulting in an additional cost, meaning the hedge wasn't favorable.

By locking in a rate with the FRA, XYZ Corporation limits its exposure to rising rates, securing certainty at the cost of potentially missing lower interest rates.

$$\text{FRA Payoff} = \text{Notional Principal} \times \frac{(\text{Rate at Maturity} - \text{FRA Rate}) \times \frac{\text{Days}}{360}}{1 + \text{Rate at Maturity} \times \frac{\text{Days}}{360}}$$

- **Notional Principal** is \$10,000,000 (the loan amount).
- **Rate at Maturity** is the actual market interest rate when the FRA matures.
- **FRA Rate** is the agreed rate in the FRA (5.5%).
- **Days** is the number of days the FRA is active (360 days in this case).

If the rate at maturity is 7%:

$$\text{FRA Payoff} = \$10,000,000 \times \frac{(0.07 - 0.055) \times \frac{360}{360}}{1 + 0.07 \times \frac{360}{360}} = \$140,187$$

$$\text{Pay back} = (\$10,000,000 - \$140,187) \times (1 + 0.07 \times \frac{360}{360}) = \$10,550,000$$

$$\text{Effective rate} = \left(\frac{\$10,550,000}{\$10,000,000} - 1\right) \times \frac{360}{360} \approx 5.5\%$$

This shows that even though the market rate rose to 7%, the FRA has effectively hedged XYZ Corporation's borrowing costs to approximately 5.5%.

## 3.3 Managing Equity Market Risk

**Definition**

Equity market risk refers to the potential financial loss or variability in returns resulting from fluctuations in the prices of equity securities, such as stocks.

Stock market volatility is generally higher than bond market volatility, although the stock market tends to be more liquid. Futures contracts are often used to manage the risks associated with stock market volatility, with these contracts typically tied to stock market indices rather than individual stocks. Diversified investors, managing risk at the portfolio level, tend to favor futures on broad-based indices.

### 3.3.1 Measuring and Managing the Risk of Equities

**Definition**

Beta ($\beta$) measures a stock's or portfolio's volatility relative to the market, indicating how returns respond to market swings.

It's a key component of the Capital Asset Pricing Model (CAPM), which estimates expected returns based on beta and market returns. The formula for beta is:

$$\beta_i = \frac{Cov(R_i, R_m)}{Var(R_m)}$$

Where $\beta_i$ is the beta of the stock or portfolio, $Cov(R_i, R_m)$ is the covariance between the returns of the stock/portfolio and the market, and $Var(R_m)$ is the variance of market returns.

A beta of 1 means the security moves with the market; less than 1 indicates lower volatility than the market, and greater than 1 suggests higher volatility.

To adjust portfolio beta using futures contracts, the number of contracts ($N$) is calculated as:

$$N = \frac{(\text{Target Beta} - \text{Current Beta}) \times \text{Portfolio Value}}{\text{Futures Price} \times \text{Futures Beta}}$$

Key elements:

- **Target Beta**: Desired beta post-hedging.
- **Current Beta**: Portfolio's initial beta.
- **Portfolio Value**: Value of the portfolio.
- **Futures Price**: Price of one futures contract.
- **Futures Beta**: Beta of the futures contract.

A **positive result** indicates **buying futures** (long position); a **negative result** means **selling futures** (short position). This calculation assumes the futures contract and portfolio are closely correlated.

**Example**

Here's a refined version:

- Current Portfolio Beta: 1.2
- Target Beta: 0.8
- Portfolio Value: $1,000,000
- Futures Price: $50,000
- Futures Beta: 0.95

The number of futures contracts needed is:

$$N = \frac{(0.8 - 1.2) \times \$1,000,000}{\$50,000 \times 0.95} \approx -8$$

We need to short 8 futures contracts to reduce the portfolio beta to the target of 0.8. The negative value indicates shorting futures to decrease market risk.
When the market rises by 3.7%, the portfolio increases by 4.4%, and the futures price increases by 3.45%. The overall value of the position and the effective beta are:

$$\text{Portfolio value} = \$1,000,000 \times 1.044 = \$1,044,000$$

$$\text{Futures value} = -8 \times (\$50,000 \times 1.0345 - \$50,000) = -\$13,800$$

$$\text{Total return} = \frac{\$1,044,000 + (-\$13,800)}{\$1,000,000} - 1 = 0.0302$$

$$\text{Effective beta} = \frac{0.0302}{0.037} = 0.816$$

The effective beta of 0.816 is very close to the target of 0.8.

## 3.4 Synthetic Positions

Definition

A synthetic position refers to a financial strategy that replicates the risk and return profile of an asset or portfolio using a combination of other financial instruments, usually involving derivatives, instead of directly holding the asset.

These positions are "constructed" by combining different market positions, allowing investors to mimic the behavior of a desired asset class without actually owning it. Synthetic positions can be used to manage risk, reduce transaction costs, or increase flexibility in investment strategies, providing exposure to various market movements in a cost-efficient or strategically beneficial manner.

Common synthetic positions are as follows:

$$\text{Long Stock} = \text{Long Risk-Free Bond} + \text{Long Futures}$$

$$\text{Short Futures} = \text{Long Stock} - \text{Long Risk-Free Bond}$$

$$\text{Long Risk-Free Bond} = \text{Long Stock} + \text{Short Futures}$$

These combinations allow investors to create synthetic positions, altering exposure between equities and risk-free returns, depending on their market outlook or risk preferences.

### 3.4.1 Creating Equity out of Cash

Stock index futures can be used to create **synthetic equity positions**, reducing transaction costs and preserving liquidity. By combining risk-free bonds and futures, cash can be "equitized," effectively converting it into an equity position. This strategy is ideal for funds seeking equity exposure but needing to maintain liquidity or meet regulatory constraints, allowing for equity market participation without significant liquidation risk.

$$\text{Long Stock} = \text{Long Risk-Free Bond} + \text{Long Futures}$$

This synthetic index fund is created by combining risk-free bonds with futures on a desired index. The number of futures contracts required ($N$) is calculated using:

$$N = \frac{V(1+r)^t}{q \times f}$$

- $V$ ... amount to be invested
- $f$ ... futures price
- $t$ ... time to futures expiration
- $r$ ... risk-free rate
- $q$ ... contract multiplier
- $\delta$ ... dividend yield on the index.

**Example**

On January 1st, an investment firm wants to construct a synthetic index fund with \$200 million invested in the S&P 500. The index's dividend yield is 1.5%, and a futures contract on the S&P 500 is priced at \$3,000 with a \$50 multiplier. The position will be held for three months, and the US risk-free rate is 2%, compounded annually.
The number of futures contracts needed is:

$$N = \frac{\$200,000,000 \times (1.02)^{0.25}}{\$50 \times 3,000} = 1,339.95$$

Rounding up, the firm buys 1,340 contracts. The effective investment is:

$$\frac{1,340 \times 50 \times 3,000}{1.02^{0.25}} = \$200,007,377$$

The money is invested in risk-free bonds, growing to \$201,000,000 after three months. The number of stock units effectively purchased at the start is:

$$\frac{1,340 \times 50}{(1.015)^{0.25}} = 66,751.08$$

With dividends reinvested, the number of shares grows to:

$$66,751.08 \times (1.015)^{0.25} = 69,574.35$$

At expiration, the futures payoff is:

$$\text{Futures payoff} = 1,340 \times 50 \times (S_T - 3,000) = 69,574.35 \times S_T - 201,000,000$$

This reflects the effective equity exposure created through the synthetic index fund.

### 3.4.2 Creating Cash out of Equity

Converting a long stock position into synthetic cash involves selling futures. This approach minimizes transaction costs and avoids the need to sell a large volume of stocks at once. The relationship can be expressed as:

$$\text{Long Risk-Free Bond} = \text{Long Stock} + \text{Short Futures}$$

This strategy is similar to creating a synthetic index fund, with two key differences: a negative sign at the start because futures are being sold, and $V$ represents the market value of the stock investment.

$$N = -\frac{V(1+r)^t}{q \times f}$$

This method is equivalent to setting $\beta$ to zero. In this scenario, the stock portfolio must mirror the index (with the same $\beta$) for a full hedge. The formula for managing $\beta$ is a more general approach to eliminating systematic risk in any portfolio, though the number of futures contracts required to transform stocks into a risk-free asset may differ.

It's important to note that eliminating systematic risk does not mean the portfolio is entirely risk-free; diversifiable risk can still remain. Both the $\beta$ reduction and full hedging formulas align when applied to a portfolio identical to the index on which the futures are based. Derivatives provide a cost-effective and flexible means for adjusting $\beta$, particularly for temporary adjustments.

### 3.4.3 Adjusting Allocation among Asset Classes

Consider a portfolio worth \$300 million, with 80% allocated to stocks (\$240 million) and 20% to bonds (\$60 million). If the goal is to shift the allocation to 50% stocks (\$150 million) and

50% bonds ($150 million), we need to reduce the stock allocation by $90 million and increase the bond allocation by the same amount.

To achieve this, we sell stock index futures contracts to neutralize the beta on $90 million of stock, effectively converting it to cash. Simultaneously, bond futures contracts are purchased to increase the portfolio's duration, aligning it with the new desired bond exposure.

### 3.4.4 Pre-Investing in an Asset Class

Pre-investing with futures contracts allows investors to gain exposure to an asset class before having the available cash. This strategy essentially involves borrowing against future cash inflows and investing in the underlying asset via futures. By taking long positions in futures with the appropriate beta and duration, investors can closely replicate the performance of a direct investment.

The relationship is represented by:

$$\text{Long Stock} = \text{Long Futures} + \text{Long Risk-Free Bond}$$

Alternatively adding a loan to both sides gives:

$$\text{Long Stock} + \text{Short Risk-Free Bond} = \text{Long Futures}$$

Here, Short Risk-Free Bond represents a loan. When the anticipated cash is received, the futures position is closed, and the cash is invested, which effectively repays the loan. However, this speculative strategy carries risks if the market underperforms and fails to cover the borrowing costs.

## 3.5 Managing Foreign Currency Risk

Definition

Foreign currency risk, also known as exchange rate risk, is the potential for financial loss due to fluctuations in the exchange rates between different currencies.

Companies engaged in international business are exposed to exchange rate risks, which can affect financial planning, income targets, and balance sheets.

- **Transaction exposure** arises when future cash flows must be converted from foreign to domestic currency.

- **Translation exposure** occurs when foreign subsidiaries' financial statements are consolidated into the parent company's domestic currency.
- **Economic exposure** reflects the impact of exchange rate changes on a company's overall competitiveness, even for businesses not directly involved in foreign trade, such as when a strong domestic currency reduces tourism.

Managing these risks requires tailored approaches: **forward contracts are typically used to mitigate transaction exposure**, accounting strategies handle translation exposure, and demand forecasting addresses economic exposure. Forward contracts are particularly useful for managing specific cash flow conversions from one currency to another, as futures contracts may be too standardized for such purposes.

### 3.5.1 Managing the Risk of a Foreign Currency Receipt/Payment

When a company expects to receive cash flows in a foreign currency, it is considered "long" on that currency. If the domestic currency appreciates while awaiting the cash flow, the company will receive fewer domestic currency units for the same amount of foreign currency. To hedge this risk, the company can sell the foreign currency in the forward market, effectively going "short" on a currency forward contract. Hedging a foreign currency cash outflow follows a similar process but in reverse.

### 3.5.2 Managing the Risk of a Foreign-Market Asset Portfolio

Foreign currency risk is a key factor when investing in foreign markets. Simultaneously hedging foreign market risk and currency risk is challenging, and estimating the future portfolio value helps determine hedging strategies. It is generally difficult to hedge currency risk while keeping the local equity market exposure intact. Companies may choose to hedge local market returns or leave both market and currency risks unhedged.

If only the foreign market returns are hedged, the gain equals the foreign risk-free rate. If both market returns and exchange rates are hedged, the gain equals the domestic risk-free rate. Neither strategy is optimal for long-term investments, but in the short run, this approach can be beneficial for investors who are already in foreign markets and want to adopt a more defensive position without liquidating the portfolio.

## 3.6 Final Comments

Betas and durations are inherently unstable and difficult to measure, complicating the accurate assessment of risk sensitivities in stocks and bonds. Derivatives should not be criticized for failing to deliver precise hedging outcomes or for their speculative applications, as they play a crucial role in hedging strategies.

Futures and forwards typically incur significantly lower transaction costs compared to stock indices, which is a key reason for their continued relevance as risk management tools. They are less disruptive to portfolio managers and can be effectively used to alter a portfolio's risk profile or asset allocation.

Although futures and forwards require less capital to trade, they are not immune to liquidity concerns, especially for contracts with longer expirations. Organizational policies on the use of futures, forwards, and options often hinge on leverage and credit-risk factors. Unlike options, which require an upfront investment and provide non-linear payoffs, futures and forward contracts have zero initial value and offer linear payoffs.

## 3.7 Practice Questions and Problems

1. An investment management firm wishes to increase the beta for one of its portfolios under management from 0.95 to 1.20 for a three-month period. The portfolio has a market value of $175,000,000. The investment firm plans to use a futures contract priced at $105,790 in order to adjust the portfolio beta. The futures contract has a beta of 0.98.

    - A. Calculate the number of futures contracts that should be bought or sold to achieve an increase in the portfolio beta.
    - B. At the end of three months, the overall equity market is up 5.5%. The stock portfolio under management is up 5.1%. The futures contract is priced at $111,500. Calculate the value of the overall position and the effective beta of the portfolio.

2. Consider an asset manager who wishes to create a fund with exposure to the Russell 2000 stock index. The initial amount to be invested is $300,000,000. The fund will be constructed using the Russell 2000 Index futures contract, priced at 498.30 with a $500 multiplier. The contract expires in three months. The underlying index has a dividend yield of 0.75%, and the risk-free rate is 2.35% per year.

    - A. Indicate how the money manager would go about constructing this synthetic index using futures.
    - B. Assume that at expiration, the Russell 2000 is at 594.65. Show how the synthetic position produces the same result as investment in the actual stock index.

3. An investment management firm has a client who would like to temporarily reduce her exposure to equities by converting a $25 million equity position to cash for a period of four months. The client would like this reduction to take place without liquidating her equity position. The investment management firm plans to create a synthetic cash position using an equity futures contract. This futures contract is priced at 1170.10, has a multiplier of $250, and expires in four months. The dividend yield on the underlying index is 1.25%, and the risk-free rate is 2.75%.

- A. Calculate the number of futures contracts required to create synthetic cash.
- B. Determine the effective amount of money committed to this risk-free transaction and the effective number of units of the stock index that are converted to cash.
- C. Assume that the stock index is at 1031 when the futures contract expires. Show how this strategy is equivalent to investing in the risk-free asset, cash.

4. Consider a portfolio with a 65% allocation to stocks and 35% to bonds. The portfolio has a market value of $200 million. The beta of the stock position is 1.15, and the modified duration of the bond position is 6.75. The portfolio manager wishes to increase the stock allocation to 85% and reduce the bond allocation to 15% for a period of six months. In addition to altering asset allocations, the manager would also like to increase the beta on the stock position to 1.20 and increase the modified duration of the bonds to 8.25. A stock index futures contract that expires in six months is priced at $157,500 and has a beta of 0.95. A bond futures contract that expires in six months is priced at $109,000 and has an implied modified duration of 5.25. The stock futures contract has a multiplier of one.

   - A. Show how the portfolio manager can achieve her goals by using stock index and bond futures. Indicate the number of contracts and whether the manager should go long or short.
   - B. After six months, the stock portfolio is up 5% and bonds are up 1.35%. The stock futures price is $164,005 and the bond futures price is $110,145. Compare the market value of the portfolio in which the allocation is adjusted using futures to the market value of the portfolio in which the allocation is adjusted by directly trading stocks and bonds.

5. A pension fund manager expects to receive a cash inflow of $50,000,000 in three months and wants to use futures contracts to take a $17,500,000 synthetic position in stocks and $32,500,000 in bonds today. The stock would have a beta of 1.15 and the bonds a modified duration of 7.65. A stock index futures contract with a beta of 0.93 is priced at $175,210. A bond futures contract with a modified duration of 5.65 is priced at $95,750.

   - A. Calculate the number of stock and bond futures contracts the fund manager would have to trade in order to synthetically take the desired position in stocks and bonds today. Indicate whether the futures positions are long or short.
   - B. When the futures contracts expire in three months, stocks have declined by 5.4% and bonds have declined by 3.06%. Stock index futures are priced at $167,559, and bond futures are priced at $93,586. Show that profits on the futures positions are essentially the same as the change in the value of stocks and bonds during the three-month period.

6. Consider a US company, GateCorp, that exports products to the United Kingdom. GateCorp has just closed a sale worth 200,000,000 GBP. The amount will be received in two months. Because it will be paid in pounds, the us company bears the exchange risk. In

order to hedge this risk, GateCorp intends to use a forward contract that is priced at $1.4272 per pound. Indicate how the company would go about constructing the hedge. Explain what happens when the forward contract expires in two months.

7. ABCorp is a US-based company that frequently imports raw materials from Australia. It has just entered into a contract to purchase 175,000,000 AUD worth of raw wool, to be paid in one month. ABCorp fears that the australian dollar will strengthen, thereby raising the US dollar cost. A forward contract is available and is priced at $0.5249 per australian dollar. Indicate how ABCorp would go about constructing a hedge. Explain what happens when the forward contract expires in one month.

# 4 Options Strategies

**References:**

- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 8 - Risk Management Applications of Option Strategies

**Learning outcomes:**

- Compare the use of covered calls and protective puts for risk management.
- Calculate/interpret values for option strategies including bull spread, bear spread, butterfly spread, collar, straddle.
- Determine the effective annual rate for an anticipated loan managed with interest rate options.
- Calculate payoffs for a floating rate loan combined with an interest rate cap, floor, or collar.
- Explain delta hedging for dealers, including adjustments and why delta changes.
- Interpret the gamma of a delta-hedged portfolio and explain how gamma changes with options moving toward expiration.

## 4.1 Introduction

Definition

Options are financial contracts that provide the buyer with the right, but not the obligation, to buy (call option) or sell (put option) an underlying asset at a predetermined price (strike price) within a specific time frame. The seller is obligated to fulfill the contract if the buyer exercises the option.

- Options feature non-linear payoffs and require an upfront premium.
- They allow for gains from favorable price movements while limiting losses from unfavorable ones.
- Sellers bear the risk of substantial losses and must be adequately compensated.

### 4.1.1 Notation and Assumptions

**Assumptions:**

- The option user has a view on underlying asset movements or volatility.
- Options are European-style and customized.
- Strategies cannot be terminated early.

**Time notation:**

- Time 0 ... Strategy initiation.
- Time $t$ ... Option expiration, expressed as a fraction of a year.
- Time until expiration ... $t - 0 = t = \frac{\text{Days to expiration}}{365}$.

**Symbol definitions:**

- $c_0, c_t$ ... Call option prices at time 0 and time $t$.
- $p_0, p_t$ ... Put option prices at time 0 and time $t$.
- $X$ ... Strike price.
- $S_0, S_t$ ... Underlying asset prices at time 0 and time $t$.
- $V_0, V_t$ ... Value of the position at time 0 and time $t$.
- $\Pi$ ... Profit from the transaction $(V_t - V_0)$.
- $r$ ... Risk-free rate.

## 4.2 Option Strategies for Equity Portfolios

While these strategies can be applied to various asset classes, they are primarily discussed in the context of equity investments or equity indices.

### 4.2.1 Standard Long and Short Positions

From the course Financial Derivatives, you should be able to:

- Define a long and short position in a stock.
- Define long and short positions in call and put options.
- Calculate the profit and value of an option.
- Determine the maximum potential profit or loss for each position.

### 4.2.2 Covered Call

**Definition**

A covered call strategy involves owning the underlying asset (e.g., stocks) and selling call options on that asset. The primary goal is to generate additional income from the option premium, which the investor retains whether the option is exercised or not.

- Covered calls reduce the risk of a position, countering the assumption that options always increase risk.
- While reducing risk, this strategy also limits the potential for returns compared to simply holding the stock.
- In bull markets, covered calls cap gains, but in bear markets, the option premium provides some protection against losses.

The payoff structure for a covered call can be summarized as follows:

$$
\begin{aligned}
V_t &= S_t - \max(0, S_t - X) \\
\Pi &= V_t - S_0 + c_0 \\
\text{Maximum profit} &= X - S_0 + c_0 \\
\text{Maximum loss} &= S_0 - c_0 \\
\text{Breakeven: } S_t &= S_0 - c_0
\end{aligned}
$$

**Example**

- You own 100 shares of Company XYZ, currently trading at \$50 per share.
- You sell one call option (covering 100 shares) with a strike price of \$55, earning a premium of \$2 per share.

$$\text{Profit} = \text{Profit from shares} + \text{Profit from option}$$

- If the share price rises to \$60, the option is exercised:

$$\text{Profit} = (\$60 - \$50) \times 100 + (\$2 - \$5) \times 100 = \$1000 - \$300 = \$700$$

- If the share price stays below \$55, the option is not exercised:

$$\text{Profit} = (\$55 - \$50) \times 100 + \$2 \times 100 = \$0 + \$200 = \$200$$

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```

### 4.2.3 Protective Put

**Definition**

A protective put strategy involves owning an asset (e.g., stocks) and purchasing a put option on that same asset. The goal is to limit downside risk by ensuring that if the asset's price falls, the gains from the put option will offset the loss from the asset.

- This strategy provides downside protection while allowing for upside potential, though the cost of the put option reduces the net profit.
- Protection against large losses can be expensive, and the buyer must be willing to sacrifice some upside.
- Protective puts are often likened to an insurance policy for the underlying asset.

The protective put's payoff structure is summarized as follows:

$$V_t = S_t + \max(0, X - S_t)$$
$$\Pi = V_t - S_0 - p_0$$
$$\text{Maximum profit} = \infty$$
$$\text{Maximum loss} = S_0 + p_0 - X$$
$$\text{Breakeven: } S_t = S_0 + p_0$$

**Example**

Suppose you own 100 shares of Company XYZ, currently trading at \$50 per share. You decide to implement a protective put strategy by purchasing one put option (covering 100 shares) with a strike price of \$45 for a premium of \$2 per share.

$$\text{Profit} = \text{Profit from shares} + \text{Profit from option}$$

- If the share price rises to \$60, the put option is not exercised:

$$\text{Profit} = (\$60 - \$50) \times 100 + (-\$2) \times 100 = \$1000 - \$200 = \$800$$

- If the share price drops to \$40, the put option is exercised:

$$\text{Profit} = (\$40 - \$50) \times 100 + (-\$2) \times 100 + (\$45 - \$40) \times 100$$

$$\text{Profit} = -\$1000 - \$200 + \$500 = -\$700$$

```
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```

### 4.2.4 Money Spreads

**Definition**

In options trading, "money spreads" or "spreads" refer to strategies that involve buying and selling options of the same class (e.g., calls or puts) on the same underlying asset but with different strike prices and/or expiration dates. These strategies aim to reduce risk or lower the initial cost of entering a position.

Spreads are categorized based on how they generate profits or losses, with common types being:

1. **Vertical Spreads**:
   - **Bull Call Spread**: Buy a call with a lower strike price and sell a call with a higher strike price, both expiring at the same time.
   - **Bear Call Spread**: Sell a call with a lower strike price and buy a call with a higher strike price.
   - **Bull Put Spread**: Sell a put with a higher strike price and buy a put with a lower strike price.
   - **Bear Put Spread**: Buy a put with a higher strike price and sell a put with a lower strike price.
2. **Horizontal Spreads (Calendar Spreads)**:
   - Involve options with the same strike price but different expiration dates.
3. **Diagonal Spreads**:
   - Combine vertical and horizontal elements, using different strike prices and expiration dates.
4. **Iron Condor**:
   - A combination of two vertical spreads (one using calls, the other using puts) with the same expiration but different strike prices.
5. **Butterfly Spread**:

- Constructed using three strike prices, typically with calls or puts. Can be a call butterfly or put butterfly.

The profit and loss from these strategies depend on the underlying asset's price at expiration relative to the options' strike prices. Money spreads help target a specific range of price movement, generally involving lower risk but also offering limited profit potential compared to outright options positions.

### 4.2.5 Collars

> Definition
>
> A collar strategy is a defensive option strategy used by investors with a long position in an underlying asset, aiming to protect against potential losses.

It involves three key components:

1. **Owning the Underlying Asset**: The investor holds shares of a stock or another asset.
2. **Buying a Put Option**: The investor buys a put option with a strike price below the asset's current price, setting a floor for potential losses. This put acts as insurance.
3. **Selling a Call Option**: Simultaneously, the investor sells a call option with a strike price above the asset's current price, generating premium income that can offset the cost of the put. However, this caps the upside if the asset's price rises above the call's strike price.

The combination of these elements forms a "collar" around potential profits and losses. This strategy is commonly used by investors with significant gains in the underlying asset who wish to protect those gains without selling the asset. By selecting different strike prices and expiration dates, investors can tailor the collar to their risk preferences and market outlook.

### 4.2.6 Straddle

> Definition
>
> A straddle is a neutral options strategy that involves buying or selling a call option and a put option with the same strike price and expiration date on the same underlying asset. It is used when an investor expects significant price movement but is uncertain about the direction.

- **Long Straddle**: This strategy is suited for investors who anticipate high volatility but are unsure whether the price will rise or fall. The cost is the total premium paid for both options, and the profit potential is unlimited.

- **Short Straddle**: This strategy is for investors who expect minimal price movement. The profit is limited to the premiums received, but the risk can be significant if the asset's price moves sharply in either direction.
- **Strangle**: A variation of the straddle where the call and put options have different strike prices—typically, the call's strike price is above and the put's strike price is below the current asset price. It can also be implemented as a long or short strategy.

The straddle is commonly employed around events like earnings announcements or other significant news that may cause substantial price swings. It allows investors to potentially profit from volatility itself, without needing to predict the direction of the price movement.

## 4.3 Interest Rate Option Strategies

Definition

Interest rate options use an interest rate as the underlying asset, typically based on a 90- or 180-day instrument. These options are primarily used to hedge against changes in interest rates.

- A call option is exercised if the underlying interest rate exceeds the exercise rate.
- A put option is exercised if the underlying interest rate falls below the exercise rate.
- The notional principal determines the payoff when the option is exercised.
- Payoff is delayed by a period corresponding to the life of the underlying instrument.
- The interest rate is determined on the option's expiration date, but the actual payment occurs $m$ days later, a standard practice for managing the risk of floating-rate loans.

**Interest Rate Call Option Payoff**:

$$\text{Notional principal} \times \max(0, \text{Underlying rate at expiration} - \text{Exercise rate}) \times \frac{\text{Days in underlying rate}}{360}$$

**Interest Rate Put Option Payoff**:

$$\text{Notional principal} \times \max(0, \text{Exercise rate} - \text{Underlying rate at expiration}) \times \frac{\text{Days in underlying rate}}{360}$$

The term "Days in underlying" refers to the remaining maturity of the underlying instrument from which the interest rate is derived.

Example

| Loan Details | Value |
|---|---|
| Loan Amount | \$40,000,000 |
| Underlying | 180-day LIBOR |
| Spread | 200 basis points over LIBOR |
| Current LIBOR | 5.5% |
| Expiration | 128 days later |
| Exercise Rate | 5% |
| Call Premium | \$100,000 |
| LIBOR at expiration | 8% |

- **Call Payoff at Expiration** (received 180 days later):

$$\$40,000,000 \times \max(0, 0.08 - 0.05) \times \frac{180}{360} = \$600,000$$

- **Premium Compounded** from option initiation to expiration (128 days) at 5.5% LIBOR plus 200 basis points. It reflects the opportunity cost of paying the premium at the beginning of the period:

$$\$100,000 \times \left[1 + (0.055 + 0.02) \times \frac{128}{360}\right] = \$102,667$$

- **Effective Loan Proceeds**:

$$\$40,000,000 - \$102,667 = \$39,897,333$$

- **Loan Interest**:

$$\$40,000,000 \times (0.08 + 0.02) \times \frac{180}{360} = \$2,000,000$$

- **Effective Interest**: Loan interest minus the call payoff
- **Effective Rate on the Loan**:

$$\left(\frac{\$40,000,000 + \$2,000,000 - \$600,000}{\$39,897,333}\right)^{365/180} - 1 = 7.79\%$$

For different LIBOR rates at expiration, the maximum effective rate is 7.79% (if the option is exercised). If the option is not exercised, the effective rate decreases.
A similar analysis applies to lending using interest rate **put options**. A lender can buy a put that pays off if the interest rate falls below a specified level, compensating for the lower interest on the loan. Equations for puts are adjusted slightly, but the overall structure remains the same.

### 4.3.1 Floating-Rate Cap, Floor, and Collar

Many corporate loans are structured as floating-rate loans, with periodic interest payments that reset based on market rates, creating several distinct risks.

- **Interest Rate Calls (Caplets)**:

  - Borrowers can use multiple call options (caplets) to place a ceiling on their borrowing rates.
  - Each caplet has its own expiration date, although the exercise rates are usually uniform.
  - A combination of these caplets forms **a cap**.

- **Interest Rate Puts (Floorlets)**:
  - Lenders may be concerned about declining rates and can purchase puts (floorlets) that expire on rate reset dates.
  - A series of these puts is referred to as **a floor**.

- **Interest Rate Collars**:
  - A collar involves a long position in a cap and a short position in a floor.
  - A **zero-cost collar** occurs when the premium received from selling the floor equals the cost of buying the cap, making it appear cost-neutral, though the borrower forfeits potential gains from falling rates.
  - Borrowers typically buy caps and sell floors, while lenders might do the opposite, purchasing floors and selling caps.
  - Collars are frequently initiated by borrowers to hedge against fluctuating interest rates.

## 4.4 Option Portfolio Risk Management Strategies

Dealers provide liquidity in options markets by taking on risk temporarily and hedging positions to earn the bid-ask spread without bearing long-term exposure. When a customer buys a call option, the dealer sells it, creating a short call position, which carries significant risk. The dealer may attempt to offset this risk by finding a counterparty or using put-call parity, but often they rely on **delta hedging** to manage the position.

### 4.4.1 Delta Hedging

- **Delta** measures the sensitivity of an option's price to changes in the underlying asset price, typically between 0.0 and 1.0.
- A short call position can be hedged by holding a long position in the underlying asset, with the size of the long position proportional to the option's delta.
- For every call option sold, the dealer must hold an equivalent number of shares multiplied by the option's delta.

**Challenges with Delta Hedging**:

- **Approximation**: Delta is an approximation that changes as the underlying price or time to expiration changes, leading to potential mismatches.
- **Rounding**: The number of shares to hedge per option may not be an exact match, leading to imprecision.
- **Continuous Monitoring**: Delta hedging requires frequent adjustments, especially when the underlying price fluctuates significantly.

### 4.4.2 Gamma Hedging

- **Gamma** measures the rate of change of delta relative to changes in the underlying asset's price.
- High gamma indicates that delta will change more rapidly with price movements, increasing the risk of the delta hedge becoming ineffective.
- A delta-hedged position is only risk-free when gamma is zero.
- Large gamma implies greater sensitivity to price changes, which makes the position more difficult to manage.
- Gamma is highest for at-the-money options nearing expiration and lower for deep in-the-money or out-of-the-money options.

To reduce gamma risk, dealers often use **gamma hedging**, which involves adjusting the portfolio by adding another option to balance both delta and gamma.

### 4.4.3 Vega and Volatility Risk

- **Vega** measures an option's sensitivity to changes in the volatility of the underlying asset.
- Volatility is a crucial factor in option pricing, and changes in volatility can significantly impact an option's value, even if the position is delta-hedged.
- An increase in volatility raises the price of options, which can lead to losses for a dealer holding the underlying asset and selling options.
- **At-the-money** options are more sensitive to volatility changes, making vega a key concern for dealers managing such positions.
- Dealers hedge vega risk by taking positions in other options, but managing delta, gamma, and vega simultaneously requires careful monitoring and adjustments to maintain a balanced portfolio.

## 4.5 Practice Questions and Problems

Stanley Singh, CFA, is the risk manager at SS Asset Management. Singh works with individual clients to manage their investment portfolios. One client, Sherman Hopewell, is worried

about how short-term market fluctuations over the next three months might impact his equity position in Walnut Corporation. While Hopewell is concerned about short-term downside price movements, he wants to remain invested in Walnut shares as he remains positive about its long-term performance. Hopewell has asked Singh to recommend an option strategy that will keep him invested in Walnut shares while protecting against a short-term price decline. Singh gathers the information in the following table to explore various strategies to address Hopewell's concerns.

- **Current stock Price**: $67.79
- **Walnut Corporation European options:**

| Exercise Price | Market Call Price | Call Delta | Market Put Price | Put Delta |
|---|---|---|---|---|
| $55.00 | $12.83 | 4.7 | $0.24 | –16.7 |
| $65.00 | $3.65 | 12.0 | $1.34 | –16.9 |
| $67.50 | $1.99 | 16.5 | $2.26 | –15.3 |
| $70.00 | $0.91 | 22.2 | $3.70 | –12.9 |
| $80.00 | $0.03 | 35.8 | $12.95 | –5.0 |

**Note:** Each option has 106 days remaining until expiration.

Another client, Nigel French, is a trader who does not currently own shares of Walnut Corporation. French has told Singh that he believes that Walnut shares will experience a large move in price after the upcoming quarterly earnings release in two weeks. However, French tells Singh he is unsure which direction the stock will move. French asks Singh to recommend an option strategy that would allow him to profit should the share price move in either direction.

A third client, Wanda Tills, does not currently own Walnut shares and has asked Singh to explain the profit potential of three strategies using options in Walnut: a bull call spread, a straddle, and a butterfly spread. In addition, Tills asks Singh to explain the gamma of a call option. In response, Singh prepares a memo to be shared with Tills that provides a discussion of gamma and presents his analysis on three option strategies:

- **Strategy 1:** A straddle position at the $67.50 strike option.
- **Strategy 2:** A bull call spread using the $65 and $70 strike options.
- **Strategy 3:** A butterfly spread using the $65, $67.50, and $70 strike call options.

1. The option strategy Singh is most likely to recommend to Hopewell is a:
    - a. collar
    - b. covered call
    - c. protective put
2. The option strategy that Singh is most likely to recommend to French is a:
    - a. straddle

- b. butterfly
- c. box spread

3. Based upon the table, strategy 1 is profitable when the share price at expiration is *closest* to:

   - a. $63.24
   - b. $65.24
   - c. $69.49

4. Based upon the table, the maximum profit, on a per share basis, from investing in strategy 2, is *closest* to:

   - a. $2.26
   - b. $2.74
   - c. $5.00

5. Based upon the table, and assuming the market price of Walnut's shares at expiration is $66, the profit or loss, on a per share basis, from investing in strategy 3, is *closest* to:

   - a. -$1.57
   - b. $0.42
   - c. $1.00

6. Based on the data in the table, Singh would advise Tills that the call option with the largest gamma would have a strike price *closest* to:

   - a. $55
   - b. $67.50
   - c. $80

# 5 Volatility, Correlations, and Copulas

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 8 - Volatility
  - Chapter 9 - Correlations and Copulas
  - Chapter 10 - Valuation and Scenario Analysis

**Learning outcomes:**

- Understand the concept of volatility and implied volatility.
- Become familiar with standard volatility models like EWMA and GARCH(1,1).
- Explore basic concept of correlation and copulas.
- Describe the funcioning of a simple copula.
- Be aware of copulas shortcomings and criticism.
- Describe the process of valuation and scenario analysis.

## 5.1 Volatility

Monitoring the volatility of market variables like exchange rates, equity prices, and commodity prices is essential for financial institutions to manage portfolio value effectively. Common models for capturing volatility include the Exponentially Weighted Moving Average (EWMA), Autoregressive Conditional Heteroscedasticity (ARCH), and Generalized Autoregressive Conditional Heteroscedasticity (GARCH). These models recognize that volatility is dynamic, varying across time, with periods of low and high fluctuation.

> Definition
>
> Realized volatility, $\sigma$, measures how much an asset's price has moved over a certain time period. It is calculated using squared returns or the standard deviation of returns over shorter intervals, giving a clear picture of past price fluctuations.

- Volatility is typically measured annually for option pricing and daily for risk management.
- Volatility is higher when the market is open, so time is measured in trading days (252 per year) instead of calendar days.

$$\sigma_{year} = \sigma_{day} \times \sqrt{252}$$

$$\sigma_{day} = \sigma_{year} \times \frac{1}{\sqrt{252}}$$

### 5.1.1 Implied Volatility

**Definition**

Implied volatility is the market's forecast of a security's future price fluctuations, derived from the price of an option. It reflects the expected volatility over the life of the option, based on the assumption that the option's current price reflects market expectations.

- Implied volatility is not directly observable and must be derived from market prices using an option pricing model.
- It tends to rise during periods of market uncertainty or stress, reflecting increased expectations of future price swings.
- Unlike historical volatility, which looks at past price movements, implied volatility is forward-looking and reflects the market's consensus on future price movements.
- Different options on the same asset may have different implied volatilities, a phenomenon known as the "volatility smile" or "volatility skew."
- The VIX S&P 500 Volatility Index.

```
import pandas as pd
import yfinance as yf
import plotly.graph_objects as go

# Download historical data for VIX
data = yf.download("^VIX", period="30y", progress=False)

# Create a plotly line chart
fig = go.Figure()

# Add trace for VIX closing prices
fig.add_trace(go.Scatter(x=data.index, y=data["Close"], mode="lines", name="VIX Close"))

# Update layout for better readability
fig.update_layout(
    title="VIX Index",
    xaxis_title="Date",
```

```
    yaxis_title="Close Price",
    hovermode="x",
    template="plotly_white",
    width=745,
    height=380,
)

# Display the chart
fig.show()
```

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### 5.1.2 Are Daily Percentage Changes in Financial Variables Normal?

This section evaluates whether daily percentage changes in financial variables, specifically the SPY index, follow a normal distribution. Let's begin with a summary of the descriptive statistics for SPY daily returns.

```
import pandas as pd
import yfinance as y
from IPython.display import Markdown
from tabulate import tabulate

# Download historical data for SPY
spy_data = yf.download("SPY", period="10y", progress=False)

# Calculate daily returns
spy_data['Returns'] = spy_data['Close'].pct_change()
spy_data = spy_data.dropna()

# Calculate descriptive statistics
desc_table = pd.DataFrame(spy_data['Returns'].describe())
desc_table.reset_index(inplace=True)
Markdown(tabulate(desc_table, headers=['Metric', 'Value'], floatfmt=".6f", showindex=False))
```

| Metric | Value |
|---|---|
| count | 2515.000000 |

| Metric | Value |
|---|---|
| mean | 0.000487 |
| std | 0.011151 |
| min | -0.109424 |
| 25% | -0.003803 |
| 50% | 0.000588 |
| 75% | 0.005886 |
| max | 0.090603 |

The data show that the mean daily return is close to zero, as expected for financial time series, while the standard deviation provides an indication of volatility. However, the presence of significant minimum and maximum values suggests potential deviations from normality. Next, we examine how frequently SPY daily returns exceed different standard deviation levels, comparing real data with theoretical normal and t-distributions (5 degrees of freedom).

```
import pandas as pd
import numpy as np
import scipy.stats as stats
from IPython.display import Markdown
from tabulate import tabulate

# Standard deviations levels
std_devs = np.arange(1, 7)

# Theoretical probabilities for normal distribution in pct
theoretical_probs = [2 * (1 - stats.norm.cdf(sd)) * 100 for sd in std_devs]

# Real SPY probabilities in pct
real_probs = [(np.abs(spy_data['Returns']) > (spy_data['Returns'].std() * sd)).mean() * 100 f

# Estimate degrees of freedom from SPY data for t-distribution
degrees_of_freedom = 5  # Example value, you can adjust based on fit or empirical estimation

# Theoretical probabilities for Student's t-distribution in pct
t_dist_probs = [2 * (1 - stats.t.cdf(sd, df=degrees_of_freedom)) * 100 for sd in std_devs]

# Create a table combining all three distributions
prob_table = pd.DataFrame({
    'Normal Dist (%)': np.round(theoretical_probs, 6),
    'T Dist (%)': np.round(t_dist_probs, 6),
    'Real SPY Data (%)': np.round(real_probs, 6)
```

```
}, index=std_devs)

# Rename the index to 'Standard Deviation'
prob_table.index.name = 'Std'

# Display the adjusted table
Markdown(tabulate(prob_table, headers="keys", floatfmt=".6f", numalign="center"))
```

| Std | Normal Dist (%) | T Dist (%) | Real SPY Data (%) |
|---|---|---|---|
| 1 | 31.731051 | 36.321747 | 21.073559 |
| 2 | 4.550026 | 10.193948 | 4.691849 |
| 3 | 0.269980 | 3.009925 | 1.351889 |
| 4 | 0.006334 | 1.032342 | 0.675944 |
| 5 | 0.000057 | 0.410472 | 0.318091 |
| 6 | 0.000000 | 0.184614 | 0.238569 |

The results indicate that the real SPY data exceed these standard deviation thresholds more frequently than predicted by the normal distribution, especially at higher levels. The t-distribution, with its heavier tails, provides a better fit for the real data, particularly for extreme events.

Next, we can check normality visually. The following figure presents a histogram of SPY daily returns with a normal distribution curve fitted as well.

```
import pandas as pd
import plotly.graph_objs as go
import numpy as np
import scipy.stats as stats

# Generate histogram data
hist_data = np.histogram(spy_data['Returns'], bins=50, density=True)

# Plotly figure for histogram
fig = go.Figure()

# Add histogram trace
fig.add_trace(go.Bar(
    x=hist_data[1][:-1],
    y=hist_data[0],
    name="SPY Returns Histogram",
    marker=dict(color='blue'),
```

```
    opacity=0.75
))

# Mean and standard deviation of the returns
mean, std_dev = spy_data['Returns'].mean(), spy_data['Returns'].std()

# Generate data for the normal distribution curve
x = np.linspace(min(spy_data['Returns']), max(spy_data['Returns']), 100)
normal_dist_curve = stats.norm.pdf(x, mean, std_dev)

# Add normal distribution curve trace
fig.add_trace(go.Scatter(
    x=x,
    y=normal_dist_curve,
    mode='lines',
    name='Normal Distribution',
    line=dict(color='black', width=2)
))

# Update layout for better readability
fig.update_layout(
    title="Histogram of SPY Daily Returns with Normal Distribution Curve",
    xaxis_title="Daily Returns",
    yaxis_title="Density",
    hovermode="x",
    template="plotly_white",
    legend=dict(
        orientation="v",
        yanchor="bottom",
        y=0.15,
        xanchor="center",
        x=1
    ),
    width=745,
    height=600
)

# Show plot
fig.show()
```

```
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```

The histogram shows a clear deviation from normality, particularly in the tails, where more extreme values occur more frequently than expected. The normal curve fails to capture this behavior, reinforcing the finding that SPY returns are not normally distributed.

To formally assess normality, we apply two statistical tests: the Shapiro-Wilk test and the Kolmogorov-Smirnov test. The results are presented below:

```
import pandas as pd
from scipy.stats import shapiro, kstest, norm
from IPython.display import Markdown
from tabulate import tabulate

# Shapiro-Wilk Test
shapiro_stat, shapiro_p_value = shapiro(spy_data['Returns'].dropna())

# Kolmogorov-Smirnov Test
mean = spy_data['Returns'].mean()
std = spy_data['Returns'].std()
ks_stat, ks_p_value = kstest(spy_data['Returns'].dropna(), 'norm', args=(mean, std))

# Compile results into a DataFrame
normality_test_results = pd.DataFrame({
    'Test': ['Shapiro-Wilk', 'Kolmogorov-Smirnov'],
    'Statistic': [shapiro_stat, ks_stat],
    'P-Value': [shapiro_p_value, ks_p_value]
})

# Display the results
Markdown(tabulate(normality_test_results, headers="keys", floatfmt=".6f", showindex=False))
```

| Test | Statistic | P-Value |
|---|---|---|
| Shapiro-Wilk | 0.890170 | 0.000000 |
| Kolmogorov-Smirnov | 0.101267 | 0.000000 |

Both tests reject the null hypothesis of normality, with p-values far below the conventional threshold of 0.05. These findings further confirm that SPY daily returns do not follow a normal distribution, aligning with the visual evidence from the histogram and the probability table.

The analysis shows that daily percentage changes in SPY returns deviate significantly from a normal distribution. Both visual inspection and formal tests indicate that financial variables like SPY exhibit fat tails and extreme events more frequently than the normal distribution predicts. A Student's t-distribution offers a better fit for the real data, highlighting the importance of using alternative models for risk management and pricing in financial markets.

### 5.1.3 Exponentially Weighted Moving Average (EWMA) Volatility Model

Definition

The **Exponentially Weighted Moving Average (EWMA)** model is a time-series model used to estimate volatility. It assigns exponentially decreasing weights to historical observations, meaning that more recent returns have a greater influence on the volatility estimate than older returns.

The EWMA model is frequently used in **Value at Risk (VaR)** calculations and **risk management** due to its simplicity and ability to reflect changing market conditions in real time. It is part of the **RiskMetrics** methodology developed by J.P. Morgan. The model is a good balance between responsiveness to new information and smoothness, making it a commonly used approach for tracking market volatility.

The EWMA volatility at time $t$, denoted as $\sigma_t$, is calculated using the following recursive equation:

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$$

Where:

- $\sigma_t^2$ ... is the variance (or squared volatility) at time $t$.
- $\lambda$ ... the decay factor, which determines the weighting of past data, typically close to 1 (e.g., 0.94 for daily returns as recommended by RiskMetrics).
- $r_{t-1}$ ... the return at time $t-1$. The more extreme the return (e.g., a large price change), the greater its impact on the volatility estimate.
- $\sigma_{t-1}^2$ ... the variance at time $t-1$.

#### 5.1.3.1 Properties of EWMA

- **Recency Bias**: EWMA gives more weight to recent data, which allows it to adapt more quickly to changing market conditions.
- **No Mean Reversion**: Unlike other models such as GARCH, EWMA does not assume that volatility will revert to a long-term mean.
- **Simple to Implement**: It only requires one parameter (the decay factor) and is computationally straightforward.
- **Stationarity**: EWMA does not assume stationarity of returns, allowing for changing volatility over time.
- **Adaptability**: The model adjusts more quickly to large market events (e.g., shocks), making it well-suited for capturing periods of volatility clustering.

### 5.1.4 ARCH Model

The **ARCH** model was developed by Robert Engle in 1982 to model time-varying volatility by relating the current variance of a time series to the squared errors of previous periods. In an ARCH(1) model (the simplest form):

$$\sigma_t^2 = \alpha_0 + \alpha_1 r_{t-1}^2$$

Where:

- $\sigma_t^2$ ... the variance at time $t$.
- $r_{t-1}$ ... the return at time $t-1$.
- $\alpha_0$ ... a constant (long-term average variance).
- $\alpha_1$ ... the coefficient applied to the lagged squared return.

However, the ARCH model has a limitation in that it only relies on past squared returns, making it less flexible when volatility persistence needs to be captured over long periods.

### 5.1.5 GARCH Model

The **Generalized Autoregressive Conditional Heteroskedasticity (GARCH)** model, developed by Tim Bollerslev in 1986, extends the ARCH model by incorporating both lagged squared returns and past variances. The most commonly used version is **GARCH(1,1)**, which takes the following form:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

Where:

- $\sigma_t^2$ ... the variance at time $t$.
- $\omega$ ... is a constant (long-term average level of volatility).
- $\alpha$ ... is the coefficient applied to the lagged squared return ($r_{t-1}^2$, controls the impact of recent returns on volatility).
- $\beta$ ... is the coefficient applied to the lagged variance ($\sigma_{t-1}^2$, representing volatility persistence).

The GARCH model is widely used for risk management, option pricing, and forecasting financial market volatility, making it a key tool for market participants who need to understand volatility dynamics in real time.

#### 5.1.5.1 Properties of GARCH

- **Volatility Clustering**: GARCH captures the tendency for volatility to cluster, as seen in financial markets where volatility tends to remain high after market shocks.
- **Mean Reversion**: The model assumes that volatility will eventually revert to a long-term mean, though it can remain elevated for some time following a market event.
- **Leverage Effects**: GARCH can be extended to asymmetric models (e.g., GJR-GARCH, EGARCH) to capture the fact that negative returns often have a larger impact on volatility than positive returns.

### 5.1.6 Why Financial Institutions Still Use Traditional Models?

Volatility is central to financial risk management as it measures the uncertainty of asset price movements. Financial institutions rely on volatility estimates for tasks like **Value at Risk (VaR)**, **capital allocation**, and **option pricing**. Accurate volatility modeling helps manage exposures, comply with regulations, and allocate capital efficiently.

Despite the advancements in **machine learning** and **AI**, traditional models like **EWMA** and **GARCH** still dominate because:

- **Simplicity**: These models are easy to implement and interpret, making them suitable for real-time risk management.
- **Regulatory Acceptance**: Widely used and recognized by regulators, these models align with frameworks like **Basel III**.
- **Efficiency**: They provide reliable volatility estimates with limited data and computational power.
- **Stability**: EWMA and GARCH are less prone to overfitting and work well with limited historical data.

While machine learning offers potential for improved volatility modeling, financial institutions continue to favor **EWMA** and **GARCH** due to their simplicity, regulatory compliance, and interpretability. For now, traditional models remain the standard, with AI techniques gradually integrating into risk management as regulatory acceptance grows.

## 5.2 Correlations

Risk managers must estimate both volatilities and correlations between changes in market variables to accurately assess risk exposures. Monitoring correlations is crucial, as it helps identify relationships between variables that can impact portfolio risk. Similar techniques used for monitoring volatilities, such as historical data analysis or time series models (EWMA, GARCH), can be applied to correlations.

**Definition**

The correlation between two variables, $V_1$ and $V_2$, is quantified by the correlation coefficient $\rho$, which measures the strength and direction of their linear relationship.

It is calculated as follows:

$$\rho = \frac{E(V_1V_2) - E(V_1)E(V_2)}{SD(V_1)SD(V_2)}$$

- Here, $E(.)$ represents the expected value, and $SD(.)$ stands for standard deviation.
- If there is no linear relationship between $V_1$ and $V_2$, then $E(V_1V_2) = E(V_1)E(V_2)$, resulting in $\rho = 0$.
- A perfect positive correlation occurs when $V_1 = V_2$, yielding $\rho = 1$.

It is important to note that **zero correlation does not imply statistical independence**. Variables may still be dependent through non-linear relationships, which correlation does not capture. Correlation strictly measures linear dependence, but variables can exhibit other forms of dependence.

```
import numpy as np
import plotly.graph_objects as go
from plotly.subplots import make_subplots

# Generate a range of x values
x = np.linspace(-10, 10, 400)

# Linear Relationship
y_linear = 2 * x + 3

# Non-linear Relationship (Quadratic)
y_nonlinear = x**2

# Absolute Value Function
y_abs = np.abs(x)

# Create subplots
fig = make_subplots(rows=1, cols=3, subplot_titles=("Linear Relationship", "Non-Linear Relati

# Add the linear relationship plot
fig.add_trace(go.Scatter(x=x, y=y_linear, mode='lines', name='Linear'), row=1, col=1)

# Add the non-linear (quadratic) plot
```

```
fig.add_trace(go.Scatter(x=x, y=y_nonlinear, mode='lines', name='Quadratic'), row=1, col=2)

# Add the absolute value function plot
fig.add_trace(go.Scatter(x=x, y=y_abs, mode='lines', name='Absolute Value'), row=1, col=3)

# Update layout for aesthetics
fig.update_layout(height=400, width=1000, showlegend=False)

# Show the plot
fig.show()
```

```
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```

### 5.2.1 Covariance and Its Role in Correlation

Covariance between two variables $V_1$ and $V_2$ is defined as:

$$\text{cov}(V_1, V_2) = E(V_1 V_2) - E(V_1)E(V_2)$$

This provides a direct measure of the joint variability of $V_1$ and $V_2$ and can be used to express correlation:

$$\rho = \frac{\text{cov}(V_1, V_2)}{SD(V_1)SD(V_2)}$$

While correlation is a normalized measure (bounded between -1 and 1) and often more intuitive, covariance remains fundamental for deeper risk analysis. Covariance is crucial in risk management for understanding how assets co-vary and contribute to overall portfolio volatility. For example, covariance is used to compute portfolio variance, where each asset's contribution depends on its covariance with other assets.

## 5.3 Copulas

Definition

A **copula** is a way to "link" or "couple" different events (or variables) and understand how they are connected—**even if they don't follow the same kind of distribution**.

- A copula links univariate marginal distributions to form a joint multivariate distribution.

- By using copulas, risk managers can separately analyze the behavior of **individual risk factors** (via their marginal distributions) and **their dependencies** (via the copula function).
- This separation is especially useful because it allows for the independent fitting of marginal distributions and dependency structures.
- **Limits and Criticisms:** Despite their advantages, copulas have limitations. The use of Gaussian copulas in the 2007-2008 financial crisis was criticized because they failed to capture extreme tail dependence, leading to an underestimation of risk in mortgage-backed securities.

> **Example**
>
> Let's say a bank holds both stocks and bonds. Normally, these two assets might not be closely related, meaning if stocks go down, bonds might stay steady. However, in a financial crisis, **both assets could drop sharply at the same time**. A **copula** model helps risk managers figure out how likely it is for extreme events to happen **together**, helping them prepare for scenarios where the usual diversification strategies might fail.

### 5.3.1 Applications in Risk Management

1. **Portfolio Risk Management:** Copulas model the dependence between assets, improving risk measures like Value-at-Risk (VaR) and Expected Shortfall (ES) by capturing extreme joint movements.
2. **Credit Risk Management:** Widely used in pricing credit derivatives like CDOs, copulas model joint default probabilities (default correlations).
3. **Operational Risk Management:** Copulas model dependencies among low-frequency, high-impact events (e.g., legal risks, fraud), enhancing the accuracy of capital requirements.
4. **Insurance Risk Management:** Copulas model dependencies between claims across sectors or regions, helping insurers manage and diversify risk.
5. **Stress Testing:** Copulas improve stress testing by capturing dependencies between risk factors, leading to more realistic extreme scenario modeling.

### 5.3.2 Common types of copulas

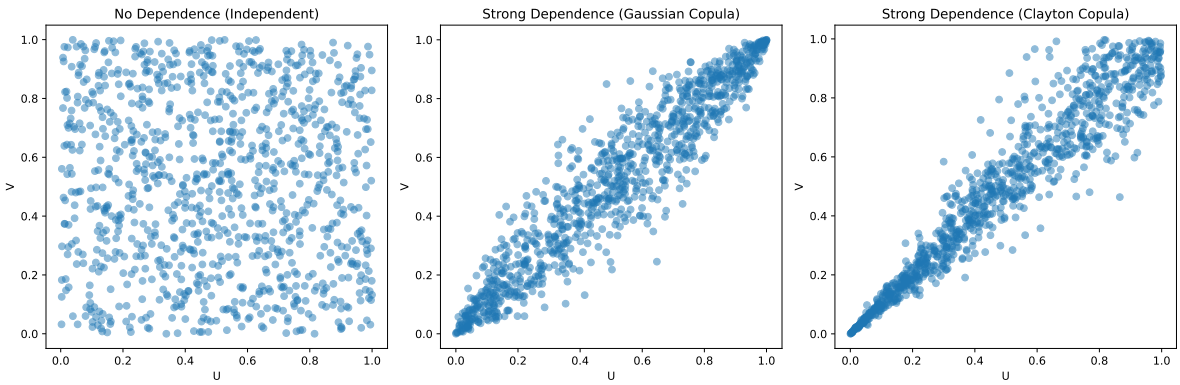
1. **Gaussian Copula**: Constructs dependence structures based on the multivariate normal distribution. It assumes linear correlations between variables but does not model tail dependencies well.

2. **t-Copula**: Similar to the Gaussian copula but incorporates tail dependencies, making it better suited for modeling extreme events in financial markets.

3. **Archimedean Copulas**: A family of copulas that are flexible and useful for modeling asymmetric dependence. Examples include:

   - **Clayton Copula**: Captures asymmetric lower tail dependence, focusing on strong joint downward movements.
   - **Gumbel Copula**: Models asymmetric upper tail dependence, ideal for scenarios with joint extreme upward movements.
   - **Frank Copula**: Symmetric copula that does not emphasize tail dependence, often used when the dependence structure is uniform across the distribution.

4. **Vine Copulas**: A more complex class of copulas that allow the construction of high-dimensional copulas by combining several bivariate copulas in a hierarchical structure. They are highly flexible and adaptable for modeling complex dependence structures across many variables, often used in finance and risk management.




### 5.3.3 How to Calculate Copula

1. **Transform Marginals to Uniform**:

   - You take the original data points of $X$ and $Y$ (which can come from any univariate distribution, not necessarily the same), and apply their respective CDFs (cumulative distribution function). This transforms them into uniform variables $U$ and $V$, both between 0 and 1.

2. **Apply the Copula**:

   - Once the data is in the uniform scale, you use the copula (e.g., Gaussian, t, or Archimedean) to capture the dependency structure between the variables. The copula function combines the uniform marginals to provide a joint distribution.

3. **Interpret the Result**:

- The result is a **multivariate joint distribution** that captures the dependency between the original variables. This joint distribution retains the original marginal distributions of $X$ and $Y$, but now you have modeled how these variables are dependent on each other (e.g., correlated or exhibiting tail dependence).

### 5.3.4 Gaussian Copula Example

Assume $X$ follows a **standard normal distribution** $N(0, 1)$, and $Y$ follows an **exponential distribution** with a rate parameter $\lambda = 1$.

Let's use the following data points:

- $X_1 = -0.5$, $X_2 = 0.8$ (from the standard normal distribution)
- $Y_1 = 0.5$, $Y_2 = 1.5$ (from the exponential distribution)

#### 5.3.4.1 1. Convert to uniform marginals using the CDF

To apply a copula, we need to convert the values of $X$ and $Y$ to uniform marginals using their respective **cumulative distribution functions (CDFs)**.

For $X$, which follows a standard normal distribution, the CDF is given by:

$$F_X(x) = \Phi(x)$$

where $\Phi(x)$ is the standard normal CDF.

$$U_1 = F_X(-0.5) = \Phi(-0.5) \approx 0.3085$$
$$U_2 = F_X(0.8) = \Phi(0.8) \approx 0.7881$$

It means that about 30.85% of the values in a standard normal distribution lie below $-0.5$ and 78.81% of values lie below 0.8.

For $Y_1 = 0.5$ and $Y_2 = 1.5$, where $Y$ follows an exponential distribution with rate parameter $\lambda = 1$, the CDF is:

$$F_Y(y) = 1 - e^{-\lambda y}$$

$$V_1 = F_Y(0.5) = 1 - e^{-0.5} \approx 0.3935$$
$$V_2 = F_Y(1.5) = 1 - e^{-1.5} \approx 0.7769$$

It means that about 39.35% of the values in the exponential distribution with $\lambda = 1$ are below 0.5 and about 77.69% of values are below 1.5.

Using Python, we can calculate the CDF for the given values of $X$ and $Y$:

```
import numpy as np
from scipy.stats import norm, expon

# Values for X and Y
u1 = norm.cdf(-0.5)
u2 = norm.cdf(0.8)
v1 = expon.cdf(0.5)
v2 = expon.cdf(1.5)

print(f"U1 = {u1}, U2 = {u2}")
print(f"V1 = {v1}, V2 = {v2}")
```

```
U1 = 0.3085375387259869, U2 = 0.7881446014166034
V1 = 0.3934693402873666, V2 = 0.7768698398515702
```

#### 5.3.4.2 2. Apply the copula to model dependence

Let's use a **Gaussian copula** to model the dependency between the variables, assuming a correlation $\rho = 0.5$. The Gaussian copula is defined as:

$$C_\rho(U, V) = \Phi_\rho\left(\Phi^{-1}(U), \Phi^{-1}(V)\right)$$

Where $\Phi^{-1}$ is the inverse CDF (quantile function) of the standard normal distribution, and $\Phi_\rho$ is the bivariate normal CDF.

Use the inverse CDF of the standard normal distribution to transform $U_1$, $U_2$ and $V_1$, $V_2$ back into normal space.

$$\Phi^{-1}(U_1) = \Phi^{-1}(0.3085) \approx -0.5$$
$$\Phi^{-1}(U_2) = \Phi^{-1}(0.7881) \approx 0.8$$
$$\Phi^{-1}(V_1) = \Phi^{-1}(0.3935) \approx -0.270$$
$$\Phi^{-1}(V_2) = \Phi^{-1}(0.7769) \approx 0.762$$

```python
import numpy as np
from scipy.stats import norm

# Values for X and Y
u1_inv = norm.ppf(u1)
u2_inv = norm.ppf(u2)
v1_inv = norm.ppf(v1)
v2_inv = norm.ppf(v2)

print(f"U1_inv = {u1_inv}, U2_inv = {u2_inv}")
print(f"V1_inv = {v1_inv}, V2_inv = {v2_inv}")
```

```
U1_inv = -0.5000000000000001, U2_inv = 0.8000000000000002
V1_inv = -0.27028802073873587, V2_inv = 0.7616644126392736
```

#### 5.3.4.3 3. Calculate the bivariate normal CDF

Now, you would compute the CDF of the bivariate normal distribution with correlation $\rho = 0.5$. This can be done numerically in Python, using libraries like `scipy.stats`.

$$C_\rho(U, V) = \Phi_\rho\left(\Phi^{-1}(U), \Phi^{-1}(V)\right)$$
$$C_\rho(0.3085, 0.3935) \approx 0.194$$
$$C_\rho(0.7881, 7769) \approx 0.663$$

```python
import numpy as np
from scipy.stats import multivariate_normal

rho = 0.5
cov_matrix = [[1, rho], [rho, 1]]

# Calculate joint probability using bivariate normal CDF
copula_1 = multivariate_normal.cdf([u1_inv, v1_inv], cov=cov_matrix)
copula_2 = multivariate_normal.cdf([u2_inv, v2_inv], cov=cov_matrix)
print(f"Copula_1 = {copula_1}, Copula_2 = {copula_2}")
```

```
Copula_1 = 0.19398256651785944, Copula_2 = 0.6632872800416401
```

The result from the copula function gives the joint probability for the variables $X$ and $Y$, adjusted for their dependency structure.

- The **copula value of 0.194** for the pair $X_1 = -0.5$ and $Y_1 = 0.5$ reflects that there's a 19.4% chance that both variables simultaneously fall below these values.
- The **copula value of 0.663** for the pair $X_2 = 0.8$ and $Y_2 = 1.5$ reflects a chance of 66.3% that both variables fall below these values.

## 5.4 Valuation and Scenario Analysis

**Valuation** estimates the current value of financial assets or liabilities, essential for assessing risk exposure. It is typically calculated as the present value of future cash flows, adjusted for risk factors.

**Scenario analysis** assesses how hypothetical or historical events impact asset values and risk metrics, focusing on adverse outcomes to answer, "How bad can things get?"

**Types of Scenarios**:

- **Historical**: Based on real past events (e.g., the 2008 financial crisis).
- **Hypothetical**: Extreme market shocks or stress events.
- **Regulatory**: Scenarios required by regulators for stress testing.

**Real-World vs. Risk-Neutral Projections**:

- **Real-World Projections**: Used in scenario analysis to reflect actual risk and investor behavior.
- **Risk-Neutral Projections**: Applied in valuation, especially for time-evolving variables or events like company defaults.

## 5.5 Practice Questions and Problems

### 5.5.1 Volatility

1. The volatility of an asset is 2% per day. What is the standard deviation of the percentage price change in three days?
2. The volatility of an asset is 25% per annum. What is the standard deviation of the percentage price change in one trading day? Assuming a normal distribution with zero mean, estimate 95% confidence limits for the percentage price change in one day.
3. What is implied volatility? What does it mean if different options on the same asset have different implied volatilities?
4. Explain the exponentially weighted moving average (EWMA) model for estimating volatility from historical data.
5. What is the difference between the exponentially weighted moving average model and the GARCH(1,1) model for updating volatilities?

6. The most recent estimate of the daily volatility of an asset is 1.5% and the price of the asset at the close of trading yesterday was \$30.00. The parameter $\lambda$ in the EWMA model is 0.94. Suppose that the price of the asset at the close of trading today is \$30.50. How will this cause the volatility to be updated by the EWMA model?
7. A company uses an EWMA model for forecasting volatility. It decides to change the parameter $\lambda$ from 0.95 to 0.85. Explain the likely impact on the forecasts.
8. Assume that an index at close of trading yesterday was 1,040 and the daily volatility of the index was estimated as 1% per day at that time. The parameters in a GARCH(1,1) model are $\omega = 0.000002$, $\alpha = 0.06$, and $\beta = 0.92$. If the level of the index at close of trading today is 1,060, what is the new volatility estimate?
9. A company uses the GARCH(1,1) model for updating volatility. The three param- eters are $\omega$, $\alpha$, and $\beta$. Describe the impact of making a small increase in each of the parameters while keeping the others fixed.

### 5.5.2 Correlations and Copulas

1. If you know the correlation between two variables, what extra information do you need to calculate the covariance?
2. What is the difference between correlation and dependence? Suppose that $y = x^2$ and $x$ is normally distributed with mean zero and standard deviation one. What is the correlation between $x$ and $y$?
3. Explain what is meant by tail dependence. How can you vary tail dependence by the choice of copula?
4. Suppose that two variables $V_1$ and $V_2$ have uniform distributions where all values between 0 and 1 are equally likely. Use a Gaussian copula to define the correlation structure between $V_1$ and $V_2$ with a copula correlation of 0.3. Produce a table for all combinations of values of 0.25, 0.50, and 0.75 for $V_1$ and $V_2$.

### 5.5.3 Valuation and Scenario Analysis

1. A stock price has an expected return of 12% and a volatility of 20%. It is currently \$50. What is the probability that it will be greater than \$70 in two years?
2. Explain the principle of risk-neutral valuation.
3. An analyst calculates the expected future value of a stock index in (a) the real world and (b) the risk-neutral world. Which would you expect to be higher? Why?
4. The value of a derivative that pays off \$100 after one year if a company has defaulted during the year is \$3. The value of a derivative that pays off \$100 after one year if a company has not defaulted is \$95. What is the risk-free rate? What is the risk neutral probability of default?

5. Explain why it is sometimes necessary to work in both the real world and the risk-neutral world when carrying out a scenario analysis to determine a confidence interval for the value of a portfolio in one year.

# 6 Market Risk I

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 11 - Value at Risk and Expected Shortfall
  - Chapter 14 - Interest Rate Risk
  - Chapter 15 - Derivatives Risk
  - Chapter 16 - Scenario Analysis and Stress Testing
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management

**Learning outcomes:**

- Explain the use of Value at Risk (VaR) in measuring portfolio risk.
- Describe advantages and limitations of VaR.
- Explain the use of Expected Shortfall (ES) in measuring portfolio risk.
- Understand conditions for coherent risk measures and provide examples.
- Define how to choose parameters for VaR and ES estimation.

## 6.1 Introduction

Definition

Market risk refers to the risk associated with fluctuations in stock prices, interest rates, exchange rates, and commodity prices.

Managing this risk involves identifying and measuring potential exposures and ensuring they align with an institution's risk tolerance. Effective management of market risk requires sound judgment, experience, and an understanding of the limitations and strengths of various risk models.

Compared to other types of financial risk, market risk **is generally easier to analyze** due to the abundance of data and accumulated industry expertise. However, estimating potential losses remains challenging because they rely on historical data and probabilistic assessments

of future conditions. Risk management models incorporate historical information alongside forward-looking judgments, establishing a framework to assess and test risk exposure.

Key metrics such as **Value at Risk (VaR)** and **Expected Shortfall (ES)** are commonly used to quantify market risk in a single, interpretable measure. VaR, originally introduced by JPMorgan, has become a standard tool among corporate treasurers, fund managers, and financial institutions. Although regulators have traditionally relied on VaR for capital requirement calculations, recent shifts have seen a preference for ES in certain areas of market risk evaluation.

## 6.2 Understanding Value at Risk (VaR)

Definition

**Value at Risk (VaR)** is a statistical measure that estimates the minimum expected loss of a portfolio over a specified time period and confidence level, under normal market conditions.

VaR has become a critical risk measure in global financial markets, offering a quantifiable estimate of potential losses over a defined time horizon. It can be expressed in currency units or as a percentage of portfolio value. Importantly, VaR reflects a minimum expected loss, not the maximum potential loss of the portfolio.

Example

A portfolio with a 5% VaR of $2.2 million over one day implies:

- A minimum loss of $2.2 million would be expected 5% of the time.
- Losses would be at least $2.2 million in 5% of cases.
- 95% of the time, the loss would not exceed $2.2 million.

A typical VaR statement specifies both the frequency of losses and the time period over which they might occur, with a **5% VaR implying a 95% confidence level**. This means that in 95% of scenarios, losses should not surpass the VaR amount. While often illustrated with a normal distribution, VaR calculations may use other distributions better suited to the unique characteristics of financial market returns.

```
import numpy as np
import plotly.graph_objects as go
from scipy.stats import norm

# Parameters
```

```
mu = 0  # Mean of the distribution
sigma = 1  # Standard deviation of the distribution
confidence_level = 0.95  # 95% confidence level
x = np.linspace(-4, 4, 1000)

# Calculate VaR at 5% level (inverse CDF for the normal distribution)
VaR = norm.ppf(1 - confidence_level, mu, sigma)

# Probability density function values
y = norm.pdf(x, mu, sigma)

# Plot the normal distribution and VaR threshold in Plotly
fig = go.Figure()

# Normal distribution curve
fig.add_trace(go.Scatter(x=x, y=y, mode='lines', name='Normal Distribution', line=dict(color=

# Highlighting the VaR area (left tail of the distribution at 5% level)
x_fill = np.linspace(-4, VaR, 1000)
y_fill = norm.pdf(x_fill, mu, sigma)

fig.add_trace(go.Scatter(
    x=np.append(x_fill, VaR), y=np.append(y_fill, 0),
    mode='lines', name='VaR (95% confidence)', fill='tozeroy',
    line=dict(color='blue'), fillcolor='rgba(0,0,255,0.3)'
))

# Update layout for the graph
fig.update_layout(
    title="Value at Risk (VaR) on a Normal Distribution",
    xaxis_title="Value",
    yaxis_title="Probability Density",
    showlegend=True,
    hovermode="x unified",
    width=745,
    height=600
)

fig.show()
```

```
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```

```
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```

### 6.2.1 Simple VaR Calculations

#### 6.2.1.1 Example 1

Suppose that the gain from a portfolio during six months is normally distributed with a mean of $2 million and a standard deviation of $10 million. How would you calculate the VaR for the portfolio with a time horizon of six months and confidence level of 99%, given the properties of the normal distribution?

#### 6.2.1.2 Example 2

Suppose that for a one-year project all outcomes between a loss of $50 million and a gain of $50 million are considered equally likely, resulting in a uniform distribution extending from –$50 million to +$50 million. How would you determine the VaR with a one-year time horizon and a 99% confidence level for this project?

#### 6.2.1.3 Example 3

A one-year project has a 98% chance of leading to a gain of $2 million, a 1.5% chance of leading to a loss of $4 million, and a 0.5% chance of leading to a loss of $10 million. How would you calculate the VaR with a confidence level of 99% and a one-year time horizon for this project, given the cumulative loss distribution?

## 6.3 Advantages and Limitations of VaR

### 6.3.1 Advantages of VaR

- **Simple and intuitive:** VaR offers an easily understood concept that answers the essential question, "How bad can things get?" making it accessible for decision-makers without technical expertise.
- **Clear communication tool:** By consolidating complex risk metrics into a single, interpretable number, VaR simplifies the communication of risk to stakeholders.
- **Risk comparability:** VaR enables the comparison of risks across various asset classes, portfolios, and trading units, fostering informed decision-making across diverse financial contexts.
- **Supports capital allocation:** VaR serves as a benchmark for allocating capital efficiently across different trading units or portfolio positions.

- **Performance evaluation:** VaR facilitates risk-adjusted performance measurement, offering a more comprehensive understanding of profitability by adjusting for the inherent risk in portfolios.
- **Backtesting capability:** The reliability of VaR can be validated through backtesting, allowing firms to assess the accuracy of their risk estimates based on historical data.
- **Regulatory acceptance:** Widely endorsed by regulators, such as the SEC, VaR is frequently featured in financial institutions' risk disclosures, fostering industry consistency and compliance.

### 6.3.2 Limitations of VaR

- **Subjectivity in estimation:** Estimating VaR involves subjective choices, such as selecting the confidence level, time horizon, and estimation method, which can lead to inconsistency.
- **Underestimates extreme events:** VaR may overlook the likelihood of severe adverse events or "left-tail events," often underrepresenting risk during extreme market downturns.
- **Ignores liquidity risk:** VaR may understate risk for illiquid assets, particularly during market stress when these assets are challenging to sell without substantial losses.
- **Sensitivity to correlation shifts:** VaR assumes stable correlations among assets, which can be misleading as correlations often spike during market crises, compounding risk.
- **Prone to regime shifts:** VaR may underestimate losses during low-volatility periods or trending markets, as it fails to account for shifts in volatility regimes.
- **Misinterpreted as a worst-case measure:** VaR does not reflect the worst possible loss and can give a false sense of security, as losses beyond VaR are possible and potentially severe.
- **Oversimplifies risk profiles:** While straightforward to communicate, VaR can obscure nuances of the underlying risk structure, providing an incomplete picture of potential outcomes.
- **Neglects right-tail events:** VaR emphasizes losses (left-tail events) and disregards potential gains (right-tail events), missing opportunities for a full distributional view of outcomes.
- **Risk of misleading compliance:** Traders can meet VaR limits without truly minimizing risk, as skewed strategies can fit within set limits but harbor large, unaccounted losses.

> **Example**
>
> A trader could structure a portfolio with a 99.1% chance of a loss under \$10 million but a 0.9% chance of a \$500 million loss, meeting a bank's one-day 99% VaR limit of \$10

million while masking potential catastrophic outcomes.

- **Encourages high-risk strategies:** Some trading strategies, such as writing out-of-the-money options, offer high returns but carry a low probability of large losses, which can appear benign within VaR limits but introduce hidden risks.
- **Incentive for risk-seeking behavior:** Traders motivated by returns may seek ways to engage in high-risk strategies that technically stay within VaR limits, thus exposing firms to heightened risk without breaching formal boundaries.

### 6.3.3 Extensions of VaR

No single risk model can address all of a risk manager's concerns, so several extensions of VaR provide additional insights into portfolio risk.

- **Conditional VaR (CVaR, or Expected Shortfall):** CVaR estimates the average loss given that the loss has exceeded the VaR threshold, offering a more complete picture of potential extreme losses.
- **Incremental VaR (IVaR):** This measures the change in the portfolio's VaR resulting from adjusting the size of a particular position, helping managers understand how modifications impact overall risk.
- **Marginal VaR (MVaR):** MVaR evaluates the effect of a small position change on total portfolio VaR, providing insights into each asset's contribution to overall risk within diversified portfolios.
- **Relative VaR (or Ex Ante Tracking Error):** This extension quantifies the potential deviation of a portfolio's performance from its benchmark, allowing managers to gauge risk relative to specific performance targets.

IVaR and MVaR are particularly valuable for assessing potential trades before execution, helping managers make proactive adjustments to maintain desired risk levels.

## 6.4 Expected Shortfall

Definition

**Expected Shortfall (ES)** is a risk measure that calculates the average loss in scenarios where losses exceed the Value at Risk (VaR) threshold, providing a more comprehensive view of potential extreme losses beyond the VaR estimate.

Expected Shortfall (ES), also known as conditional Value at Risk, conditional tail expectation, or expected tail loss, is an alternative risk measure to Value at Risk (VaR) that offers improved

incentives for traders by focusing on the severity of extreme losses. Unlike VaR, which estimates the maximum loss at a specified confidence level, ES provides a more comprehensive risk assessment by averaging losses that exceed the VaR threshold. This makes ES particularly valuable for understanding the risk of rare but significant losses.

To compute ES, it is necessary to first determine VaR, as ES is the expected loss over a time horizon $T$ given that the loss exceeds the VaR. ES depends on two parameters: the time horizon $T$ and the confidence level $X$. For example, if $X = 99\%$, $T$ is 10 days, and the VaR is \$64 million, then ES represents the average loss over those 10 days when losses exceed \$64 million.

Setting ES limits can reduce excessive risk-taking, as ES better captures the tail risks in a portfolio. It also has desirable mathematical properties, such as always acknowledging the benefits of diversification. However, ES can be more challenging to understand and implement compared to VaR, and back-testing procedures for ES are also more complex.

Alternative risk measures, such as the exponential spectral risk measure, adjust ES by using different weightings rather than equal weights across all extreme losses.

## 6.5 Coherent Risk Measures

Coherent risk measures offer a consistent and reliable framework for assessing risk by incorporating the benefits of diversification and ensuring that scaling a portfolio proportionally adjusts the risk measure. Expected Shortfall (ES) is an example of a coherent risk measure, while Value at Risk (VaR) is not coherent as it does not satisfy all coherence properties, specifically subadditivity.

A coherent risk measure is characterized by four fundamental axioms:

1. **Monotonicity**: If portfolio $A$ consistently outperforms portfolio $B$, the risk measure for $A$ should be less than or equal to that of $B$.
2. **Subadditivity**: The risk measure of a combined portfolio $A + B$ should be less than or equal to the sum of the individual risk measures of $A$ and $B$, rewarding diversification.
3. **Positive Homogeneity**: Scaling a portfolio by a factor $\lambda$ proportionally scales the risk measure. Thus, the risk measure of $\lambda A$ is $\lambda$ times that of $A$ for any $\lambda > 0$.
4. **Translation Invariance**: Adding a risk-free asset with return $r$ to a portfolio should decrease the risk measure by $r$. For portfolio $A$, the risk measure of $A + r$ is the risk measure of $A$ minus $r$.

These axioms ensure that coherent risk measures, such as ES, provide a robust assessment of risk, particularly in recognizing the advantages of diversification.

### 6.5.1 Subadditivity Example

Consider two independent projects, each with a 2% probability of incurring a \$10 million loss and a 98% probability of a \$1 million loss within one year.

- The one-year 97.5% **VaR (Value at Risk)** for each project is \$1 million.

When these projects are combined into a portfolio, the loss distribution changes as follows:

- A 0.04% probability of a \$20 million loss,
- A 3.92% probability of an \$11 million loss,
- A 96.04% probability of a \$2 million loss.

In this combined portfolio, the one-year 97.5% VaR is \$11 million, exceeding the sum of the individual VaRs (\$2 million) by \$9 million, thereby **violating the subadditivity property**.

For **Expected Shortfall (ES)**, each project's 97.5% ES considers a 2% chance of a \$10 million loss and a 0.5% chance of a \$1 million loss, leading to an expected loss of \$8.2 million per project. For the combined portfolio, the ES considers:

- A 0.04% chance of a \$20 million loss,
- A 2.46% chance of an \$11 million loss.

This results in a portfolio ES of \$11.144 million. Here, the sum of the individual project ES values (\$16.4 million) is greater than the portfolio ES (\$11.144 million), thereby **satisfying the subadditivity property**.

## 6.6 Choice of Parameters for VaR and ES

For both Value at Risk (VaR) and Expected Shortfall (ES), users must specify two key parameters: **the time horizon and the confidence level**. Typically, it is assumed that changes in portfolio value at the time horizon follow a normal distribution, though this assumption may not always hold.

When the portfolio loss distribution has a mean $\mu$ and standard deviation $\sigma$, VaR at confidence level $X$ is calculated as:

$$\text{VaR} = \mu + \sigma N^{-1}(X)$$

where $N^{-1}(\cdot)$ denotes the inverse cumulative normal distribution. For short time horizons, $\mu$ is often assumed to be zero, simplifying VaR to be directly proportional to $\sigma$.

Similarly, if the portfolio loss is normally distributed with mean $\mu$ and standard deviation $\sigma$, ES at confidence level $X$ is given by:

$$\text{ES} = \mu + \sigma \frac{e^{-(Y^2/2)}}{\sqrt{2\pi}(1-X)}$$

where $Y$ is the $X$th percentile of the standard normal distribution.

> **Example**
>
> Assume the loss from a portfolio over a 10-day horizon is normally distributed with a mean of zero and a standard deviation of \$20 million. The calculations for 10-day 99% VaR and ES are:
>
> $$\text{VaR} = 20 \times N^{-1}(0.99) = 20 \times 2.326 = \$46.5 \text{ million}$$
>
> $$\text{ES} = 20 \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi} \times 0.01} = \$53.3 \text{ million}$$
>
> These values provide a clearer assessment of potential losses, with VaR representing the threshold loss and ES giving the expected average loss in extreme scenarios.

### 6.6.1 The Time Horizon

The choice of time horizon for VaR and ES depends on the liquidity and trading frequency of the assets in the portfolio. Liquid, actively traded positions typically use shorter time horizons, while less liquid portfolios may require longer horizons. For portfolios with assets of varying liquidity, analysts may apply tailored risk measures to each asset.

A common starting point is a 1-day time horizon, from which T-day VaR and T-day ES can be calculated using the following formulas:

$$\text{T-day VaR} = \text{1-day VaR} \times \sqrt{T}$$

$$\text{T-day ES} = \text{1-day ES} \times \sqrt{T}$$

These formulas hold precisely when daily portfolio value changes are independently and identically normally distributed with a mean of zero. In other cases, these calculations serve as approximations.

### 6.6.2 Impact of Autocorrelation

Portfolio value changes are not always independent. When portfolio returns exhibit higher autocorrelation, VaR and ES calculations may underestimate risk. To adjust for autocorrelation, the standard deviation $\sigma$ can be modified when calculating VaR and ES. The adjusted standard deviation for the sum of $T$ daily changes, considering autocorrelation, is given by:

$$\sigma\sqrt{T + 2(T-1)\rho^1 + 2(T-2)\rho^2 + \cdots + 2\rho^{T-1}}$$

### 6.6.3 Confidence Level

The confidence level for VaR and ES depends on risk tolerance and objectives, such as maintaining a particular credit rating. For example, a bank aiming for an AA rating might select a 99.98% confidence level to match a 0.02% annual default probability. Estimating VaR directly at very high confidence levels can be challenging, but extreme value theory offers methods to improve accuracy in such cases.

### 6.6.4 Back-Testing

Back-testing assesses the accuracy of VaR models by comparing model predictions with actual historical losses. This process is straightforward for VaR, as it relies on tracking "exceptions" where observed losses exceed the predicted VaR. A deviation from expected exception rates may reveal model weaknesses. Statistical tests help determine if a model should be adjusted or rejected based on these exceptions, and regulators may require increased market risk capital if back-testing results over a 250-day period are unsatisfactory.

## 6.7 Additional Topics

### 6.7.1 Interest Rate Risk

Financial institutions often have exposure to multiple yield curves, necessitating the calculation of VaR and ES to manage this risk. Two common approaches include treating each interest rate as a separate risk factor or applying principal components analysis (PCA). The bank's net interest margin, which measures the difference between earned and paid interest, is a key focus of asset/liability management to maintain stability.

Duration is a crucial measure, assessing portfolio sensitivity to shifts in the zero-coupon yield curve, while convexity adjustments add precision for large parallel shifts (but less so for non-parallel shifts). Multiple duration measures (or "deltas") capture different potential yield

curve changes. PCA can serve as an alternative to multiple deltas, effectively hedging against common yield curve shifts.

### 6.7.2 Derivatives Risk

In managing derivatives risk, traders monitor Greek letters such as delta, gamma, and vega to stay within risk limits. Delta ($\Delta$) represents the rate of change in portfolio value relative to changes in the underlying asset's price. Delta hedging, a fundamental practice, involves creating a delta-neutral position by taking an offsetting position of $-\Delta$ in the underlying asset. Complex portfolios require frequent rebalancing to maintain delta neutrality.

Beyond delta, traders examine gamma (the rate of change of delta) and vega (sensitivity to volatility), which can be adjusted through options trading. Daily rebalancing is typical for delta neutrality, though maintaining gamma and vega neutrality is less feasible and is managed through corrective actions if these measures exceed limits.

### 6.7.3 Scenario Analysis and Stress Testing

Stress testing plays a critical role in risk management by evaluating the impacts of extreme scenarios typically beyond the scope of VaR and ES analysis. This approach can reveal potential systemic issues and aid in risk mitigation. Common scenario generation methods include:

- Testing the impact of extreme movements in a single market variable,
- Using historical data from periods of extreme market shocks,
- Constructing plausible extremes through a committee of senior management and economists,
- Reverse stress testing with algorithms to uncover large-loss scenarios.

Scenario analysis also considers knock-on effects, such as market turmoil leading to a liquidity shortage or flight to quality. Regulators in the US, EU, and UK develop standardized stress scenarios for large financial institutions, which may prompt capital-raising measures. Integrating stress testing with VaR through subjective probability weighting remains largely unexplored by institutions and regulators.

## 6.8 Practice Questions and Problems

1. What is the difference between expected shortfall and VaR? What is the theoretical advantage of expected shortfall over VaR?

2. A fund manager announces that the fund's one-month 95% VaR is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?

3. A fund manager announces that the fund's one-month 95% expected shortfall is 6% of the size of the portfolio being managed. You have an investment of $100,000 in the fund. How do you interpret the portfolio manager's announcement?

4. Suppose that each of two investments has a 0.9% chance of a loss of $10 million and a 99.1% chance of a loss of $1 million. The investments are independent of each other.

   - What is the VaR for one of the investments when the confidence level is 99%?
   - What is the expected shortfall for one of the investments when the confidence level is 99%?
   - What is the VaR for a portfolio consisting of the two investments when the confidence level is 99%?
   - What is the expected shortfall for a portfolio consisting of the two investments when the confidence level is 99%?
   - Show that in this example VaR does not satisfy the subadditivity condition, whereas expected shortfall does.

5. Suppose that the change in the value of a portfolio over a one-day time period is normal with a mean of zero and a standard deviation of $2 million; what is (a) the one-day 97.5% VaR, (b) the five-day 97.5% VaR, and (c) the five-day 99% VaR?

6. What difference does it make to your answer to Problem 5 if there is first-order daily autocorrelation with a correlation parameter equal to 0.16?

7. The change in the value of a portfolio in one month is normally distributed with a mean of zero and a standard deviation of $2 million. Calculate the VaR and ES for a confidence level of 98% and a time horizon of three months.

8. Suppose that each of two investments has a 4% chance of a loss of $10 million, a 2% chance of a loss of $1 million, and a 94% chance of a profit of $1 million. They are independent of each other.

   - What is the VaR for one of the investments when the confidence level is 95%?
   - What is the expected shortfall when the confidence level is 95%?
   - What is the VaR for a portfolio consisting of the two investments when the confidence level is 95%?
   - What is the expected shortfall for a portfolio consisting of the two investments when the confidence level is 95%?
   - Show that, in this example, VaR does not satisfy the subadditivity condition, whereas expected shortfall does.

9. Suppose the first-order autocorrelation for daily changes in the value of a portfolio is 0.12. The 10-day VaR, calculated by multiplying the one-day VaR by $\sqrt{10}$, is $2 million. What is a better estimate of the VaR that takes account of autocorrelation?

10. The change in the value of a portfolio in three months is normally distributed with a mean of \$500,000 and a standard deviation of \$3 million. Calculate the VaR and ES for a confidence level of 99.5% and a time horizon of three months.

# 7 Reading Week

Tip

Take some rest and revise the first six topics!

# 8 Market Risk II

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 12 - Historical Simulation and Extreme Value Theory
  - Chapter 13 - Model-Building Approach
- Pirie, W. L., and M. P. Kritzman. 2017. Derivatives. CFA Institute Investment Series.
  - Chapter 6 - Risk Management

**Learning outcomes:**

- Understand the concept of Value at Risk (VaR) and Expected Shortfall (ES) as a measure of portfolio risk, and the importance of different estimation methods.
- Understand the Historical Simulation method VaR and ES estimation.
- Recognize the benefits and limitations of using historical data to estimate future portfolio risk.
- Learn Parametric Approach to VaR and ES estimation.
- Understand the limitations of assuming a normal distribution of returns.
- Acquire knowledge of the Monte Carlo Simulation method.
- Understand the computational intensity and the assumptions involved in the Monte Carlo simulation method.
- Develop the ability to critically evaluate each of these methods in different investment contexts and scenarios.

## 8.1 Introduction

Several methods are available for estimating Value at Risk (VaR), each with distinct advantages and assumptions:

1. **Historical Simulation**: This approach uses past data to simulate potential changes in portfolio value, calculating VaR based on the worst observed losses at a specified confidence level.

2. **Parametric (Analytical) Approach**: Assuming that asset returns follow a normal distribution, this method employs the mean and standard deviation of returns to calculate VaR. It can be applied to both individual assets and portfolios, incorporating asset correlations.

3. **Monte Carlo Simulation**: By generating random price paths for assets, this method simulates thousands of possible portfolio outcomes. VaR is then determined from the distribution of simulated losses.

4. **Extreme Value Theory (EVT)**: EVT models the tail of the loss distribution, making it particularly useful for calculating VaR at very high confidence levels where extreme losses are of primary concern.

## 8.2 Historical Simulation

The historical simulation method for VaR evaluates the current portfolio by revaluing it based on historical changes in key risk factors over a specified lookback period. The process involves:

1. **Decomposition and Data Collection**: The portfolio is broken down into individual risk factors, and historical returns for each factor are collected over the lookback period.
2. **Repricing**: The current portfolio is repriced using the returns for each day in the lookback period. This generates a range of simulated portfolio values.
3. **Sorting Results**: These simulated outcomes are sorted from the largest loss to the greatest gain.
4. **VaR Calculation**: To estimate a one-day VaR at a given confidence level (e.g., 5%), a cutoff point is selected where 5% of the outcomes result in larger losses, representing the VaR threshold.
5. **Expected Shortfall (ES)**: To calculate ES, the average of all losses that exceed the VaR threshold is taken, providing a view of the tail risk beyond VaR.

This approach leverages real historical data, providing a non-parametric view of potential losses, although it assumes that historical trends will adequately represent future risk scenarios.

> **Example**
>
> Consider a 500-day time window of daily returns for the SPY ETF to calculate the 1-day VaR and ES at a 99% confidence level. The table below lists the 10 worst daily returns from this period:

| No. | Date | Log Returns |
|---|---|---|
| 1. | 2024-08-05 | -0.029556 |
| 2. | 2022-11-02 | -0.025416 |
| 3. | 2022-12-15 | -0.024766 |
| 4. | 2024-07-24 | -0.022923 |
| 5. | 2022-11-09 | -0.020817 |
| 6. | 2024-09-03 | -0.020794 |
| 7. | 2023-02-21 | -0.020265 |
| 8. | 2024-08-02 | -0.018794 |
| 9. | 2023-03-09 | -0.018622 |
| 10. | 2022-12-05 | -0.018153 |

In this example:

- **1% of 500 observations equals 5**, meaning the 5th worst loss represents the 1-day 99% VaR.
- **VaR**: The daily 99% VaR is the 5th worst return, which corresponds to -0.020817.
- **ES**: The daily 99% ES is the average of the 4 worst returns, -0.025665, providing an estimate of the average loss beyond the VaR threshold.

This method leverages historical returns to identify potential extreme losses based on recent market behavior.

### 8.2.1 Stressed VaR and Stressed ES

In the historical simulation method, VaR and ES are usually based on recent data, known as current VaR and current ES. To account for risk under extreme conditions, regulators introduced stressed VaR and ES, which use data from a particularly adverse 251-day period for the current portfolio. This period yields 250 scenarios, each reflecting changes in market variables between consecutive days.

- **Stressed VaR**: The one-day 99% stressed VaR is the midpoint loss between the second and third worst scenarios.
- **Stressed ES**: The one-day 99% stressed ES is calculated as the average of the two worst losses.

### 8.2.2 Accuracy of VaR

The historical simulation method for Value at Risk (VaR) relies on a finite set of observations, introducing estimation error. The standard error for the $q$-percentile of the distribution,

estimated as $x$, is given by:

$$\text{Standard Error} = \frac{1}{f(x)}\sqrt{\frac{(1-q)q}{n}}$$

where $n$ is the sample size and $f(x)$ represents an estimate of the probability density at $x$. Generally, the standard error for VaR estimates from historical simulation is high, decreasing as sample size grows or confidence level lowers. Loss distributions often have heavier tails than the normal distribution, so using a Pareto distribution for $f(x)$ can yield more accurate standard error estimates.

**Example**

1. We aim to estimate the 99th percentile of a loss distribution using 500 observations, so $n = 500$ and $q = 0.99$.
2. Assuming a normal distribution with a mean of 0 and a standard deviation of \$10 million.
3. In Excel, the 99th percentile is calculated as $\text{NORMINV}(0.99, 0, 10) = 23.26$.
4. The density $f(x)$ at this point is $\text{NORMDIST}(23.26, 0, 10, \text{FALSE}) = 0.0027$.
5. The standard error of the estimate is:

$$\frac{1}{0.0027} \times \sqrt{\frac{0.01 \times 0.99}{500}} = 1.67$$

6. If the historical simulation estimate of the 99th percentile is \$25 million, a 95% confidence interval is calculated as:

$$25 \pm 1.96 \times 1.67$$

This results in an interval from \$21.7 million to \$28.3 million, providing an estimate range for the 99th percentile at a 95% confidence level.

### 8.2.3 Extensions and Computational Issues

VaR and ES calculations rely on recent historical data under the assumption that this provides a reliable forecast of future behavior. Traditional historical simulation assigns equal weight to each past observation, but there are ways to refine this approach:

- **Weighting of Observations**: Recent data can be weighted more heavily to reflect current volatility and macroeconomic conditions. Exponential weighting, where weights decay over time, is commonly used for this purpose.

- **Volatility Scaling for Market Variables**: Incorporating volatility estimates into historical simulations provides additional accuracy. For instance, using the Exponentially Weighted Moving Average (EWMA) allows dynamic monitoring of the standard deviation of simulated losses.
- **Bootstrap Method**: This involves creating multiple simulated datasets by sampling observations with replacement, providing a robust estimate of potential outcomes.

Calculating VaR and ES through historical simulation requires revaluing the entire portfolio, often through multiple rounds of simulation (e.g., 500 runs), which can be computationally demanding. For portfolios with complex instruments valued via Monte Carlo simulation, this can lead to a "simulation within a simulation" scenario, further escalating computational load.

To reduce processing time, financial institutions may employ an approximation technique using a Taylor series expansion, which simplifies the calculations while preserving accuracy.

### 8.2.4 Extreme Value Theory

Extreme Value Theory (EVT) is a method for estimating the tails of a distribution, based on the power law, which can enhance Value at Risk (VaR) or Expected Shortfall (ES) calculations, especially at high confidence levels. EVT works by smoothing and extrapolating the tails of an empirical distribution, yielding more accurate tail estimates that better reflect the distribution's extreme values.

Applying EVT to the probability distribution of daily portfolio changes derived from historical data mitigates calculation uncertainties. Even with limited datasets (e.g., 500 days), EVT can provide reliable VaR or ES estimates at very high confidence levels, such as 99.9%.

### 8.2.5 Advantages of the Historical Simulation Method

- **Simplicity and Transparency**: The method is intuitive and straightforward, relying on historical data to model potential future scenarios.
- **Non-Parametric Approach**: Unlike parametric methods, historical simulation does not assume a specific distribution (e.g., normal) for returns, making it resilient to non-normal returns.
- **Captures Market Volatility**: By using actual historical data, this method naturally incorporates periods of high volatility, producing realistic VaR estimates during turbulent times.
- **Empirical Foundation**: Historical simulation relies on observed data rather than theoretical assumptions, offering a more grounded view of extreme event risk.
- **Flexibility**: The approach can easily be adjusted to accommodate various time horizons, confidence levels, and other parameters as needed.

- **Adaptability to New Data**: As new data becomes available, it can be seamlessly integrated into the model, allowing continuous updates to reflect evolving market conditions.
- **Capability with Options**: Unlike parametric methods, historical simulation can account for options and other non-linear instruments, enhancing its applicability to a broader range of portfolios.

## 8.3 Parametric Approach

The parametric method, also known as the analytical or **variance-covariance method**, is an alternative to historical simulation for estimating risk measures like VaR and ES. It assumes that the return distributions of portfolio risk factors follow a **normal distribution**, requiring only the expected value and standard deviation as parameters, which are estimated using historical data and adjusted for consistency with long-term values.

This approach is efficient when portfolio value changes are linearly dependent on market variable changes, and when these changes can be assumed to be multivariate normal, making it **computationally faster than historical simulation**. This method builds on Harry Markowitz's portfolio theory but is not suitable for portfolios with non-linear derivatives. While other distributions can be used to capture more complex behaviors, they add computational complexity. It's also possible to calculate stressed measures by choosing appropriate historical data to estimate volatilities and correlations.

### 8.3.1 The Basic Methodology

Assuming a normal distribution of returns, VaR and ES can be calculated with the following equations:

$$\text{VaR} = \mu + \sigma \times \sqrt{T} \times N^{-1}(X)$$

$$\text{ES} = \mu + \sigma \times \sqrt{T} \times \frac{e^{-(Y^2/2)}}{\sqrt{2\pi}(1-X)}$$

where:

- $\mu$ is the mean of losses, typically assumed to be zero,
- $\sigma$ is the standard deviation of losses,
- $X$ is the confidence level,
- $N^{-1}(\cdot)$ is the inverse cumulative normal distribution,
- $Y$ is the $X$ th percentile of the standard normal distribution,

- $T$ is the time horizon in days, where for extended horizons, using an appropriate return frequency is recommended.

The standard deviation of a multi-asset portfolio is calculated by:

$$\sigma_P = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} \rho_{ij} \delta_i \delta_j \sigma_i \sigma_j}$$

or equivalently:

$$\sigma_P = \sqrt{\sum_{i=1}^{n} \delta_i^2 \sigma_i^2 + \sum_{i \neq j} \rho_{ij} \delta_i \delta_j \sigma_i \sigma_j}$$

where:

- $\rho_{ij}$ is the correlation between assets $i$ and $j$,
- $\delta_i$ is the weight of asset $i$ in the portfolio,
- $\sigma_i$ is the standard deviation of asset $i$'s returns.

This approach efficiently computes portfolio risk in linear and normally distributed portfolios but is less accurate for portfolios with non-linear payoffs, such as those involving options.

### 8.3.2 Parametric Approach Example

#### 8.3.2.1 Asset A

Consider a portfolio consisting of one share of **Asset A** valued at \$10 million, with a 10-day time horizon and a 99% confidence level for VaR. The assumptions are as follows:

- **Daily volatility**: 2%, or \$200,000.
- **Expected change**: Assumed to be zero.
- **Distribution**: Changes in value are assumed to be normally distributed with a mean of zero and a standard deviation of \$200,000.

Calculations:

- **One-day 99% VaR**: $2.326 \times \$200,000 = \$465,300$.
- **10-day 99% VaR**: $2.326 \times \$200,000 \times \sqrt{10} = \$1,471,300$.
- **10-day 99% ES**:

$$\text{ES} = \$200,000 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi} \times (1 - 0.99)} = \$1,686,994$$

#### 8.3.2.2 Asset B

Now consider a portfolio with one share of **Asset B** valued at \$5 million, using the same 10-day horizon and 99% confidence level. The parameters are:

- **Daily volatility**: 1%, or \$50,000.
- **Expected change**: Assumed to be zero.
- **Distribution**: Changes in value are normally distributed, with a mean of zero and a standard deviation of \$50,000.

Calculations:

- **One-day 99% VaR**: $2.326 \times \$50,000 = \$116,300$.
- **10-day 99% VaR**: $2.326 \times \$50,000 \times \sqrt{10} = \$367,800$.
- **10-day 99% ES**:

$$\text{ES} = \$50,000 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi} \times (1 - 0.99)} = \$421,400$$

These examples illustrate the application of the parametric method for estimating VaR and ES, where the assumption of normality enables direct calculation based on volatility and confidence level.

#### 8.3.2.3 Two-Asset Case

Assume a portfolio consisting of **Asset A** and **Asset B** with a correlation of 0.3. The combined standard deviation ($\sigma_{X+Y}$) of two variables $X$ and $Y$, given their individual standard deviations ($\sigma_X$ and $\sigma_Y$) and the correlation coefficient $\rho$, is calculated as:

$$\sigma_{X+Y} = \sqrt{\sigma_X^2 + \sigma_Y^2 + 2\rho\sigma_X\sigma_Y}$$

Using the values from the previous example for Asset A (\$200,000) and Asset B (\$50,000):

$$\sigma_{A+B} = \sqrt{200,000^2 + 50,000^2 + 2 \times 0.3 \times 200,000 \times 50,000} = \$220,227$$

Calculations:

- **One-day 99% VaR**: $2.326 \times 220,227 = 512,300$
- **10-day 99% VaR**: $512,300 \times \sqrt{10} = 1,620,100$
- **10-day 99% ES**:

$$\text{ES} = 220,227 \times \sqrt{10} \times \frac{e^{-(2.326^2/2)}}{\sqrt{2\pi} \times (1 - 0.99)} = 1,856,100$$

#### 8.3.2.4 Benefits of Diversification

- **10-day 99% VaR for Asset A**: \$1,471,300
- **10-day 99% VaR for Asset B**: \$367,800
- **10-day 99% VaR for both assets combined**: \$1,620,100

The difference between the sum of individual VaRs and the combined VaR represents the diversification benefit:

$$(1,471,300 + 367,800) - 1,620,100 = 219,000$$

Thus, \$219,000 is the amount of risk diversified away due to the less-than-perfect correlation between the assets. If both assets were perfectly correlated, the combined VaR would equal the sum of the individual VaRs, with no diversification benefit. This benefit applies to both VaR and ES measures.

> Note: For non-normal distributions, the VaR of a combined portfolio can sometimes exceed the sum of the individual VaRs. Expected Shortfall (ES) does not exhibit this limitation, making it a more robust measure under such conditions.

### 8.3.3 Parametric Approach vs. Historical Simulation

#### 8.3.3.1 Advantages of Parametric Approach

- Allows for rapid calculation of VaR and ES.
- Integrates easily with procedures for updating volatility and correlation, adapting quickly to changing market conditions.
- While volatility adjustments can be applied in historical simulation, they are more naturally incorporated in the parametric approach.

#### 8.3.3.2 Disadvantages of Parametric Approach

- Assumes that changes in portfolio value are linearly related to proportional or actual changes in risk factors, which may not hold in practice.
- Assumes that daily changes in risk factors follow a multivariate normal distribution, whereas real market distributions often deviate from normality, exhibiting heavier tails.

- Highly sensitive to parameter estimates, especially covariances in portfolios, which can lead to inaccuracies.
- Less suitable for portfolios containing options due to their non-linear payoff structures, which are challenging to model with a parametric approach.

## 8.4 Monte Carlo Simulation

The Monte Carlo simulation method allows users to define custom assumptions about distribution statistics, generating random hypothetical returns to estimate a portfolio's risk. It is widely used for pricing complex options and handling investments with multiple uncertain variables due to its flexibility in accommodating non-normal distributions.

- **Applicability to Complex Portfolios**: Ideal for portfolios with multiple assets and complex risk interactions, Monte Carlo simulation does not rely on the normal distribution assumption.
- **Customizable Reliability**: Users select the number of random values to simulate; a larger number increases the accuracy of results but requires more computational time.
- **Correlation Considerations**: Simulations must account for correlations among variables to avoid independent, unrealistic scenarios.
- **Process**: The method involves simulating risk factor returns, revaluing portfolio exposures under these returns, and then sorting the results to identify a percentile for VaR estimation.
- **Accuracy for Non-linear Instruments**: The flexibility of Monte Carlo simulation allows it to handle any distribution, making it especially effective for portfolios containing options or bonds with embedded options, where parametric methods fall short.

Monte Carlo's adaptability makes it a powerful tool for accurate risk estimation, though it demands significant computational resources, particularly for highly detailed simulations.

### 8.4.1 Procedure for Monte Carlo Simulation to Calculate VaR and ES

1. **Define Assumptions**: Specify assumptions about the statistical distribution of returns for each risk factor in the portfolio (e.g., mean, volatility, skewness). Monte Carlo simulation allows for any distribution, making it suitable for complex portfolios and non-linear instruments like options.

2. **Generate Random Scenarios**: Decide on the number of random scenarios to simulate. Each scenario represents a possible outcome for the portfolio's value. A higher number of scenarios improves accuracy but increases computation time.

3. **Simulate Risk Factor Returns**: For each scenario, generate random returns for all risk factors in the portfolio, ensuring the correct distribution characteristics and incorporating the correlations between risk factors.
4. **Reprice Portfolio**: For each simulated scenario, revalue the portfolio based on the simulated returns. This step may involve complex calculations, particularly for non-linear instruments, as it requires calculating how each risk factor affects portfolio value.
5. **Calculate Portfolio Losses**: Determine the portfolio's loss or gain for each scenario by comparing its revalued amount to its original value. Sort these losses from largest to smallest.
6. **Estimate VaR**: Identify the loss at the desired percentile to estimate VaR. For example, for a 99% confidence level, VaR is the loss at the 1st percentile of sorted losses, meaning only 1% of simulated losses are expected to exceed this amount.
7. **Calculate ES**: Expected Shortfall is calculated by averaging the losses that exceed the VaR threshold. For a 99% confidence level, ES is the average of the worst 1% of simulated losses.

This procedure allows for robust risk estimation by accommodating any return distribution and accounting for complex interactions within the portfolio, making it particularly suited for portfolios with non-linear or path-dependent instruments.

## 8.5 Practice Questions and Problems

Use excel file 'daily_returns.xlsx' containing daily percentage returns in % for selected ETFs and calculate/show:

1. Show distribution of daily returns for each asset for the whole period.
2. Calculate 95% and 99% one-day **VaR** for the last 500 days using historical simulation and variance-covariance approach.
3. Calculate 95% and 99% one-day **ES** for the last 500 days using historical simulation and variance-covariance approach.
4. Convert VaR and ES to 10-day estimates.
5. Find 95% stressed VaR (251-days) for each asset.

### 8.5.1 Further Questions

1. What assumption is being made when VaR is calculated using the historical simulation approach and 500 days of data?

2. Suppose we estimate the one-day 95% VaR from 1,000 observations (in millions of dollars) as 5. By fitting a standard distribution to the observations, the probability density function of the loss distribution at the 95% point is estimated to be 0.01. What is the standard error of the VaR estimate?

3. Consider a position consisting of a \$100,000 investment in asset A and a \$100,000 investment in asset B. Assume that the daily volatilities of both assets are 1% and that the coefficient of correlation between their returns is 0.3. What are the five-day 97% VaR and ES for the portfolio?

# 9 Credit Risk

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 17 - Estimating Default Probabilities
  - Chapter 18 - xVAs
  - Chapter 19 - Credit Value at Risk

**Learning outcomes:**

- Understand default probabilities, recovery rates, and their importance in risk management.
- Explore credit ratings, CDS markets, and credit spreads for estimating defaults.
- Distinguish between real-world and risk-neutral probabilities and methods to infer them.
- Understand Credit Value Adjustment (CVA), Debit Value Adjustment (DVA), and their implications.
- Learn the risks and complexities in CVA, including wrong-way and right-way risk.
- Explore Credit Value at Risk (VaR) and methods to calculate VaR or Expected Shortfall.

## 9.1 Estimating Default Probabilities

### 9.1.1 Credit Ratings

#### 9.1.1.1 Credit Rating Agencies (e.g., Moody's, S&P, Fitch)

Credit rating agencies assign ratings to corporate bonds, providing an assessment of the issuer's creditworthiness. These ratings are typically stable and change infrequently, as agencies aim to avoid frequent reversals. They often use a "through-the-cycle" approach, focusing on long-term credit stability and disregarding short-term economic fluctuations.

#### 9.1.1.2 Market-Based Providers (e.g., Moody's KMV, Kamakura)

Unlike traditional rating agencies, these companies estimate default probabilities using market data, such as equity prices and other financial variables. Their models are more dynamic and responsive to real-time market changes, prioritizing timely insights over stability.

#### 9.1.1.3 Internal Credit Ratings in Banks

Banks use internal credit rating systems to evaluate the credit risk of small and medium-sized enterprises (SMEs) that lack publicly traded debt. These ratings incorporate financial ratios, such as:

- **Profitability ratios:** e.g., Return on Assets (ROA).
- **Leverage ratios:** e.g., Debt-to-Equity Ratio.

Additionally, banks analyze projected cash flow statements to assess a firm's ability to meet its debt obligations.

#### 9.1.1.4 Altman's Z-Score

The Z-Score is a statistical model developed to predict corporate default risk using a combination of accounting ratios. The formula is:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 0.999X_5$$

Where:

- $X_1$: Working capital / Total assets
- $X_2$: Retained earnings / Total assets
- $X_3$: Earnings before interest and taxes (EBIT) / Total assets
- $X_4$: Market value of equity / Book value of total liabilities
- $X_5$: Sales / Total assets

**Interpretation of Z-Scores:**

- $Z > 3.0$: Low probability of default.

- $2.7 < Z < 3.0$: Company is on alert.
- $1.8 < Z < 2.7$: Moderate risk of default.
- $Z < 1.8$: High likelihood of financial distress.

This framework is widely used for early warning signals and credit risk assessment in non-financial firms.

### 9.1.2 Historical Default Probabilities

Credit rating agencies compile historical data to track default experiences over time, segmented by credit ratings. This information provides valuable insights into the likelihood of default for bonds of varying creditworthiness.

**Table of Cumulative Default Probabilities (1981–2020):**

| Time (years) | 1 | 2 | 3 | 4 | 5 | 7 | 10 | 15 |
|---|---|---|---|---|---|---|---|---|
| AAA | 0.00 | 0.03 | 0.13 | 0.24 | 0.34 | 0.51 | 0.70 | 0.90 |
| AA | 0.02 | 0.06 | 0.11 | 0.21 | 0.30 | 0.49 | 0.70 | 0.99 |
| A | 0.05 | 0.13 | 0.22 | 0.33 | 0.46 | 0.76 | 1.20 | 1.84 |
| BBB | 0.16 | 0.43 | 0.75 | 1.14 | 1.54 | 2.27 | 3.24 | 4.54 |
| BB | 0.63 | 1.93 | 3.46 | 4.99 | 6.43 | 8.89 | 11.64 | 14.65 |
| B | 3.34 | 7.80 | 11.75 | 14.89 | 17.35 | 20.99 | 24.62 | 28.24 |
| CCC/C | 28.30 | 38.33 | 43.42 | 46.36 | 48.58 | 50.75 | 52.76 | 54.76 |

- For investment-grade bonds (BBB and above), the probability of default typically increases over time.
- For lower-rated bonds, default probabilities may decrease as weaker issuers default early, leaving more resilient entities in the sample.

> **Example**
>
> Default Probabilities for BBB Bonds:
>
> - A bond initially rated BBB has a 0.16% chance of defaulting in the first year, 0.43% in the second year, etc.
> - **Yearly Probability Calculation**: The probability of defaulting in a specific year can be calculated by subtraction. E.g., for the second year, $0.43\% - 0.16\% = 0.27\%$.

#### 9.1.2.1 Unconditional Default Probability

Definition

**Unconditional Default Probabilities:** Reflect the likelihood of default during a specific year as measured from the start of the observation period.

The unconditional default probability is the increase in cumulative probability during a specific year.

Example

Unconditional probability of default for CCC/C bonds during third year:

$$43.42\% - 38.33\% = 5.09\%.$$

#### 9.1.2.2 Conditional Default Probability

Definition

**Conditional Default Probabilities:** Indicate the probability of default in a given year, assuming no prior default.

The conditional probability adjusts for survival until a specific time, offering more context for ongoing risk management.

**Steps to Calculate:**

1. Calculate the probability of surviving until the end of the previous year.

2. Divide the yearly default probability by the survival probability.

Example

Conditional probability of default for CCC/C bonds during third year:

$$100\% - 38.33\% = 61.67\%$$

$$\frac{5.09\%}{61.67\%} = 8.25\%.$$

#### 9.1.2.3 Applications

- **Unconditional Probabilities:** Useful for static, long-term risk assessments.
- **Conditional Probabilities:** Crucial for dynamic risk management as they reflect updated information and ongoing contexts.

By analyzing these probabilities, financial practitioners can better understand and manage the risk profiles of bonds, making informed investment and credit decisions.

#### 9.1.2.4 Hazard Rates

Definition

The **hazard rate** ($\lambda(t)$) represents the intensity of default over a short time period, conditional on the entity's survival until that time. It bridges the concepts of conditional and unconditional probabilities, offering a dynamic measure of default risk.

- The hazard rate measures the likelihood of default in a short time interval, given no prior default.
- It incorporates survival probabilities to provide a nuanced understanding of credit risk over time.
- $V(t)$ is the cumulative probability of the company surviving to time $t$.

**Conditional Probability of Default:**

$$\frac{V(t) - V(t + \Delta t)}{V(t)} = \lambda(t)\Delta t$$

**Cumulative Survival Probability:**

$$V(t) = e^{-\int_0^t \lambda(\tau)d\tau}$$

**Cumulative Default Probability:**

$$Q(t) = 1 - e^{-\int_0^t \lambda(\tau)d\tau} = 1 - e^{-\lambda(t)t}$$

#### 9.1.2.5 Example Calculations with Hazard Rates

Assume a constant hazard rate of 1.5% per year. The default probabilities for each year are computed as follows:

$$Q(1) = 1 - e^{-0.015\times 1} = 0.0149$$

$$Q(2) = 1 - e^{-0.015\times 2} = 0.0296$$
$$Q(3) = 1 - e^{-0.015\times 3} = 0.0440$$
$$Q(4) = 1 - e^{-0.015\times 4} = 0.0582$$
$$Q(5) = 1 - e^{-0.015\times 5} = 0.0723$$

- **Unconditional Default Probability:** The probability of default in the 4th year is the difference between cumulative probabilities of the 4th and 3rd years:

$$Q(4) - Q(3) = 0.0582 - 0.0440 = 0.0142$$

- **Conditional Default Probability:** The probability of default in the 4th year, assuming survival until the end of the 3rd year:

$$\frac{0.0142}{1 - Q(3)} = \frac{0.0142}{1 - 0.0440} = 0.0149$$

- Hazard rates are essential for modeling time-dependent default risk in credit risk management.
- They provide a more granular and dynamic perspective compared to static probabilities, aiding in real-time risk assessments and pricing of credit derivatives.

### 9.1.3 Recovery Rates

Definition

The **recovery rate** of a bond is the percentage of the principal amount recovered after default, typically based on the bond's trading price about one month post-default. This metric provides critical insights into creditor losses and varies based on the bond's seniority and prevailing economic conditions.

- The recovery rate represents the value creditors reclaim through partial payments or asset liquidation.
- Recovery rates differ significantly based on bond category:
  - **First Lien Bonds:** 54.6% recovery.
  - **Junior Subordinated Bonds:** 22.3% recovery.
- Recovery rates are inversely related to default rates:

  - **Good economic conditions:** Higher recovery rates.
  - **Poor economic conditions:** Lower recovery rates.

- In challenging economic periods, the combination of high default rates and low recovery rates exacerbates creditor losses.
- Understanding recovery rate trends is crucial for credit risk assessment, pricing of distressed debt, and strategic risk management.

### 9.1.4 Credit Default Swaps (CDS)

> Definition
>
> A Credit Default Swap (CDS) is a financial contract that acts as a form of insurance against the risk of credit default.

- It involves two parties: the protection buyer, who pays periodic premiums (typically quarterly), and the protection seller, who compensates the buyer if a specified credit event, such as a default, occurs.
- CDS contracts can be linked to corporate bonds, government bonds, or other credit instruments, providing the buyer with protection against the risk of the issuer defaulting on the underlying instrument.

The pricing of a CDS is determined by several factors, including the creditworthiness of the reference entity, current market conditions, and the maturity of the swap. CDS contracts play a crucial role in price discovery for credit risk and significantly influence overall market liquidity and stability.

#### 9.1.4.1 CDS Spread

- The **CDS spread** is the annual premium, usually expressed in basis points (bps), that the protection buyer pays to the protection seller over the duration of the contract.
- The spread is calculated as a percentage of the notional amount of the underlying credit instrument.
- A wider CDS spread typically indicates a higher perceived risk of default for the underlying entity, whereas a narrower spread suggests lower perceived risk.

For example, if the CDS spread for a particular corporate bond is 200 bps, the buyer pays the seller 2% of the notional amount annually for the protection. The CDS spread serves as a valuable indicator of credit risk and market sentiment concerning the referenced entity.

#### 9.1.4.2 Credit Indices

- **Credit indices** are benchmarks that track the performance of a basket of credit instruments, such as bonds or credit default swaps.
- These indices often include a diversified portfolio of credit instruments representing various sectors, credit ratings, or geographic regions.
- Credit indices are used to gauge overall market health, providing insights into credit risk, liquidity, and investor sentiment.

Popular credit indices include the **Markit iTraxx** in Europe and the **CDX** in North America, which track CDS spreads across different categories of issuers. These indices enhance liquidity in the credit market by creating standardized products that can be traded more easily.

### 9.1.5 Credit Spreads

Definition

The **credit spread** is the additional rate of interest per annum that investors require for bearing a particular credit risk.

- Credit spreads can be measured using **CDS spreads** or **bond yield spreads** (calculated as the difference between a bond's yield and the yield on a risk-free bond).
- A CDS can effectively hedge a corporate bond, essentially converting it into a risk-free bond.
- The **N-year CDS spread** should approximate the yield difference between an N-year corporate bond and an N-year risk-free bond. However, liquidity constraints can limit the effectiveness of arbitrage opportunities between CDS and bonds.
- CDS contracts provide a direct estimate of the credit spread, while **risk-free rates** are generally derived from Treasury bonds.
- Notably, implied risk-free rates are often found to be higher than Treasury rates.
- **Asset swaps** also provide direct estimates of excess bond yields over a floating reference rate.

#### 9.1.5.1 CDS–Bond Basis

The **CDS–Bond Basis** is the difference between the CDS spread and the bond yield spread:

$$\text{CDS-Bond Basis} = \text{CDS Spread} - \text{Bond Yield Spread}$$

Various factors, such as counterparty default risk and bond price fluctuations, can cause the CDS–Bond Basis to deviate from zero. Historically, the CDS–Bond Basis was positive before the 2007 financial crisis, but it has since tended to be small and negative.

### 9.1.6 Estimating Default Probabilities from Credit Spreads

Estimating default probabilities from credit spreads involves using market indicators such as CDS spreads, bond yield spreads, or asset swap spreads to infer the likelihood of default.

#### 9.1.6.1 Approximate Calculation

This method provides a straightforward way to estimate average default probabilities when the bond price is near par value.

**Scenario**: Suppose a 5-year credit spread is 240 basis points (or 2.4%), and the recovery rate is 40%.

**Formula**: The average hazard rate ($\overline{\lambda}$) over the time horizon $T$ is given by:

$$\overline{\lambda} = \frac{s(T)}{1 - R}$$

Where:

- $s(T)$: Credit spread for maturity $T$,
- $R$: Recovery rate,
- $\overline{\lambda}$: Average hazard rate.

Using the given parameters:

$$\overline{\lambda} = \frac{0.024}{1 - 0.4} = 4\%$$

This method is effective for credit default swaps (CDS), bond yield spreads, and asset swap spreads. It assumes that the bond price is close to its par value.

### 9.1.7 Comparison of Default Probability Estimates

When estimating default probabilities, it is essential to distinguish between probabilities derived from historical data (real-world) and those inferred from credit spreads (risk-neutral).

- **Real-World Probabilities**: Derived from historical default rates, representing actual default frequencies observed in the past.

- **Risk-Neutral Probabilities**: Derived from credit spreads and market prices, reflecting the probability of default adjusted for investor risk preferences and market expectations.

Hazard rates estimated from credit spreads are generally **higher** than those derived from historical data. This disparity grows more pronounced as credit quality declines, and several factors contribute to these differences:

1. **Illiquidity of Corporate Bonds**: Corporate bonds often trade in less liquid markets, leading to higher spreads.
2. **Systematic Risk**: Defaults are often correlated, increasing perceived risk in credit spreads.
3. **Non-Systematic Risk**: Corporate bonds carry firm-specific risks that are harder to diversify compared to equity portfolios.

- Use **risk-neutral default probabilities** for:
  - Valuing credit derivatives: Required for pricing instruments like CDS or CDOs.
  - Estimating the impact of default risk: Particularly when using market-consistent measures.
- Use **real-world default probabilities** for:
  - Scenario analyses: Assessing potential future losses under various economic conditions.

The Merton (1974) model provides a theoretical foundation for calculating default probabilities. It treats a company's equity as a **call option** on its assets, with the strike price equivalent to the debt repayment amount. This approach links default risk to asset value volatility, offering a structural method for default probability estimation.

## 9.2 xVAs

Over-the-counter (OTC) derivatives are cleared through either **Central Clearing Parties (CCPs)** or bilateral agreements. Regulatory frameworks now require most standardized derivatives to be cleared via CCPs, significantly reducing credit risk. However, bilaterally cleared transactions necessitate additional collateral to manage credit risk. Despite these safeguards, certain OTC transactions, such as those with non-financial end-users, remain outside the scope of these regulations.

Assessing credit risk in derivatives is more complex than in traditional loans due to the uncertainty of future exposure and the potentially unlimited loss. To manage this complexity, various adjustments, collectively known as **xVAs**, are used. These include:

- **Credit Value Adjustment (CVA)**: Quantifies the counterparty credit risk.

- **Debit Value Adjustment (DVA)**: Reflects the counterparty's perspective of the dealer's credit risk.

- **Funding Value Adjustment (FVA)**, **Margin Value Adjustment (MVA)**, and **Capital Value Adjustment (KVA)**: Capture other costs and risks in derivative transactions.

This discussion focuses on **CVA**, the primary metric for quantifying a dealer's credit risk in bilaterally cleared transactions.

### 9.2.1 Credit Exposure on Derivatives

Credit exposure in derivatives arises from market value fluctuations, unlike loans with fixed exposures. Future exposure is uncertain, and losses can theoretically be unlimited. Consider a single bilaterally cleared transaction with no collateral posted:

1. **Liability for the dealer (e.g., short option)**: The dealer has no credit exposure.
2. **Asset for the dealer (e.g., long option)**: The dealer bears the credit exposure.

3. **Variable exposure (e.g., forward or swap)**: Exposure fluctuates over time, depending on market conditions.

For multiple transactions between a dealer and a counterparty, exposures are often **netted**. The dealer's net exposure is given by:

$$\max(V, 0)$$

where $V$ represents the net market value of all transactions. The dealer must consider both current and potential future exposure when assessing credit risk.

### 9.2.2 Credit Value Adjustment (CVA)

**CVA** measures the expected loss from a counterparty's potential default in OTC derivative transactions. It directly impacts a dealer's financials:

- An **increase in CVA** reduces profits, as it represents higher expected credit losses.

- CVA must be calculated for each counterparty to incorporate their specific default risk.

CVA is a crucial risk management tool, as it reflects the financial cost of counterparty risk, helping dealers to price derivatives more accurately and align with regulatory and accounting requirements.

#### 9.2.2.1 Calculations for CVA

To calculate **Credit Value Adjustment (CVA)**, the process involves discretizing the timeline of the longest derivative and estimating the potential losses over each interval.

- $T$: The life of the longest derivative.
- $[t_{i-1}, t_i]$: Intervals dividing $[0, T]$.
- $q_i$: Risk-neutral probability of default during the $i$th interval.
- $v_i$: Present value of net exposure at the midpoint of the $i$th interval.
- $R$: Estimated recovery rate.
- $s_i$: Credit spread for a maturity of $t_i$.

The total expected loss from counterparty default is calculated as:

$$\text{CVA} = \sum_{i=1}^{n} (1 - R) q_i v_i$$

The risk-neutral default probability ($q_i$) for the $i$th interval is given by:

$$q_i = \exp\left(\frac{-s_{i-1} t_{i-1}}{1 - R}\right) - \exp\left(\frac{-s_i t_i}{1 - R}\right)$$

The exposure ($v_i$) is typically estimated using **Monte Carlo simulations**, which account for the uncertain nature of market values over time.

#### 9.2.2.2 Collateral and Cure Periods

Collateral agreements influence the calculation of exposures. The **exposure,** $E$, is defined as:

$$E = \max(V - C, 0)$$

- $V$: Market value of outstanding transactions.
- $C$: Collateral posted at the time of default.

The **cure period**—typically 10 days—is the time between when a counterparty stops posting collateral and the close-out of the transaction. This affects the collateral's value and must be modeled in the Monte Carlo simulation.

#### 9.2.2.3 Peak Exposure

- Calculated at the midpoint of each interval and represents a high percentile (e.g., 97.5%) of potential exposures.

- **Real-world simulations** should ideally be used for this calculation, but for short time periods, risk-neutral simulations yield similar results.

#### 9.2.2.4 Downgrade Triggers

- **Downgrade triggers** in ISDA agreements require increased collateral or termination of transactions when a party's credit rating falls below a threshold.

- Overuse of these triggers can lead to liquidity crises, as seen in cases such as AIG, Enron, Xerox, and Ambac.

- While they provide some protection, they are limited in addressing sudden and severe credit rating declines.

### 9.2.3 The Impact of a New Transaction

The addition of a new transaction to an existing portfolio can impact the Credit Value Adjustment (CVA) depending on its correlation with the existing transactions:

- **Positive correlation** with existing transactions increases CVA, as it raises the potential exposure to the counterparty.
- **Negative correlation** reduces CVA, as the exposure to the counterparty becomes less correlated with existing positions, mitigating overall risk.

Companies may receive more favorable quotes from dealers with whom they already have existing transactions, due to **netting effects** and CVA considerations. However, recalculating CVA for each new transaction can be computationally intensive. Instead, **stored values** from previous Monte Carlo simulations can be leveraged to efficiently update CVA incrementally without starting from scratch.

### 9.2.4 CVA Risk and Wrong-Way Risk

CVA can be thought of as a complex derivative position, and its management requires the same hedging techniques used for other derivatives. Since a rise in CVA results in a loss for the dealer, **hedging CVA** becomes essential.

**Computing Greek letters** (such as delta, gamma, etc.) for CVA is computationally intensive, yet necessary for risk management. Regulators require that changes in credit spreads be incorporated into **market risk capital** calculations to reflect CVA risk.

- **Wrong-way risk** occurs when the probability of default is **positively correlated** with exposure, increasing risk for the dealer.
- **Right-way risk** occurs when the default probability is **negatively correlated** with exposure, reducing risk for the dealer.

Evaluating these types of risk requires an understanding of the counterparty's business and the full scope of their other transactions. To adjust for wrong-way risk, regulators require a **CVA adjustment** using an **alpha multiplier** of at least **1.2**.

### 9.2.5 Debit Value Adjustment (DVA)

**Debit Value Adjustment (DVA)** reflects the counterparty's potential cost due to the dealer's default. It is essentially the counterpart to CVA, representing the cost to the counterparty if the dealer defaults.

- DVA benefits the dealer, effectively offsetting the dealer's CVA, which aligns with the zero-sum nature of derivatives markets.
- The **book value of derivatives** is calculated as:

$$f_{nd} - \text{CVA} + \text{DVA}$$

where $f_{nd}$ is the value of the derivative assuming no defaults.

An increase in the dealer's **credit spread** results in an increase in DVA, which can boost reported profits. However, because this effect could artificially inflate profits, **regulators exclude DVA gains and losses** from the definition of **common equity** in regulatory capital calculations. This exclusion aims to prevent misrepresentation of financial stability.

## 9.3 Credit Value at Risk (Credit VaR)

**Credit VaR** is a critical tool for regulatory capital determination and internal credit risk management. It measures the potential credit loss over a specified time horizon that will not be exceeded at a given confidence level.

- Can consider only default losses or include losses from downgrades and credit spread changes.
- Banks use Credit VaR for regulatory capital (to comply with Basel requirements) and economic capital.

- Credit VaR typically uses a longer horizon (e.g., one year) than market risk VaR (e.g., one day).
- Credit risk models are more elaborate than those for market risk due to the need to capture interdependent defaults and potential downgrades.

A critical variable in credit risk modeling is **credit correlation**, which accounts for the interdependence of defaults among companies in a portfolio. During economic downturns, increased credit correlation magnifies the risk for institutions, emphasizing its importance in stress testing and capital adequacy assessments.

### 9.3.1 Ratings Transition Matrices

**Ratings transition matrices** quantify the likelihood of a company transitioning between credit ratings over a given time period, based on historical data.

**One-Year Ratings Transition Matrix, 1981–2020** (Source: S&P)

- Probabilities Expressed as Percentages and Adjustments for Transition to the NR (not rated) Category

| Initial Rating | AAA | AA | A | BBB | BB | B | CCC/C | D |
|---|---|---|---|---|---|---|---|---|
| AAA | 89.85 | 9.35 | 0.55 | 0.05 | 0.11 | 0.03 | 0.05 | 0.00 |
| AA | 0.50 | 90.76 | 8.08 | 0.49 | 0.05 | 0.06 | 0.02 | 0.02 |
| A | 0.03 | 1.67 | 92.61 | 5.23 | 0.27 | 0.12 | 0.02 | 0.05 |
| BBB | 0.00 | 0.10 | 3.45 | 91.93 | 3.78 | 0.46 | 0.11 | 0.17 |
| BB | 0.01 | 0.03 | 0.12 | 5.03 | 86.00 | 7.51 | 0.61 | 0.70 |
| B | 0.00 | 0.02 | 0.08 | 0.17 | 5.18 | 85.09 | 5.66 | 3.81 |
| CCC/C | 0.00 | 0.00 | 0.12 | 0.20 | 0.65 | 14.72 | 50.90 | 33.42 |
| D | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 100.00 |

- According to S&P's one-year transition matrix, a company rated 'A' has a 92.61% probability of maintaining its rating over one year.
- Estimate multi-period transitions via matrix multiplication (e.g., a 5-year transition is derived by raising the 1-year matrix to the 5th power).
- **Assumptions**: Transition probabilities are independent across periods, which oversimplifies real-world scenarios, as downgrades often increase the probability of further downgrades.
- Transition matrices are integral to **CreditMetrics**, enabling portfolio revaluation based on credit rating migrations.

### 9.3.2 Methods for Calculating Credit VaR

1. **Vasicek's Model**:
   - Based on a one-factor Gaussian copula model.
   - Focuses on the time to default for portfolio companies.
   - Widely used for regulatory capital calculations.
2. **Credit Risk Plus**:
   - Introduced by Credit Suisse in 1997.
   - Inspired by insurance models, focusing on default probabilities and loss rates.
   - Incorporates uncertainty in default probabilities to calculate loss distributions.
3. **CreditMetrics**:
   - Developed by JPMorgan in 1997.
   - Combines Gaussian copula models with ratings transition matrices.
   - Measures both default and downgrade risks, producing a more comprehensive loss estimate.

- CreditMetrics and Credit Risk Plus, both aim to estimate long-term loss distributions.
- Differences lie in **timing** of loss recognition:
  - **Credit Risk Plus** recognizes losses only upon default.
  - **CreditMetrics** considers revaluation losses for downgrades over time.
- **Example**: If a loan is downgraded in Years 1 and 2 before defaulting in Year 3:
  - Credit Risk Plus calculates the loss only in Year 3.
  - CreditMetrics includes incremental revaluation losses in Years 1 and 2.

### 9.3.3 Credit Spread Risk

**Credit spread risk** reflects changes in credit spreads, which can significantly impact the value of credit-sensitive products such as bonds and derivatives.

1. **Historical Simulation**:
   - Uses historical data (e.g., the last 500 days of credit spread changes).
   - Limitation: Assumes no default probability and may lack sufficient data quality.
2. **CreditMetrics**:
   - Combines rating transition matrices and Monte Carlo simulations.
   - Accounts for the distribution of credit spreads over the simulation horizon.

## 9.4 Practice Questions and Problems

### 9.4.1 Estimating Default Probabilities

1. What is a rating?
2. What rating agencies exist and how many categories do they use for investment-grade bonds?
3. What are the disadvantages of ratings with respect to credit risk for financial institutions?
4. What are alternative methods for estimating credit risk?
5. What is meant by a hazard rate?
6. Calculate the average hazard rate for a B-rated company during the first year from the data in Table in the text.
7. Calculate the average hazard rate for a BB-rated company during the third year from the data in Table in the text.
8. A credit default swap requires a semiannual payment at the rate of 60 basis points per year. The principal is $300 million and the credit default swap is settled in cash. A default occurs after four years and two months, and the calculation agent estimates that the price of the cheapest deliverable bond is 40% of its face value shortly after the default. List the cash flows and their timing for the seller of the credit default swap.
9. Explain the two ways a credit default swap can be settled.
10. Explain the difference between the default probabilities calculated from historical default experience and those calculated from credit spreads.
11. The spread between the yield on a three-year corporate bond and the yield on a similar risk-free bond is 50 basis points. The recovery rate is 30%. Estimate the implied average hazard rate per year over the three-year period.
12. The spread between the yield on a five-year bond issued by a company and the yield on a similar risk-free bond is 80 basis points. Assuming a recovery rate of 40%, estimate the implied average hazard rate per year over the five-year period. If the spread is 70 basis points for a three-year bond, what do your results indicate about the implied average hazard rate in years 4 and 5?
13. How are recovery rates usually defined?
14. "The position of a buyer of a credit default swap is similar to the position of someone who is long a risk-free bond and short a corporate bond." Explain this statement.
15. Why is there a potential asymmetric information problem in credit default swaps?

### 9.4.2 xVAs

1. Explain why a new transaction by a bank with a counterparty can have the effect of increasing or reducing the bank's credit exposure to the counterparty.
2. A company offers to post its own equity as collateral. How would you respond?
3. Can the existence of default triggers increase default risk? Explain your answer.

4. Give two examples of when (a) wrong-way risk and (b) right-way risk can be expected to be observed.
5. Explain the term "cure period."
6. "Netting means that CVA cannot be calculated on a transaction-by-transaction basis." Explain this statement.
7. "DVA can improve the bottom line when a bank is experiencing financial difficulties." Explain why this is so.
8. What part of CVA risk is considered a component of market risk by regulators?

### 9.4.3 Credit Value at Risk

1. Explain the difference between Vasicek's model, the Credit Risk Plus model, and CreditMetrics as far as the following are concerned: (a) when a credit loss is recognized and (b) the way in which default correlation is modeled.
2. Explain what is meant by the constant level of risk assumption.
3. Use the transition matrix from the text to calculate a transition matrix for three years. What is the probability of a company rated AAA being AAA at the end of two years? What is the probability of it moving to AA?
4. How can historical simulation be used to calculate a one-day 99% VaR for the credit risk of bonds in the trading book? What are the disadvantages?

# 10 Operational Risk, Liquidity Risk, and Model Risk

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 20 - Operational Risk
  - Chapter 21 - Liquidity Risk
  - Chapter 22 - Model Risk Management

**Learning outcomes:**

- Understand the evolution of operational risk capital charges, the role of key risk indicators, and the allocation of risk capital across units.
- Recognize insurance options for operational risk, their limitations, and the dynamic nature of risk management procedures.
- Comprehend liquidity risk types (trading and funding), factors affecting trading liquidity, and methods to measure liquidity like bid–ask spreads.
- Learn to forecast cash needs in varied conditions, identify "liquidity black holes," and the importance of diverse trading strategies.
- Acknowledge model risk as operational risk and the unique challenges of financial models compared to physical science models.

## 10.1 Operational Risk

Operational risk refers to the potential for losses arising from failed internal processes, human errors, system failures, or external events. Unlike credit or market risk, operational risk is inherently more challenging to quantify and manage, as it spans diverse and unpredictable events. Regulatory bodies began addressing operational risk formally in 1999 by introducing capital requirements for its management.

One prominent and growing subtype of operational risk is **cyber risk**, which necessitates robust security measures and continuous employee education. Unlike insurance companies, which can adjust premiums to offset large losses, banks operate in competitive environments that limit their ability to pass such costs to clients, making effective operational risk management critical.

Operational risk not only affects a bank's costs and revenues but also interacts with credit and market risks, amplifying their impact.

Definition

The Basel Committee defined operational risk in 2001 as:
*"The risk of loss resulting from inadequate or failed internal processes, people, and systems, or from external events."*

- **Inclusions**: Legal risk, covering lawsuits and regulatory penalties.
- **Exclusions**: Reputation risk and risks arising from strategic decisions.

Operational risks can be classified into several distinct types:

1. **Internal Fraud**
   Losses due to intentional acts by employees to deceive or circumvent laws.
   Examples: Insider trading, rogue trading, and embezzlement.
2. **External Fraud**
   Fraudulent activities by third parties, often targeting a bank's systems or assets.
   Examples: Robbery, computer hacking, and phishing attacks.
3. **Employment Practices and Workplace Safety**
   Violations related to employment laws, workplace health, and safety.
   Examples: Discrimination lawsuits, claims for wrongful termination, and workplace injury cases.
4. **Clients, Products, and Business Practices**
   Failures to meet professional standards or obligations, misuse of customer information, or ethical lapses.
   Examples: Fiduciary breaches, customer data leaks, and money laundering.
5. **Damage to Physical Assets**
   Losses stemming from external events that damage tangible assets.
   Examples: Fires, earthquakes, floods, and vandalism.

6. **Business Disruption and System Failures**
   Issues that disrupt business operations, often due to technological or utility failures.
   Examples: Hardware malfunctions, server outages, and power failures.

7. **Execution, Delivery, and Process Management**
   Failures in transaction processing, documentation, or vendor management.
   Examples: Settlement errors, missed deadlines, and supply chain disputes.

### 10.1.1 Loss Severity and Loss Frequency

Effective management of operational risk requires understanding two key statistical distributions: **loss frequency** and **loss severity**. These distributions provide a framework for estimating the potential magnitude of operational risk losses.

#### 10.1.1.1 Loss Frequency

Definition

The distribution of the number of loss events occurring within a given time period, typically one year.

- Commonly modeled using the Poisson distribution:

$$P(n) = e^{-\lambda T} \frac{(\lambda T)^n}{n!}$$

where:

- $P(n)$: Probability of observing $n$ losses in a year.
- $\lambda$: Average annual frequency of losses.
- $T$: Time period (usually $T = 1$).

#### 10.1.1.2 Loss Severity

Definition

The distribution of the magnitude of losses, given that a loss occurs.

- Often represented by the **lognormal distribution**, as it accounts for the typically skewed nature of loss sizes, with many small losses and a few large ones.

#### 10.1.1.3 Monte Carlo Simulation

Monte Carlo simulation combines the loss frequency and severity distributions to estimate the total loss distribution. This process involves:

1. Simulating the number of loss events from the loss frequency distribution.
2. Simulating the magnitude of each loss event from the loss severity distribution.
3. Aggregating simulated losses to determine the total potential loss.

Banks often face challenges due to limited internal data on operational losses. To address this:

- External data, such as vendor or consortium data, may be used but must be adjusted to reflect the institution's size and characteristics.
- Publicly available loss data often overrepresents large losses, introducing biases.

#### 10.1.1.4 Scenario Analysis

- Used to estimate risks for rare or unobserved events.
- Relies on expert judgment to assess potential loss frequency and severity.
- Encourages proactive planning for adverse events and their mitigation.

### 10.1.2 The Standardized Measurement Approach (SMA)

Definition

The **Standardized Measurement Approach** was introduced by regulators in December 2017 to provide a consistent framework for calculating operational risk capital across banks. Its implementation began in **January 2023** (source).

#### 10.1.2.1 Key Components of the SMA

1. **Business Indicator (BI)**
    - Measures a bank's size and potential exposure to operational risk.

- Components:
  - **Interest, Leases, and Dividend Component (ILDC)**
  - **Services Component (SC)**
  - **Financial Component (FC)**
- Formula:

$$\text{BI} = \text{ILDC} + \text{SC} + \text{FC}$$

2. **Business Indicator Component (BIC)**
   - Converts the BI into a base measure of operational risk capital.
   - Formulas:
     - For **small banks** (BI $<$ 1 billion euros):

$$BIC = 0.12 \times BI$$

     - For $1 \leq \text{BI} < 30$ billion euros:

$$BIC = 0.12 + 0.15 \times (\text{BI} - 1)$$

     - For ( BI  30 ) billion euros:

$$BIC = 4.47 + 0.18 \times (\text{BI} - 30)$$

3. **Internal Loss Multiplier (ILM)**
   - Adjusts the BIC based on historical operational risk losses.
   - Formula:

$$ILM = \ln\left[e - 1 + \left(\frac{LC}{BIC}\right)^{0.8}\right]$$

     where:
     - $LC$: **Loss Component**, equal to 15 times the average annual operational risk losses over the previous 10 years.
4. **Operational Risk Capital Calculation**
   - For small banks (BI $<$ 1 billion euros):

$$\text{Capital} = \text{BIC}$$

- For other banks:

$$\text{Capital} = \text{BIC} \times \text{ILM}$$

- When $LC = BIC$: $ILM = 1$, and the required capital equals the BIC.
- Banks with higher-than-average operational losses have $ILM > 1$, requiring additional capital.
- Conversely, banks with lower-than-average losses have $ILM < 1$, reducing their capital requirement.
- Supervisors may apply discretion to equalize capital requirements across banks or adjust them for smaller institutions.

### 10.1.3 Preventing Operational Risk Losses

Operational risk managers work proactively to mitigate losses by analyzing failures at other institutions and identifying patterns and causal relationships. This analysis helps design preventive measures such as improving employee qualifications, upgrading systems, or enhancing internal processes.

- **Risk Identification**:
  Business unit managers play a key role in identifying risks through tools like questionnaires and scorecards in a process known as **Risk Control and Self-Assessment (RCSA)**.
- **Key Risk Indicators (KRIs)**:
  KRIs serve as early warning signals, allowing managers to monitor and address potential vulnerabilities before they lead to losses. Examples include:
  - Staff turnover rates.
  - Number of failed transactions.
  - Percentage of temporary staff.
  - Supervisor-to-staff ratio.
  - Number of unfilled positions.
  - Percentage of employees not taking at least 10 consecutive leave days annually (an indicator of potential fraud or operational strain).

### 10.1.4 Allocation of Operational Risk Capital

Operational risk capital must be allocated strategically to encourage effective risk management across business units.

- **Incentivization**: Reducing operational risk should decrease the allocated capital, thereby improving the unit's **return on capital**.

- **Cost-Benefit Consideration**: In some cases, the costs of mitigating certain risks may outweigh the potential benefits, requiring careful evaluation of risk-reduction strategies.

- **Governance**: Active support and oversight by senior management and the board of directors are essential. The Basel Committee on Banking Supervision recommends periodic board reviews of risk management programs to ensure their effectiveness.

### 10.1.5 Insurance in Operational Risk Management

Insurance can play a role in managing operational risk, but it introduces its own challenges:

- **Moral Hazard**: Insuring against operational risks may lead to complacency in risk management, as employees and managers might rely on insurance rather than prevention.

- **Adverse Selection**: Insurance providers must carefully assess potential clients, as higher-risk institutions are more likely to seek coverage.

- **Risk Mitigation by Insurers**: To counter these challenges, insurers use mechanisms like:
  - Deductibles: Ensuring that the insured institution shares part of the loss.
  - Coinsurance: Requiring the insured to cover a percentage of the loss.
  - Policy Limits: Capping the maximum amount payable by the insurer for a claim.

## 10.2 Liquidity Risk

Liquidity risk gained prominence during the Global Financial Crisis of 2007, highlighting its critical role in financial stability.

Definition

- **Solvency**: A firm is solvent if its assets exceed its liabilities, ensuring positive equity value.
- **Liquidity**: Refers to the firm's ability to meet its cash payment obligations as they come due.

Even solvent financial institutions can fail due to liquidity problems, underscoring the distinction between solvency and liquidity. Liquidity risk is divided into two key categories:

1. **Liquidity Funding Risk**: The inability to meet short-term obligations.
2. **Liquidity Trading Risk**: Challenges related to marketability of assets and associated transaction costs, such as widening bid-ask spreads.

Regulatory liquidity requirements aim to ensure institutions can survive worst-case scenarios, such as sudden market shocks or systemic events. The concept of **"liquidity black holes"** describes situations where market liquidity vanishes, often triggered by widespread, simultaneous attempts to liquidate positions.

### 10.2.1 Liquidity Trading Risk

Liquidity trading risk involves challenges in converting assets to cash without significant losses. Several factors affect an asset's liquidity:

- **Asset Type**: Market liquidity varies greatly by asset. For example, IBM shares are typically more liquid than non-investment-grade bonds.
- **Trade Size**: Larger trades can strain market liquidity and widen bid-ask spreads.
- **Speed of Liquidation**: Quickly liquidating positions often requires accepting lower prices or higher costs.
- **Economic Environment**: Adverse economic conditions exacerbate liquidity risks by reducing market participants' willingness to trade.

#### 10.2.1.1 Bid-Ask Spread

The **bid-ask spread** is a primary measure of liquidity and can be expressed in two forms:

- **Cash Bid-Ask Spread**:

$$p = \text{Ask price} - \text{Bid price}$$

- **Proportional Bid-Ask Spread**:

$$s = \frac{\text{Ask price} - \text{Bid price}}{\text{Mid-market price}}$$

#### 10.2.1.2 Cost of Liquidation

The cost of liquidation reflects the expense of unwinding a portfolio under different market conditions:

- **Normal Market Conditions**:

$$\text{Cost} = \sum_{i=1}^{n} \frac{s_i \alpha_i}{2}$$

  - $s_i$: Proportional bid-ask spread for the $i$-th asset.
  - $\alpha_i$: Value of the position in the $i$-th asset.

- **Stressed Market Conditions**:

$$\text{Cost} = \sum_{i=1}^{n} \frac{(\mu_i + \lambda \sigma_i)\alpha_i}{2}$$

  - $\mu_i$: Mean of the proportional bid-ask spread for the $i$-th asset.
  - $\sigma_i$: Standard deviation of the proportional bid-ask spread for the $i$-th asset.
  - $\lambda$: Multiplier for the desired confidence level.

#### 10.2.1.3 Key Considerations in Liquidity Management

- Diversification alone does not guarantee a reduction in liquidity risk.
- Liquidity monitoring is sensitive to the assumed liquidation timeframe.
- Holding multiple small positions generally reduces liquidity risk compared to a few large positions.

- Limiting the size of any single position is a practical method for managing liquidity trading risk.

#### 10.2.1.4 Liquidity-Adjusted Value at Risk (L-VaR)

**Liquidity-Adjusted VaR** extends traditional VaR by incorporating liquidity risk. It combines market risk with the **cost of liquidation**, providing a comprehensive risk metric. L-VaR can be calculated for both normal and stressed market conditions, enhancing its applicability during periods of market stress.

#### 10.2.1.5 Optimal Position Unwinding

Unwinding large positions presents a tradeoff between:

1. **Quick Liquidation**: Minimizes market risk but incurs higher bid-ask spreads.

2. **Slow Liquidation**: Reduces bid-ask spreads but increases exposure to market risk.

Almgren and Chriss (2001) proposed a mathematical framework to optimize this tradeoff, balancing execution cost and risk exposure.

#### 10.2.1.6 Additional Measures of Market Liquidity

- **Trading Volume and Price Impact**: High trading volumes indicate better liquidity, while large price changes for small trades reflect poor liquidity.
- **Amihud's Illiquidity Measure**: Proposed by Amihud (2002), this metric links daily returns to trading volume:

$$\text{Amihud's Measure} = \frac{\text{Absolute daily return}}{\text{Daily dollar volume}}$$

  - Lower liquidity results in higher values of this measure.

  - Empirical evidence shows that assets with lower liquidity require higher expected returns as compensation for liquidity risk.

### 10.2.2 Liquidity Funding Risk

Liquidity funding risk pertains to a financial institution's ability to meet its cash needs as they arise. Even solvent institutions can fail if they encounter liquidity constraints. The primary causes of liquidity funding problems are:

1. **Economic Liquidity Stresses**: A lack of willingness among investors to fund credit-risk-bearing activities, often observed during financial crises.

2. **Aggressive Funding Decisions**: Relying on short-term funding to meet long-term needs, creating a mismatch in liquidity horizons.

3. **Poor Financial Performance**: Weak performance can lead to a loss of market confidence, triggering deposit withdrawals and funding difficulties.

In severe cases, all these factors may converge, exacerbating liquidity issues. Effective management involves predicting cash needs—some are predictable (e.g., bond coupon payments), while others, such as deposit withdrawals or margin calls, are not. Complex financial instruments add further unpredictability, particularly in scenarios involving downgrade triggers or counterparty defaults.

The ultimate goal is to ensure that institutions can meet cash demands even during adverse scenarios.

#### 10.2.2.1 Sources of Liquidity

Institutions rely on multiple liquidity sources, each with unique characteristics and challenges:

1. **Cash and Treasury Securities**: Highly liquid and reliable but expensive to maintain in large quantities.
2. **Liquidating Trading Book**: Requires monitoring under both normal and stressed market conditions to assess liquidity accurately.
3. **Ability to Borrow**: Borrowing is accessible during stable periods but becomes challenging during times of market stress.
4. **Retail and Wholesale Deposits**: Highly volatile and susceptible to market-wide conditions, making them an unreliable liquidity source in crises.
5. **Securitization**: Once a dependable source of liquidity, its risks became evident after the 2007 financial crisis.
6. **Central Bank Borrowing**: Serves as a lender of last resort but is subject to collateral requirements and potential haircuts.

#### 10.2.2.2 Key Considerations

- **Stressed vs. Normal Conditions**: Liquidity sources behave differently in stress scenarios; stressed conditions often constrain borrowing capacity or require rapid liquidation of assets.

- **Collateral and Haircuts**: Borrowing from central banks depends on the quality and value of collateral, which may be discounted (haircut).

- **Hedging Liquidity Risk**: Derivative hedging may create additional liquidity pressures through margin requirements.

### 10.2.3 Reserve Requirements and Regulation

Liquidity management is tightly regulated to ensure institutions can withstand short-term and long-term liquidity challenges:

1. **Reserve Requirements**: Banks must hold a fixed percentage of their deposits in reserve, limiting their ability to lend but ensuring liquidity. Requirements vary by country and influence the broader money supply.
2. **Basel III Liquidity Risk Requirements**:
   - **Liquidity Coverage Ratio (LCR)**: Ensures that banks hold enough high-quality liquid assets to cover net cash outflows for a 30-day stress scenario.

     $$\text{LCR} \geq 100\%$$

   - **Net Stable Funding Ratio (NSFR)**: Promotes stability by requiring sufficient stable funding to meet long-term funding needs.

     $$\text{NSFR} \geq 100\%$$

### 10.2.4 Liquidity Black Holes

A **liquidity black hole** occurs when market liquidity evaporates, often during periods of extreme stress. This phenomenon arises when most market participants simultaneously seek to sell (or buy) assets, with few or no counterparties willing to trade. While transactions may still occur, they do so at significantly widened bid-ask spreads, reflecting the market's illiquidity.

1. **Negative Feedback Traders**: These traders stabilize markets by buying when prices fall and selling when prices rise.

2. **Positive Feedback Traders**: These traders destabilize markets by buying when prices rise and selling when prices fall, amplifying price movements and contributing to liquidity black holes.

The dominance of **positive feedback traders** creates unstable markets prone to sudden liquidity crises.

#### 10.2.4.1 Reasons for Positive Feedback Trading

1. **Trend Trading**: Traders buy or sell based on observed price trends, reinforcing upward or downward movements.

2. **Stop-Loss Rules**: Automatic sell orders triggered when prices fall below a predetermined level can accelerate price declines.

3. **Dynamic Hedging**: Adjusting hedge positions for short options increases selling pressure in falling markets, reducing liquidity.

4. **Synthetic Options**: Commercial programs that mimic put options can cause large-scale selling during market downturns.

5. **Margin Calls**: Sharp market moves trigger margin requirements, forcing leveraged traders to sell assets, further depressing prices.

6. **Predatory Trading**: Opportunistic traders exploit large forced sales by driving prices down further.

7. **Historical Example – LTCM Failure**: The collapse of Long-Term Capital Management illustrates how excessive leverage and margin calls can destabilize markets.

#### 10.2.4.2 Leveraging and Deleveraging Cycles

- **Leveraging**: During periods of abundant liquidity, credit availability increases, fueling asset price growth.

- **Deleveraging**: When liquidity contracts, credit availability tightens, driving asset prices downward.

Hedge funds are especially vulnerable to these cycles due to their reliance on leverage. The transition from leveraging to deleveraging often triggers liquidity black holes, as asset sales cascade.

#### 10.2.4.3 Irrational Exuberance

The term **"irrational exuberance"**, coined by Alan Greenspan in 1996, describes overvaluation driven by speculative behavior and short-term profit-seeking.
- Overvalued markets can create self-reinforcing cycles of buying and overconfidence.
- These cycles often end in sharp corrections or crashes, accompanied by liquidity black holes.

#### 10.2.4.4 The Role of Regulation

1. **Uniform Regulations and Synchronous Behavior**: A one-size-fits-all regulatory framework can lead to synchronized actions among institutions, amplifying market stress during crises.

2. **Need for Differentiated Capital Requirements**: Financial institutions experience different cycles, and uniform capital requirements may exacerbate systemic risks. Tailored regulations can mitigate these issues.

#### 10.2.4.5 The Importance of Market Diversity

A lack of diversity in market behavior is a key contributor to liquidity black holes. When participants act homogeneously, markets become fragile and prone to crises.

1. **Regulatory Differentiation**: Different types of institutions should face varied regulatory requirements to promote heterogeneity in market behavior.

2. **Contrarian Investment Strategies**: Contrarian strategies, which involve acting against prevailing market trends, can stabilize markets. However, they are difficult to implement under short-term, VaR-based risk management frameworks, which discourage deviations from the norm.

Promoting diversity in market participants and strategies is crucial to mitigating liquidity black holes and fostering resilient financial systems.

## 10.3 Model Risk Management

Models have become indispensable tools in modern finance, supporting tasks such as credit decisions, liquidity management, and risk assessment. Advances in analytics, particularly machine learning, have automated tasks traditionally performed by humans. However, the increasing reliance on models brings with it significant **model risk**, a subset of operational risk, necessitating dedicated model risk management functions in large institutions.

Models are approximations of real-world phenomena, relying on assumptions and simplifications. Understanding these assumptions, their limitations, and their potential implications is essential. Regulatory systems require that banks develop, validate, and use models responsibly to minimize the risk of adverse outcomes.

### 10.3.1 Regulatory Guidance

One key regulatory framework is **SR 11-7** (Federal Reserve SR 11-7 Guidance), which establishes principles for effective model risk management.

> Definition
>
> Model risk arises primarily from:
> 1. **Fundamental Errors**: Mistakes in data, calculations, or design.
> 2. **Improper Usage**: Misapplication or misunderstanding of a model's assumptions and limitations.

#### 10.3.1.1 Key Components of Model Risk Management

1. **Model Development**:
   - Thorough documentation is essential for continuity, regulatory compliance, and validation.
   - Testing should be rigorous, including:
     - **Out-of-Sample Testing**: Evaluates performance on unseen data.
     - **Stress Testing**: Assesses performance under extreme or unlikely scenarios.
2. **Model Validation**:
   - Validation should be conducted by independent groups, fostering objectivity and critical evaluation.
   - Core elements of validation:

- **Conceptual Soundness**: Ensures the model is theoretically robust.
- **Ongoing Monitoring**: Tracks model performance over time to detect deviations or failures.
- **Outcomes Analysis**: Compares predicted outcomes with actual results to assess accuracy.

- Corporate culture should support independence and transparency in validation.

3. **Vendor Models**:
   - Vendor-supplied models must undergo the same validation rigor as internally developed models.
   - Institutions should demand thorough documentation and continuous monitoring from vendors.
   - Contingency plans should be in place for vendor unavailability to ensure business continuity.

#### 10.3.1.2 Models in Physics vs. Finance

While models in physics describe fundamental and consistent natural laws, financial models are approximations influenced by human behavior and market dynamics. This distinction creates unique challenges in finance:

- **Stability vs. Volatility**: Parameters like gravitational pull in physics are constant, but financial parameters (e.g., volatility) fluctuate frequently.
- **Precision vs. Approximation**: Physical models achieve high precision; financial models often introduce significant errors due to their reliance on assumptions and approximations.
- **Behavioral Impact**: Financial models are sensitive to human decisions, market sentiment, and external shocks.

Incorrect assumptions or inadequate validation in financial models can lead to substantial errors, resulting in significant financial losses. Institutions must actively monitor model performance, trading patterns, and competitive positioning across products to detect and mitigate potential risks early.

### 10.3.2 Models for Pricing Actively Traded Products

In the over-the-counter (OTC) market, financial instruments often require valuation using models because they are not identical to actively traded products, even if they are standard in nature (e.g., options).

- **Market-Derived Prices**: Actively traded instruments have transparent, market-determined prices.

- **Model Use**: Models like the Black–Scholes–Merton framework ensure pricing consistency for products that are similar but not identical to market-traded instruments.

- **Role of Models**: Primarily serve as interpolation tools, meaning they help fill in gaps for non-standard instruments without significantly altering prices under normal conditions.

- **Model Risk**: Becomes critical during extreme market conditions, requiring careful validation to ensure reliability.

- **Hedging and Risks**:
  - Hedging strategies account for both variables within and outside the model's scope.
  - Portfolio value fluctuations may arise due to unhedged risks, model imperfections, or new trades.

- **Risk Diversification**: Large institutions often diversify risks across portfolios, mitigating the effects of inaccurate hedging strategies.

### 10.3.3 Models for Less Actively Traded Products

For less actively traded products, valuation and risk management pose greater challenges:

- **Calibration for Traded Products**: Actively traded products benefit from calibration to observed market prices, minimizing model risk.

- **Valuation Challenges for Non-Traded Products**: Model choice plays a significant role in pricing and hedging, as direct market calibration is not possible.

- **Relative Valuation Approach**: Institutions focus on relative valuation, referencing similar market-traded instruments rather than absolute valuation.

- **Multiple Model Usage**: Using several models helps quantify model risk by providing a range of valuations.

  - Institutions should price conservatively at the higher end of this range.
  - Conservative profit recognition avoids overestimating gains from illiquid or complex instruments.

- **Accounting Standards**: Valuations are categorized into:
  - **Level 1**: Prices from active markets.
  - **Level 2**: Prices based on observable inputs but requiring some modeling.
  - **Level 3**: Prices based on unobservable inputs, heavily reliant on assumptions.

### 10.3.4 Successful Pricing Models and Common Pitfalls

The aim of a pricing model is to provide insights into market behavior and support decision-making while avoiding unnecessary complexity.

#### 10.3.4.1 Characteristics of Successful Pricing Models

1. **Consistency**: Aligns with observed market data, facilitating reliable interpolation.
2. **Communication**: Produces stable and interpretable metrics, such as implied volatility, which are easier for traders and stakeholders to understand.
3. **Simplicity**: Offers intuitive frameworks that traders can grasp and apply effectively.

#### 10.3.4.2 Model-Building Best Practices

1. **Avoid Overfitting**:
   - Complex models may fit historical data well but fail in new or changing market conditions.
   - Overfitting leads to unrealistic predictions and poor decision-making.
2. **Limit Over-Parameterization**:
   - Excessive parameters increase model complexity without adding significant value.
   - Simplified models are more robust across different market regimes.

3. **Prioritize Simplicity**:
   - Simple models are easier to explain, validate, and maintain.
   - They reduce reliance on "black box" mechanisms, which can obscure risks and undermine trust.

## 10.4 Practice Questions and Problems

### 10.4.1 Operational Risk

1. What risks are included by regulators in their definition of operational risks? What risks are not included?
2. Discuss how moral hazard and adverse selection are handled in car insurance.
3. When is a trading loss classified as a market risk and when is it classified as an operational risk?
4. Discuss whether there is (a) moral hazard and (b) adverse selection in life insurance contracts.
5. Give two examples of key risk indicators that might be monitored by a central operational risk management group within a bank.
6. Consider two banks, each with a BIC of 200 million euros. Bank A has suffered 10 operational risk losses of 20 million euros in the past 10 years. Bank B has suffered one 200 million euro loss in the past 10 years. What is the operational risk regulatory capital for each bank under SMA?

### 10.4.2 Liquidity Risk

1. An asset is quoted bid 50, offer 55. What does this mean? What is the proportional bid–offer spread?
2. Suppose that an investor has shorted shares worth \$5,000 of Company A and bought shares worth \$3,000 of Company B. The proportional bid–offer spread for Company A is 0.01 and the proportional bid–offer spread for Company B is 0.02. What does it cost the investor to unwind the portfolio?
3. Suppose that in Problem 2. the bid–offer spreads for the two companies are normally distributed. For Company A the bid–offer spread has a mean of 0.01 and a standard deviation of 0.01. For Company B the bid–offer spread has a mean of 0.02 and a standard deviation of 0.03. What is the cost of unwinding that the investor is 95% confident will not be exceeded?
4. Explain the difference between the liquidity coverage ratio (LCR) and the net stable funding ratio (NSFR).
5. Why is it risky to rely on wholesale deposits for funding?

6. What is meant by (a) positive feedback trading and (b) negative feedback trading? Which is liable to lead to liquidity problems?
7. What is meant by liquidity-adjusted VaR?
8. Explain how liquidity black holes occur. How can regulation lead to liquidity black holes?
9. Why is it beneficial to the liquidity of markets for traders to follow diverse trading strategies?

### 10.4.3 Model Risk Management

1. How is a financial institution likely to find that it is using a model different from its competitors for valuing a particular type of derivatives product?
2. Distinguish between within-model and outside-model hedging.
3. What is meant by P&L decomposition?
4. What is meant by over-fitting and over-parameterization in model building?
5. "For structured products, traders mark to model. They do not mark to market." Explain this remark.

# 11 Climate Risk and Enterprise Risk Management

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 23 - Climate Risk, ESG, and Sustainability
  - Chapter 24 - Enterprise Risk Management

**Learning outcomes:**

- Understand the growing importance of environmental risks in risk management practices.
- Recognize the terms ESG, sustainability, greenwashing.
- Understand the holistic approach of Enterprise Risk Management (ERM).
- Distinguish between quantitative and qualitative aspects of risk appetite.
- Emphasize the significance of a strong risk culture, especially in medium- to long-term risk assessment.
- Be aware of the potential pitfalls in focusing solely on short-term profits, as evident from historical crises.

## 11.1 Climate Risk, ESG, and Sustainability

Environmental risks threaten the habitability of Earth for future generations, placing immense responsibility on companies and financial institutions to mitigate these risks. Societal and regulatory pressures to address environmental concerns have intensified, and the prominence of these issues is expected to grow. Among environmental risks, climate risk stands out as the most critical challenge.

### 11.1.1 Climate Risk

Climate risk is one of the most urgent environmental concerns facing humanity. Human activities release tens of billions of tons of greenhouse gases (GHGs) annually, with carbon dioxide (CO2) being the most significant contributor. These gases can persist in the atmosphere for hundreds of years, intensifying global warming and its associated impacts.

Efforts to combat climate change have been codified in international agreements like the Paris Agreement (2015) and the Glasgow Climate Pact (2021), which set ambitious targets for achieving net-zero emissions by 2050. However, achieving these goals faces uncertainties due to the uneven implementation of global policies and the challenge of balancing immediate economic needs with long-term environmental sustainability.

https://www.worldometers.info/co2-emissions/co2-emissions-by-country/

The primary sectors contributing to greenhouse gas emissions in the United States are transportation, electricity generation, and industry. Globally, China, the United States, and India rank as the largest emitters, though significant disparities exist in per capita emissions. The effects of climate change include rising sea levels caused by melting ice and the thermal expansion of water, as well as an increase in extreme weather events such as hurricanes, heatwaves, and floods. These extreme events already have significant economic costs; for example, Hurricane Ida caused an estimated $75 billion in damages in 2021.

#### 11.1.1.1 Physical and Transition Risks

Climate risks are often categorized into **physical risks** and **transition risks**:

1. **Physical Risks**: These refer to the direct costs of climate change, such as damages from extreme weather events like hurricanes, floods, and wildfires. Such events disrupt economic activity, destroy infrastructure, and result in substantial recovery costs.
2. **Transition Risks**: These arise from policy changes and societal shifts toward lower-emission economies. Policies such as carbon taxes or cap-and-trade systems may compel companies to close fossil fuel plants prematurely, affecting profitability and operations. However, these transitions also open opportunities in green technologies and renewable energy sectors.

The estimated cost of transitioning to a low-carbon economy is between $100 and $150 trillion over the next 30 years. Tools like **carbon taxes** and **cap-and-trade systems** have been developed to regulate emissions, but their adoption and implementation vary widely across countries. These mechanisms are critical for incentivizing reductions in greenhouse gas emissions while funding innovations in sustainable technologies.

As global efforts intensify, understanding the nuances of climate risk and its economic implications remains crucial for businesses and policymakers alike.

#### 11.1.1.2 Insurance Companies

The insurance industry is deeply affected by climate change, as it disrupts traditional risk models by increasing the correlation between climate-related disasters. The "law of large numbers," which assumes risks are independent and geographically diverse, is less effective in the face of widespread and simultaneous climate events.

While short-term insurance contracts allow for pricing adjustments, life insurance policies—typically longer-term—are less adaptable to evolving climate risks. To address these challenges, insurers must allocate resources to enhance their understanding of climate risk and meet stricter regulatory requirements. Additionally, many insurance companies are increasingly hesitant to provide coverage in high-risk areas prone to natural disasters, leading to coverage gaps for vulnerable populations.

#### 11.1.1.3 Banks

Banks are taking proactive measures to quantify and mitigate climate risks. Initiatives like the Partnership for Carbon Accounting Financials (PCAF) help institutions measure and disclose their financed emissions. Regulatory bodies have also introduced climate stress tests to assess banks' resilience to climate-related shocks.

Climate risks affect multiple traditional risk categories for banks:

- **Credit Risk**: Reduced corporate profitability, impaired debt servicing, and declining collateral values due to climate impacts.
- **Market Risk**: Asset prices may decline as markets adjust to climate realities.
- **Liquidity Risk**: Volatility in funding markets can disrupt access to stable capital.
- **Operational Risk**: Rising legal and compliance risks related to climate policies.
- **Reputational Risk**: Public sentiment and scrutiny can damage reputations, particularly for institutions associated with high carbon emissions.

Long-term risks, such as declining real estate values due to rising sea levels or extreme weather events, threaten mortgage portfolios. Moreover, climate-driven economic disruptions, including mass migrations and shifts in agriculture, challenge global financial stability. Sovereign debt, particularly in developing countries, faces heightened risk from climate-related vulnerabilities. The transition to a low-carbon economy will also create winners and losers, influencing bank lending strategies. Increasing reputational risks further discourage banks from financing high-carbon industries.

#### 11.1.1.4 Institutional Investors

Institutional investors are increasingly focused on integrating climate risk into their investment strategies. Companies are incentivized to align their operations with emission targets to attract funding from climate-conscious investors. **Green bonds** have emerged as key financial instruments, specifically designed to support environmentally friendly projects, often underwritten by investment banks.

Efforts to standardize climate-related financial reporting are underway, although significant progress is still required. Some asset managers, like BlackRock, are leading the shift towards sustainability, while others, such as Berkshire Hathaway, remain resistant. Political counter-pressures, including legislation aimed at curbing environmental considerations in state pension investments, highlight the contentious nature of these changes. Despite these obstacles, the emphasis on emissions reduction and climate risk disclosure continues to grow, reflecting a broader societal shift towards sustainable finance.

### 11.1.2 ESG (Environmental, Social, and Governance)

The focus in corporate decision-making has shifted from a purely profit-driven approach to one that incorporates **environmental, social, and governance (ESG)** factors. These elements reflect a broader commitment to ethical, sustainable, and socially responsible practices:

- **Environmental Aspect**: Emphasizes resource efficiency, waste reduction, and climate change mitigation.
- **Social Aspect**: Prioritizes diversity, equity, inclusion, and worker well-being. However, metrics for assessing social impact are still evolving, leading to inconsistencies.
- **Governance Aspect**: Focuses on transparent corporate accounting, accountability to shareholders, and the avoidance of unethical practices or conflicts of interest.

Despite the growing adoption of ESG frameworks, the lack of regulated reporting standards creates challenges. Multiple ESG data providers often produce conflicting assessments, allowing companies to seek favorable scores. This inconsistency poses risks to investors, who may face misleading signals about a company's true ESG performance.

### 11.1.3 Sustainability

**Sustainability** is defined as meeting present needs without compromising the ability *of future generations to meet their own. Unsustainable practices, such as excessive* greenhouse gas emissions, ocean pollution, and soil contamination, highlight the *urgency of addressing environmental challenges.

Efforts to standardize sustainability practices include guidelines from organizations like the **Sustainability Accounting Standards Board (SASB)** and the **Global Reporting Initiative (GRI)**. Sustainability is often categorized into three interconnected pillars:

1. **Environmental Sustainability**: Focuses on conserving natural resources and reducing ecological harm.
2. **Economic Sustainability**: Aims for long-term economic growth while integrating environmental and social considerations.
3. **Social Sustainability**: Supports the creation of livable, equitable communities for both current and future generations.

Sustainable investment funds, designed to invest in companies with strong ESG and sustainability practices, have seen increased demand. However, research indicates they do not consistently outperform traditional funds, contradicting some exaggerated claims about their low-risk, high-return potential. (See https://onlinelibrary.wiley.com/doi/abs/10.1111/jofi.12841 for more details.) These inflated expectations often overlook the basic principles of finance.

### 11.1.4 Greenwashing

**Greenwashing** occurs when companies falsely portray themselves as environmentally responsible to appeal to eco-conscious consumers. It undermines genuine sustainability efforts and erodes trust. Examples of greenwashing include:

- **Innocent Drinks**: Misleading claims about recyclable plastics.
- **Keurig**: Exaggerated recycling claims led to a $3 million fine.
- **IKEA**: Used illegally sourced wood despite its sustainability credentials.
- **Ryanair**: Unsubstantiated claim of being Europe's "lowest emissions airline."

Greenwashing also extends to financial institutions. For instance, Deutsche Bank's DWS exaggerated its green investment credentials, resulting in significant reputational damage and the resignation of its CEO.

The implications of greenwashing are severe, ranging from legal penalties to a loss of consumer trust and corporate credibility. As awareness grows, regulators and stakeholders are demanding greater accountability from companies making environmental claims.

## 11.2 Enterprise Risk Management (ERM)

Risk managers are tasked with identifying, assessing, and mitigating risks in a way that considers the interconnected nature of risks across the organization. Enterprise Risk Management (ERM) emphasizes a holistic approach, integrating both **top-down** and **bottom-up** perspectives for effective risk management:

- **Top-down**: Sets the overall risk appetite and aligns it with the organization's strategic objectives.

- **Bottom-up**: Identifies and assesses risks at the business unit level, ensuring detailed insights are aggregated into the organization-wide framework.

ERM has evolved from a compliance-focused framework to one that is strategically oriented, aiming to proactively align risk management with broader business goals.

### 11.2.1 Risk Appetite

Risk appetite defines the level of risk an organization is willing to accept to achieve its strategic objectives and business plans. It is a cornerstone of ERM, guiding decision-making and ensuring risks are aligned with organizational goals.

For financial institutions, defining risk appetite is complex, as it encompasses both quantitative and qualitative dimensions:

- **Quantitative Measures**:
  - **Fund Managers**: Risk appetite is more straightforward to quantify using metrics like beta ($\beta$), expected return, and risk. For example, $\beta$ consistent with a specific **Value at Risk (VaR)** objective can be calculated as:

    $$\beta = \frac{R - R_F}{R_M - R_F + N^{-1}(1-p)\sigma_M}$$

    Where:
    * $R$: The return that has a probability $p$ of being exceeded in one year.
    * $R_M$: The expected market return.
    * $R_F$: The risk-free rate.
    * $N^{-1}(1-p)$: The inverse of the cumulative normal distribution at $1-p$.
    * $\sigma_M$: The market standard deviation.

- **Other Financial Institutions**: For banks and insurance companies, risk appetite often includes **VaR** or **Expected Shortfall (ES)** to quantify worst-case losses. However, these institutions face additional challenges in quantifying risks such as reputational, credit rating, and legal risks.

- **Qualitative Measures**:
  Elements like reputational risk, regulatory risk, and strategic considerations often require qualitative evaluation.

#### 11.2.1.1 Addressing Risk Concentration

A robust risk appetite framework must address **risk concentration**, ensuring that exposure to specific sectors, companies, or market variables is appropriately diversified. Concentrated risks can magnify losses during adverse events, so setting exposure limits is critical.

#### 11.2.1.2 Monitoring and Adherence

Ongoing monitoring of risk limits and adherence to established thresholds are essential for maintaining the effectiveness of ERM. Clear communication of risk policies and regular reviews of risk metrics ensure that the organization remains aligned with its risk appetite.

### 11.2.2 Risk Culture

Risk culture reflects how an organization approaches decision-making, particularly under uncertainty. In a strong risk culture, decisions are made in a disciplined manner, carefully weighing potential risks and rewards. This culture aligns with the institution's risk appetite and long-term objectives, ensuring that short-term profits do not overshadow long-term risks.

Compensation structures play a critical role in shaping risk culture. Reforms, such as spreading bonuses over multiple years or including clawback provisions for adverse outcomes, are designed to mitigate excessive short-term risk-taking. Regulatory guidelines, such as those issued by the Financial Stability Board, provide a framework for fostering robust risk cultures across financial institutions.

### 11.2.3 Identifying Major Risks

Identifying key risks is a fundamental aspect of ERM, akin to stress testing, and involves analyzing historical trends and emerging risks. Major categories include:

- **Regulatory Changes**: Increased regulations for banks often shift activities to less-regulated entities like shadow banks.

- **Cognitive Biases**: Human biases can impair risk assessments, including:
  - **Wishful Thinking**: Confusing desired outcomes with realistic expectations.
  - **Anchoring**: Relying too heavily on initial estimates while ignoring new information.
  - **Availability Bias**: Overweighting recent or easily recalled events.
  - **Representativeness**: Misjudging situations by relying on past stereotypes.
  - **Sunk Costs Bias**: Letting irrecoverable past investments influence current decisions.

Tail risks are often underestimated due to these biases. Methods to address this include improving estimation techniques and fostering diverse viewpoints, which help counteract individual biases and enhance risk identification.

### 11.2.4 Strategic Risk Management

Strategic risk management evaluates the vulnerabilities in a company's strategy and its assumptions. Key considerations include the impact of competitors, customer responses, and external events. Risk is tied to how poorly a strategy performs under adverse or "wrong" scenarios.

Once risks are identified, organizations can take several courses of action:

1. **Avoid**: Exit the business or investment entirely.

2. **Reduce**: Implement measures to lower the likelihood or impact of adverse events.

3. **Adjust**: Modify strategic plans to mitigate risks.

4. **Share/Insure**: Transfer or share the risk with other parties through insurance or partnerships.

5. **Accept**: Take no action and acknowledge the risk.

The choice of action depends on the context but requires thorough analysis to ensure alignment with broader strategic goals.

## 11.3 Practice Questions and Problems

### 11.3.1 Climate Risk

1. Define physical risks and transition risks related to climate change. Which type of risk do you think is harder for companies to manage, and why?
2. What is carbon tax and cap-and-trade system? Which one is more effective at reducing emissions? How it addresses either physical or transition risks?
3. List three economic challenges posed by the transition to a low-carbon economy.

4. Describe how green bonds work and their role in combating climate change.

5. How can climate change affect banks and insurance companies? Provide real-world examples.

### 11.3.2 ESG and Sustainability

1. Rank the three pillars of ESG in order of importance for global sustainability efforts and justify your ranking.

2. Why do inconsistent ESG ratings pose risks to investors? Suggest one way to improve standardization in ESG assessments.

3. Provide examples of greenwashing.
4. Should investors accept lower returns in exchange for supporting sustainable practices?
5. Should governments mandate ESG disclosures for all companies, or should it remain voluntary?

### 11.3.3 Enterprise Risk Management

1. Define the top-down and bottom-up approaches in ERM and explain which one is more important.
2. Imagine you are a risk manager for a bank. Identify three types of risk concentrations your institution might face and propose strategies to mitigate them.

3. Which cognitive bias (e.g., anchoring, availability bias) do you think is the most common in financial risk management, and why?

4. What steps can organizations take to reduce the impact of these biases on decision-making?
5. Are tail risks underestimated due to biases, or are they inherently difficult to predict?

# 12 Regulation, Economic Capital, and RAROC

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 25 - Basel I, Basel II, and Solvency II
  - Chapter 26 - Basel II.5, Basel III, and Other Post-Crisis Changes
  - Chapter 27 - Fundamental Review of the Trading Book
  - Chapter 28 - Economic Capital and RAROC

**Learning outcomes:**

- Trace bank capital requirements from the '80s to 2007.
- Note Basel Accord's shift to risk-weighted assets in 1988.
- Identify Basel II features: credit and operational risk.
- Understand Solvency II for insurance; relate to Basel II.
- Post-2007: grasp reasons for regulatory change.
- Uncover Basel II.5 and Basel III changes.
- Acknowledge Dodd–Frank's impact in the U.S.
- Understand "unintended consequences" in regulations.
- Note shift from VaR to ES in FRTB.
- Identify Basel's two approaches: standardized, internal.
- Grasp economic capital concept.
- Identify risks considered in economic capital.
- Understand enterprise-wide aggregation.
- Explain RAROC's role in performance, strategy.

## 12.1 Basel I, Basel II, and Solvency II

- The Basel Accord of 1988 initiated international standards for bank regulation.
- Regulatory evolution has seen new rules modify old ones while retaining certain approaches.
- Understanding history is crucial for grasping the current regulatory landscape.
- The chapter outlines Basel I, netting provisions, and the 1996 Amendment.
- Basel II, introduced around 2007, significantly reformed the regulatory framework.

- Solvency II, akin to Basel II, was adopted by the European Union in 2016 for insurance companies.

### 12.1.1 The Reasons for Regulating Banks

- The primary aim of bank regulation is to ensure sufficient capital against risks.
- Governments aim to minimize the probability of bank defaults for economic stability.
- Self-regulation by banks is not historically supported; formal regulation enhances capital and risk awareness.
- Deposit insurance programs, while protecting depositors, can inadvertently encourage risk-taking ("moral hazard").
- Systemic risk is a major concern, as a default by one large bank can trigger a financial system collapse.
- Governments face a dilemma between bailing out struggling institutions and risking systemic collapse.
- The 2008 market turmoil involved both bailouts and allowed failures, exemplified by Lehman Brothers, to send market signals.

### 12.1.2 Bank Regulation Pre-1988

- Pre-1988 bank regulation focused on setting minimum capital-to-asset ratios within individual countries.
- Regulatory stringency and definitions varied globally, leading to competitive imbalances.
- Major international loans and complex transactions, like OTC derivatives, exposed the inadequacy of existing capital requirements.
- Basel Committee formed in 1974 to standardize regulations.
- 1988 BIS Accord, later Basel I, was the Committee's first major output.

### 12.1.3 The 1988 BIS Accord

- First international risk-based standards for capital adequacy.
- Signed by all 12 Basel Committee members.
- Increased resources for risk measurement and management in banks.
- The Accord set the groundwork for modern risk management but faced criticisms for being too simplistic.
- **Cooke Ratio**: Focused on total risk-weighted assets (credit risk exposures).
- Categories of Credit Risk Exposures:

    1. **On-Balance Sheet Assets**: Excludes derivatives.
    2. **Off-Balance Sheet Items**: Excludes derivatives.
    3. **Over-the-Counter Derivatives**: Includes swaps, forwards, etc.

#### 12.1.3.1 Key Equations:

- **Risk-Weighted Assets (On-Balance)**:

$$\sum_{i=1}^{N} w_i L_i$$

- where $L_i$ is the principal amount and $w_i$ is its risk weight.
- **Credit Equivalent Amount (Off-Balance)**:
- Conversion factors applied to principal amounts of instruments like bankers' acceptances, guarantees, and loan commitments.
- **Credit Equivalent Amount (Derivatives)**:

$$\max(V, 0) + aL$$

- where $V$ is current value, $a$ is an add-on factor, and $L$ is the principal amount.
- **Total Risk-Weighted Assets**:

$$\sum_{i=1}^{N} w_i L_i + \sum_{j=1}^{M} w_j^* C_j$$

- where $C_j$ is the credit equivalent amount for off-balance-sheet items, and $w_j^*$ is the risk weight of the counterparty.

#### 12.1.3.2 Capital Requirements

- Required banks to maintain capital equal to at least 8% of total risk-weighted assets.
- **Tier 1 Capital**:
  - Includes equity and noncumulative perpetual preferred stock.
  - Goodwill is subtracted from equity.
  - Must constitute at least 50% of required capital (i.e., 4% of risk-weighted assets).
  - Must include 2% of risk-weighted assets as common equity.
- **Tier 2 Capital**:
  - Also known as Supplementary Capital.
  - Includes cumulative perpetual preferred stock, certain 99-year debentures, and subordinated debt with original life over five years.
  - Provides cushion for depositors in case of insolvency.
- Some countries and banks maintain capital ratios above the Basel Committee's minimum requirements.

### 12.1.4 Netting

- Standard in over-the-counter (OTC) derivatives markets.
- Netting clause stating that all transactions are a single transaction in case of default.
- Netting effectively reduces credit risk and modifies regulatory capital requirements.
- **Basel Accords**: Modified in 1995 to include netting.
- **Without Netting**: Loss to the bank = Sum of positive value transactions.
  - Example: Bank's transactions worth +\$24M, -\$17M, +\$8M. Loss = \$24M + \$8M = \$32M.
- **With Netting**: Loss to the bank = Sum of all transactions.
  - Example: Loss = \$24M - \$17M + \$8M = \$15M.
- **General Formula**:
  - Without Netting: $\sum_{i=1}^{N} \max(V_i, 0)$
  - With Netting: $\max(\sum_{i=1}^{N} V_i, 0)$

### 12.1.5 The 1996 Amendment

- The 1996 Amendment to the 1988 Basel Accord added capital charges for market risks in trading activities.
- It mandated fair value accounting, also known as marking to market, for a bank's trading book.
- Two approaches for calculating market risk: standardized and internal model-based.
- Value-at-risk (VaR) metrics and Specific Risk Charge (SRC) introduced.
- Total capital combines credit and market risk, with Tier options.
- Back-testing required; multiplier adjusted based on model performance.

### 12.1.6 Basel II

- Follow-up to the 1988 Basel Accord to address its weaknesses.
- **Overall Goal**: More sophisticated and comprehensive regulatory framework to manage risks in banking.
- Initial proposal in June 1999, revised in January 2001 and April 2003.
- Final rules published in June 2004, updated in November 2005.
- Implementation started in 2007.
- **U.S.**: Applies to internationally active banks; small banks follow Basel IA.
- **EU**: Applies to all banks and securities companies.

#### 12.1.6.1 Three Pillars of Basel II

1. **Minimum Capital Requirements**:

- More risk-sensitive approach to credit risk.
- Operational risk included.
- Market risk remains unchanged.
- 8% of risk-weighted assets (RWA) still the base requirement.

2. **Supervisory Review**:

- Emphasizes qualitative and quantitative risk management.
- Regulatory flexibility but seeks global consistency.
- Focuses on early intervention.

3. **Market Discipline**:

- Greater disclosure requirements.
- Aims to improve risk management through transparency.

$$\text{Total Capital} = 0.08 \times (\text{credit risk RWA} + \text{market risk RWA} + \text{operational risk RWA})$$

### 12.1.7 Solvency II

- Solvency II is the regulatory framework for insurance companies in the European Union, replacing Solvency I in 2016.
- Unlike Solvency I, which focused solely on underwriting risks, Solvency II also considers investment and operational risks.
- Structure resembles Basel II with three pillars:
    1. Pillar 1: Minimum Capital Requirements (MCR) and Solvency Capital Requirement (SCR) calculated. Breach triggers supervisory intervention.
    2. Pillar 2: Supervisory review process for internal capital adequacy.
    3. Pillar 3: Disclosure of risk management information to the market.
- SCR can be calculated using the standardized approach or internal models involving Value-at-Risk (VaR) with a one-year time horizon and 99.5% confidence limit.
- SCR includes charges for investment risk (market and credit), underwriting risk (life, non-life, health), and operational risk.
- Capital adequacy tests include various adverse event scenarios (e.g., 32% stock market decrease, 20% real estate price decrease).
- Internal models must pass three tests: statistical quality, calibration, and use.
- Types of capital:

1. Tier 1: Equity capital, retained earnings.
2. Tier 2: Liabilities subordinated to policyholders, available in wind-down scenarios.
3. Tier 3: Liabilities subordinated to policyholders, not meeting Tier 2 criteria.

- Capital composition rules are specified for Tier 1, Tier 1 + Tier 2, and Tier 1 + Tier 2 + Tier 3.

## 12.2 Basel II.5, Basel III, and Other Post-Crisis Changes

- Basel II's implementation roughly coincided with the onset of the Global Financial Crisis.
- Critics argue Basel II encouraged self-regulation, allowing banks to use their own model inputs for regulatory capital.
- Basel II may not be solely to blame:
  - The origins of the crisis predate Basel II.
  - U.S. was late in implementing Basel II, affecting capital levels.
- Basel II.5 was introduced post-crisis to amend market risk capital calculations.
  - Implementation date: December 31, 2011.
- Basel III was a more comprehensive post-crisis regulatory overhaul.
  - First set of regulations: December 2010.
  - Finalized: December 2017.
  - Implementation phased over several years.
- Other post-2008 regulations:
  - Dodd–Frank Act in the U.S., signed July 21, 2010.
  - New rules in the European Union and the United Kingdom.

### 12.2.1 Basel II.5

- Implemented on December 31, 2011, it introduced changes to the calculation of market risk capital.
- **Three Main Changes**:
  1. Stressed Value-at-Risk (VaR)
  2. Incremental Risk Charge (IRC)
  3. Comprehensive Risk Measure (CRM)
- **Impact**: Significantly increased the capital large banks need to hold for market risk.

#### 12.2.1.1 Details:

- **Stressed VaR**:
  - Originally based on 10-day, 99% VaR.
  - Introduces a stressed VaR, based on 250-day stressed market conditions.

    $$\text{Total Capital Charge} = \max(VaR_{t-1}, m_c \times VaR_{\text{avg}}) + \max(sVaR_{t-1}, m_s \times sVaR_{\text{avg}})$$

  - $VaR_{t-1}$ and $sVaR_{t-1}$ are the Value-at-Risk and stressed Value-at-Risk calculated on the previous day with a 10-day time horizon and a 99% confidence level.
  - $VaR_{\text{avg}}$ and $sVaR_{\text{avg}}$ are the averages of Value-at-Risk and stressed Value-at-Risk calculated over the previous 60 days with the same confidence level and time horizon.
  - $m_c$ and $m_s$ are multiplicative factors determined by bank supervisors, with a minimum value of three.
  - Effect: At least doubles the previous capital requirement.
- **Incremental Risk Charge (IRC)**:
  - Addresses discrepancy between trading and banking book capital requirements.
  - Calculates a one-year 99.9% VaR, considering both credit rating changes and defaults.
  - Includes a "liquidity horizon" for each instrument.
  - Employs the "constant level of risk assumption."
- **Comprehensive Risk Measure (CRM)**:
  - Focuses on correlation-dependent instruments like ABSs and CDOs.
  - Replaces the Incremental Risk Charge and specific risk charge for these instruments.
  - Includes both standardized and internal model-based approaches.

### 12.2.2 Basel III

- Introduced post-2007–2009 Global Financial Crisis to overhaul Basel II.
- First set of regulations published in December 2010, finalized in 2017.
- Implementation delayed to January 2023.
- **Six Key Components**:
  1. Capital Definition and Requirements
  2. Capital Conservation Buffer
  3. Countercyclical Buffer
  4. Leverage Ratio
  5. Liquidity Risk
  6. Counterparty Credit Risk

#### 12.2.2.1 Capital Definition and Requirements

- Consists of Tier 1 equity capital, Additional Tier 1 capital, and Tier 2 capital. No Tier 3 capital.
- Each tier has different characteristics and is intended to absorb losses to varying degrees, thereby contributing to a bank's overall financial stability.

1. **Tier 1 Equity Capital (Common Equity Tier 1 or CET1)**
   - Includes share capital and retained earnings.
   - Excludes goodwill, deferred tax assets, and defined benefit pension plan deficits.
2. **Additional Tier 1 (AT1) Capital**
   - Items like non-cumulative preferred stock.
   - Previously considered Tier 1 but not common equity.
3. **Tier 2 Capital**
   - Includes subordinated debt with an original maturity of at least five years.

- **Capital Requirements**:
  - Tier 1 equity capital: At least 4.5% of risk-weighted assets.
  - Total Tier 1 capital: At least 6% of risk-weighted assets.
  - Total capital (Tier 1 + Tier 2): At least 8% of risk-weighted assets.

#### 12.2.2.2 Capital Conservation Buffer

- Further CET1 capital equal to 2.5% of risk-weighted assets.
- Dividend restrictions based on capital ratios.
- To ensure banks accumulate additional common equity Tier 1 (CET1) capital during normal economic conditions.
- This buffer serves as a reserve for absorbing losses during financial stress.
- Banks must maintain a CET1 capital conservation buffer equal to 2.5% of risk-weighted assets.
- **Dividend Restrictions**: If the buffer is partially or wholly used, dividend payouts are constrained until the capital is replenished.

#### 12.2.2.3 Countercyclical Buffer

- To counteract the cyclicality of bank earnings and protect the financial system during economic downturns.
- The buffer can range from 0% to 2.5% of a bank's total risk-weighted assets and must be met with Tier 1 equity capital (CET1).
- The implementation level is determined by each country's regulatory authorities, allowing for regional economic conditions.

#### 12.2.2.4 Leverage Ratio

- **Purpose**: To limit banks' discretion in calculating risk-weighted assets and to provide additional information to regulators.
- **Potential Issue**: Could encourage banks to hold riskier assets if this becomes the constraining ratio.
- The leverage ratio is the ratio of Tier 1 capital to the exposure measure.
- Basel III specifies a minimum leverage ratio of 3%.
- **Capital Measure**: Total Tier 1 capital.
- **Exposure Measure**: Sum of (a) on-balance-sheet exposures, (b) derivatives exposures, (c) securities financing transaction exposures, and (d) off-balance-sheet items.
- No risk-weighting adjustments are made.
- Some countries have proposed or implemented higher minimums.

#### 12.2.2.5 Liquidity Risk

- Basel III introduces the Liquidity Coverage Ratio (LCR) and the Net Stable Funding Ratio (NSFR).
- **LCR**: Focuses on a bank's ability to survive a 30-day liquidity disruption.

$$\text{LCR} = \frac{\text{High-Quality Liquid Assets}}{\text{Net Cash Outflows in a 30-Day Period}}$$

- **NSFR**: Focuses on liquidity management over one year.

$$\text{NSFR} = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}}$$

- **Liquidity Management**: Designed to ensure that banks can survive liquidity pressures.

#### 12.2.2.6 Counterparty Credit Risk

- Banks calculate a Credit Value Adjustment (CVA) for each derivatives counterparty.
- CVA measures expected loss from counterparty default.
- Changes in CVA can be due to shifts in risk factors or credit spreads.
- Basel III includes CVA risk in market risk capital requirements.

#### 12.2.2.7 G-SIBs, SIFIs, and D-SIBs

- Regulators aim to ensure that large, systemically important institutions hold sufficient capital.
- G-SIBs: Global Systemically Important Banks
- SIFIs: Systemically Important Financial Institutions (includes both banks and non-banks).
- Importance is determined through scoring methodologies and other approaches like network theory.
- G-SIBs required to hold extra Tier 1 equity capital, based on categorization.
- Financial Stability Board publishes annual G-SIB list.
- Total Loss-Absorbing Capacity (TLAC) rules have been set for G-SIBs.
- Non-compliance with TLAC can limit a bank's dividend issuance.
- Some countries impose stricter capital requirements (e.g., Switzerland).
- Domestic Systemically Important Banks (D-SIBs) have their own sets of rules and capital requirements.

### 12.2.3 Contingent Convertible Bonds (CoCos)

- CoCos are bonds that automatically convert into equity under certain conditions, generally when a bank faces financial difficulties.
- Offer higher interest rates than regular debt.
- Benefits for banks: High return on equity in normal times, and an equity cushion during financial stress.
- Benefits for regulators: Reduces the need for public bailouts.
- Specifying the trigger for conversion, often based on Tier 1 equity capital ratios.

### 12.2.4 Dodd–Frank Act

- The Dodd-Frank Wall Street Reform and Consumer Protection Act was signed into U.S. law in July 2010.
- Aimed to prevent future bailouts and protect consumers.
- Its primary objectives are to:

  - Enhance regulation and oversight of financial markets and institutions.
  - Reduce systemic risks in the financial sector.
  - Protect consumers through the creation of the Consumer Financial Protection Bureau (CFPB).
  - Increase transparency in financial products and services.

## 12.3 Fundamental Review of the Trading Book

- FRTB was introduced by the Basel Committee in May 2012, final rules published in January 2019, implementation slated for January 2023, but delays are expected.
- **Previous Approaches**:
  - Basel I used Value at Risk (VaR) with a 99% confidence level.
  - Basel II.5 added "stressed VaR" based on historical periods of financial stress.
- **Key Features of FRTB**:
- Shifts from VaR to Expected Shortfall (ES) with a 97.5% confidence level (and stressed ES).
- Liquidity horizons introduced, varying from 10 to 120 days, based on the risk factor considered.
- Stricter definitions to counter regulatory arbitrage between the trading book and the banking book.
- The Basel Committee outlines two approaches for banks: Standardized Approach and Internal Models Approach.
- Banks approved for Internal Models Approach must also implement the Standardized Approach.
- Standardized Approach calculates regulatory capital using delta, vega, and gamma exposures in the trading book.
- Internal Models Approach uses stressed expected shortfall for regulatory capital calculation.
- Calculations are performed separately for each trading desk.

### 12.3.1 Standardized Approach

- Capital requirement is a sum of risk charge, default risk charge, and residual risk add-on.
- **Risk Charge**:
  - Seven Risk Classes: General interest rate, foreign exchange, commodity, equity, and three credit spread risks.
  - Types of Risk Charges: Delta, Vega, and Curvature.
- **Default Risk Charge**:

  - Calculated by Loss Given Default (LGD) and a default risk weight.

- **Residual Risk Add-On**:
  - For risks not captured by delta/vega/curvature approach, e.g., exotic options.
- **Simplified Approach**:
  - For smaller banks; excludes vega and gamma risks.

### 12.3.2 Trading Book vs. Banking Book

- **Trading Book**: Contains financial instruments intended for trading. Subject to market risk capital.
- **Banking Book**: Consists of instruments intended to be held to maturity. Subject to credit risk capital.
- Regulatory Arbitrage: Past practices allowed banks to choose book placement based on advantageous capital requirements.
- FRTB aims to clarify and standardize criteria for book classification.
- 'Intent to trade' insufficient; must be manageable on a trading desk and pose risks to solvency.
- Instruments are initially assigned to a specific book with limited ability to transfer later.
- Allow for book transfers, but any capital benefits from such moves are disallowed.

## 12.4 Economic Capital and RAROC

- **Economic Capital**: Internal estimate of the capital a financial institution requires for its risk exposure.
- It varies from regulatory capital, which follows standard rules.
- Acts as a "currency" for risk-taking within an institution.

### 12.4.1 Definition of Economic Capital

- **Definition**: Amount of capital needed to absorb losses over one year at a specific confidence level, tailored to maintain credit ratings like AA or BBB.
- Economic capital fills the gap between expected loss and unexpected loss.
- **Case Example**: An AA-rated bank with expected annual losses of 1% and a one-year 99.98% VaR of 5% would require $4 in economic capital per $100 of loans.
- **Approaches to Measure Economic Capital**:
  - **Top-Down**: Uses asset volatility to estimate risk, often using Merton's model.
  - **Bottom-Up**: Aggregates loss distributions by risk types and business units.

- **Risk-Adjusted Return on Capital (RAROC)**:
  - Measures profitability relative to allocated economic capital.
  - Used for assessing past and forecasting future performance, also aids in capital allocation decisions.
- **Types of Risks Considered**:
  - Regulatory Capital covers Credit, Market, and Operational risks.
  - Some institutions also consider Business risk, which is not mandated for regulatory capital.

### 12.4.2 Components of Economic Capital

- **Market Risk Economic Capital**:
  - Time horizon is usually one year; confidence level can go up to 99.98%.
  - Standard deviation for one-year loss or gain derived from annualized daily metrics.
- **Credit Risk Economic Capital**:
  - Banks have freedom to choose their credit correlation models.
  - Popular approaches include CreditMetrics and Credit Risk Plus.
  - Can be conditional (cycle-specific) or unconditional (cycle-neutral) based on economic conditions.
- **Operational Risk Economic Capital**:
  - Regulatory procedure often adopted; some banks use more advanced models.
  - Aims to create a complete distribution of losses.
- **Business Risk Economic Capital**:
  - Involves strategic and reputational risks.
  - Difficult to quantify, largely subjective.
  - Quantification can inform the impact of new strategic initiatives on total risk.
- Characteristics of Loss Distributions for Different Risk Types

| | Standard Deviation | Skewness | Kurtosis |
|---|---|---|---|
| Market risk | High | Zero | Low |
| Credit risk | Moderate | Moderate | Moderate |
| Operational risk | Low | High | High |

### 12.4.3 Aggregating Economic Capital

- To calculate the total economic capital for a financial institution by aggregating various types of risks such as market, credit, and operational risks.

1. **Simple Summation (Basel Committee Method)**
   - Formula: $E_{\text{total}} = \sum_{i=1}^{n} E_i$
   - Assumes perfect correlation between risks.
   - Overestimates capital requirement by approx. 40%.
2. **Normal Distributions**
   - Formula: $\sigma_{\text{total}} = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} \sigma_i \sigma_j \rho_{ij}}$
   - Underestimates capital due to disregard for skewness and kurtosis.
3. **Using Copulas**
   - Maps loss distributions to a standard distribution.
   - Gaussian copula as the simplest case.
4. **Hybrid Approach**
   - Formula: $E_{\text{total}} = \sqrt{\sum_{i=1}^{n} \sum_{j=1}^{n} E_i E_j \rho_{ij}}$
   - Offers a balance between underestimation and overestimation of capital.

### 12.4.4 RAROC

- Risk-adjusted performance measurement (RAPM) is crucial for assessing business units.
- RAROC (Risk-Adjusted Return on Capital) is a widely-used approach.
- Compares expected return with economic capital, adjusted for risk.

$$\text{RAROC} = \frac{\text{Revenues} - \text{Costs} - \text{Expected losses}}{\text{Economic capital}}$$

- Numerator can be pre-tax or post-tax.
- An alternative includes adding a risk-free rate of return to the numerator.
- Ex-ante calculations: For business expansion/contraction decisions.
- Ex-post calculations: For performance evaluation and bonuses.
- Decisions should not solely be based on ex-post results; consider long-term expectations.

## 12.5 Practice Questions and Problems

### 12.5.1 Basel I, Basel II, and Solvency II

1. "When a steel company goes bankrupt, other companies in the same industry benefit because they have one less competitor. But when a bank goes bankrupt other banks do not necessarily benefit." Explain this statement.
2. "The existence of deposit insurance makes it particularly important for there to be regulations on the amount of capital banks hold." Explain this statement.
3. An interest rate swap involves the exchange of a fixed rate of interest for a floating rate of interest with both being applied to the same principal. The principals are not exchanged. What is the nature of the credit risk for a bank when it enters into a five-year interest rate swap with a notional principal of $100 million? Assume the swap is worth zero initially.
4. In a currency swap, interest on a principal in one currency is exchanged for interest on a principal in another currency. The principals in the two currencies are exchanged at the end of the life of the swap. Why is the credit risk on a currency swap greater than that on an interest rate swap?
5. A four-year interest rate swap currently has a negative value to a financial institution. Is the financial institution exposed to credit risk on the transaction? Explain your answer.
6. All the derivatives transactions a bank has with a corporate client have a positive value to the bank. What is the value to the bank of netting provisions in its master agreement with the client?
7. What is the difference between the trading book and the banking book for a bank? A bank currently has a loan of $10 million to a corporate client. At the end of the life of the loan, the client would like to sell debt securities to the bank instead of borrowing. How does this potentially affect the nature of the bank's regulatory capital calculations?
8. Under Basel I, banks do not like lending to highly creditworthy companies and prefer to help them issue debt securities. Why is this? Do you think this changed as a result of Basel II?
9. Banks sometimes use what is referred to as regulatory arbitrage to reduce their capital. What do you think this means?

### 12.5.2 Basel II.5, Basel III, and Oter Post-Crisis Changes

1. What are the three major components of Basel II.5?
2. What are the six major components of Basel III?
3. What is the difference between VaR as it has been traditionally measured and stressed VaR?
4. Explain how the incremental risk charge is calculated. Why was it introduced by the Basel Committee?
5. Explain how CoCo bonds work. Why are they attractive to (a) banks and (b) regulators?

### 12.5.3 Fundamental Review of the Trading Book

1. Outline the differences between the way market risk capital is calculated in (a) Basel I, (b) Basel II.5, and (c) the FRTB.
2. What are the advantages of expected shortfall over value at risk?
3. What is the difference between the trading book and the banking book? Why are regulators concerned about specifying whether an instrument should be one or the other in the FRTB?

### 12.5.4 Economic Capital and RAROC

1. What is the difference between economic capital and regulatory capital?
2. What determines the confidence level used by a AA-rated financial institution in its economic capital calculations?
3. What types of risk are included in business risk?
4. A bank is considering expanding its asset management operations. The main risk is operational risk. It estimates the expected operational risk loss from the new venture in one year to be \$2 million and the 99.97% VaR (arising from a small chance of an large investor lawsuit) to be \$40 million. The expected fees it will receive from investors for the funds under administration are \$12 million per year, and administrative costs are expected to be \$3 million per year. Estimate the before-tax RAROC.
5. RAROC can be used in two different ways. What are they?

# 13 Financial Innovation and Risk Management Mistakes

**References:**

- Hull, J. C. 2023. Risk Management and Financial Institutions. Wiley Finance.
  - Chapter 29 - Financial Innovation
  - Chapter 30 - Risk Management Mistakes to Avoid

**Learning outcomes:**

- Understand tech's impact on financial services by 2040.
- Recognize shift to digital payments.
- Learn machine learning's role in credit and fraud.
- Know blockchain's impact on record-keeping.
- Assess changes in the insurance industry.
- Analyze trends in investing and borrowing.
- Evaluate banks' adaptation to tech changes.
- Formulate risk management mistakes to avoid.
- Understand common misconceptions about risk.

## 13.1 Financial Innovation

Technological advances have revolutionized the financial industry, driving both disintermediation (reduction of traditional intermediaries) and reintermediation (emergence of new tech-based intermediaries). FinTech disrupts traditional financial institutions like banks, requiring them to adapt to a rapidly changing landscape. Future developments in FinTech are expected to significantly impact digitization, customer behavior, and organizational strategies—key areas for risk management professionals to monitor and understand.

### 13.1.1 Technological Advances

Technological progress, particularly since the 1960s, has been marked by exponential growth in computing power and the rise of automation tools like Robotics Process Automation (RPA). These tools streamline processes such as customer onboarding and recruitment, reducing inefficiencies and operational risks.

#### 13.1.1.1 Machine Learning

The advent of machine learning marks a shift from traditional decision-making to data-driven insights, distinguishing it from earlier industrial revolutions. Applications in finance include automating credit decisions, detecting fraud, and optimizing risk strategies.

**Types of Machine Learning**:

1. **Unsupervised Learning**: Identifies patterns and clusters within data, such as customer segmentation or portfolio grouping.

2. **Supervised Learning**: Focuses on prediction and classification tasks, like forecasting loan defaults or assessing property values.

3. **Reinforcement Learning**: Excels in sequential decision-making, such as optimizing hedging strategies for derivatives.

**Challenges**:

- Overfitting occurs when models fail to generalize to unseen data.

- The most common solution involves splitting data into training, validation, and test sets to ensure robust performance evaluation.

#### 13.1.1.2 Blockchain

Traditional ledger systems in finance are prone to fraud, hacking, and inconsistencies, especially when multiple parties maintain independent records. Blockchain technology addresses these risks by introducing a distributed ledger system, where all participants share a synchronized, tamper-proof record.

**Core Features**:

- Cryptographic hashing ensures data integrity and prevents unauthorized alterations.

- Miners validate transactions and maintain the blockchain through decentralized consensus mechanisms, as seen in systems like Bitcoin.

**Types of Blockchains**:

- **Permissionless**: Open to the public, allowing unrestricted participation (e.g., Bitcoin).
- **Permissioned**: Restricted access, favored by financial institutions for secure and efficient operations.

**Applications in Finance**:

- Blockchain facilitates near real-time settlement of securities trades and back-office transactions, reducing costs and delays.
- It offers opportunities to lower operational risks, streamline regulatory compliance, and enhance transparency.

While blockchain presents significant opportunities, it also poses challenges, such as disrupting traditional fee structures and requiring widespread adoption to unlock its full potential. Its role in financial innovation is still evolving but shows promise in transforming operations across sectors.

### 13.1.2 Payment Systems

The evolution of payment systems from cash to digital platforms has been transformative, with developing countries experiencing rapid adoption. Key attributes such as speed, convenience, security, simplicity, and cost drive this shift, supported by innovations like instant fund transfers (e.g., PayPal), digital wallets (e.g., Apple Pay), and the integration of cryptocurrencies.

**Security and Privacy**:

- Biometric authentication enhances security and reduces fraud, but raises concerns over privacy and data protection.
- Data collection for analytics and credit decisions introduces risks of misuse and data selling, necessitating robust regulatory oversight.

**Credit Facilities and Future Trends**:

- Traditional credit cards, with high interest rates, face competition from FinTech solutions offering flexible credit terms, such as PayPal's 14-day free credit.

- Anticipated advancements include streamlined foreign exchange services and expanded use of data analytics for personalized offerings.

- Despite growing acceptance of cryptocurrencies, their volatility remains a barrier to widespread use. Central Bank Digital Currencies (CBDCs) could reshape financial control by integrating the benefits of digital currencies within regulatory frameworks.

The decline of cash continues, although political resistance may hinder full digitization in some regions.

### 13.1.3 Lending

#### 13.1.3.1 P2P Lending

Peer-to-peer (P2P) lending bypasses traditional banks by connecting borrowers and lenders directly, though intermediaries often remain for risk assessment and transaction facilitation. P2P platforms cater to underserved borrowers, offering rates lower than credit cards but higher than conventional loans. Effective risk management requires lenders to diversify and leverage secondary markets.

#### 13.1.3.2 Crowdfunding

Crowdfunding democratizes investment opportunities but faces challenges in valuation and regulatory compliance. While the JOBS Act in the U.S. has eased some restrictions, limits on fundraising persist. Initial Coin Offerings (ICOs) represent a high-risk, speculative crowdfunding method with significant potential for both innovation and fraud.

### 13.1.4 Wealth Management

Wealth management has traditionally been a lucrative business for banks, with fees ranging from 1% to 2% annually. Services include risk profiling and personalized investment recommendations aimed at achieving client-specific financial goals.

- The introduction of index funds by John Bogle in 1975 revolutionized investment by offering low-cost alternatives (fees around 0.15%) that eliminated the need for active human management.

- Robo-advisers, emerging around 2010, further disrupted the industry by automating risk assessment and portfolio rebalancing. These digital platforms typically charge lower fees (~0.25%) and help mitigate cognitive biases, making wealth management accessible to a broader audience.

While the popularity of robo-advisers may fluctuate with market performance, they represent a key trend in reducing costs and increasing efficiency in wealth management.

### 13.1.5 InsurTech

The insurance industry has been relatively resistant to disruption due to its reliance on stability, strict regulatory frameworks, and high capital requirements for new entrants. Traditionally dominated by brokers and agents facilitating customer interactions, the industry is gradually shifting toward disintermediation, with online platforms enabling direct customer-insurer engagement.

**Technological Innovations**:

- Online quote comparison tools have enhanced transparency and customer convenience.
- Advanced data collection improves risk assessment, enabling more precise premium adjustments and potentially streamlining claims processing.

While InsurTech is unlikely to disrupt the industry entirely, the integration of technology offers significant advantages for improving efficiency and customer experience.

### 13.1.6 Regulation and Compliance

Financial institutions face stringent regulatory requirements, which can pose barriers for small FinTech startups. Regulators aim to balance fostering innovation with ensuring client safety.

**Key Trends**:

- "Sandbox" programs allow new financial products to be tested in controlled environments, minimizing risks to the broader system.
- Challenges such as jurisdictional conflicts and the evaluation of code-based assets are prominent in the FinTech space.
- Technology simplifies compliance by automating processes and enabling real-time monitoring, reducing the risk of fines and ensuring adherence to international standards.

### 13.1.7 How Should Financial Institutions Respond?

FinTech represents a strategic risk and an opportunity for established financial institutions, necessitating proactive adaptation.

**Strategies for Adaptation**:

- In-house development of innovative solutions.
- Acquisitions of or partnerships with FinTech startups to integrate disruptive technologies.
- Responding to shifting consumer behavior with greater flexibility and adaptability.

Embracing these strategies will help financial institutions navigate the evolving landscape and maintain competitive relevance.

## 13.2 Risk Management Mistakes to Avoid

Effective risk management requires clear, board-level risk limits that cascade down to individual managers. These limits should be specific, unambiguous, and integrated into daily reporting to ensure alignment between actual gains or losses and the institution's risk appetite.

- **Monitoring Derivatives**: Derivatives must be closely managed to distinguish between hedging, speculation, or arbitrage.
- **Exceeding Risk Limits for Profit**: Allowing profit-driven violations of risk limits fosters a dangerous culture. Penalties should apply equally to profitable and unprofitable breaches to enforce discipline.
- **Overconfidence in Market Predictability**: Trading success often reflects luck, not skill. Risk limits should remain firm, even for consistently profitable traders.
- **Neglecting Diversification**: Concentrating risks on a single market variable or asset undermines risk reduction. Diversification across at least 20 assets significantly reduces portfolio risk.
- **Ignoring Stress Tests**: Measures like VaR are insufficient on their own. Scenario analyses and stress tests must complement them, with results influencing strategic decisions, even during stable periods.

### 13.2.1 Managing the Trading Room

Managing a trading room requires robust oversight and structural safeguards to mitigate risks and promote long-term trust.

- Scrutinize all traders, including high performers, to ensure compliance with risk limits and ethical standards.
- Separate front, middle, and back offices to minimize conflicts of interest and ensure checks and balances.
- Recognize that financial models are prone to errors; blind reliance can lead to catastrophic losses.
- Adopt conservative profit recognition for complex instruments, spreading revenue over the life of the deal to curb aggressive trading behaviors.
- Avoid selling inappropriate high-risk products to clients, as this erodes trust and exposes the institution to legal and reputational risks.
- Be skeptical of "easy profit" opportunities, as these often hide significant risks.

### 13.2.2 Liquidity Risk

Liquidity risk arises when institutions assume they can sell assets close to theoretical prices, a misconception that becomes perilous during market shocks.

- Actively traded instruments often anchor the pricing of illiquid assets, but this link can break during crises.
- Scenario analyses and stress tests should evaluate the impact of market-wide adoption of similar trading strategies, which can exacerbate volatility.
- Overreliance on short-term funding for long-term commitments heightens liquidity risks, as demonstrated in past financial crises.
- Basel III liquidity requirements aim to address such vulnerabilities by encouraging better liquidity coverage.

**Lessons from 2007 Credit Crisis**:

- Structured products lacked transparency, amplifying the crisis.

- Providing more detailed information about underlying assets can reduce systemic risks and prevent flight-to-quality scenarios. Institutions should adopt transparency measures to bolster market stability.

### 13.2.3 Lessons for Nonfinancial Corporations

#### 13.2.3.1 Understand Your Trades

Nonfinancial corporations must thoroughly understand any trade or trading strategy before execution. Trades lacking comprehension, especially at the senior management level, should not proceed. Valuing financial instruments internally, rather than relying solely on investment bankers, enhances understanding and reduces potential risks.

#### 13.2.3.2 Hedgers vs. Speculators

Traders may begin with hedging intentions but can drift toward speculation over time. Speculative losses often lead to increasingly risky attempts at recovery, risking financial disaster. Clear risk limits and stringent controls set by senior management are essential to prevent this drift.

#### 13.2.3.3 Treasury as Profit Center: Proceed with Caution

While transforming the treasury department into a profit center may seem attractive, it introduces significant risks. Treasurers incentivized by profit may take excessive risks, contradicting the primary purpose of a hedging program: risk reduction, not profit generation. Treating the treasury as a profit center risks speculative activities that could lead to severe financial consequences.

### 13.2.4 A Final Point: Understanding Risks

- **Known Risks**: Quantifiable and measurable, such as market or credit risks.
- **Unknown Risks**: Events whose probabilities are indeterminable, often tied to operational uncertainties.
- **Unknowable Risks**: Unpredictable events, commonly referred to as "black swans."

Knight (1921) distinguished "risk" (quantifiable) from "uncertainty" (unquantifiable). Effective preparation for these categories can help mitigate catastrophic losses.

#### 13.2.4.1 Managing Unknown Risks

- Use subjective probabilities where objective data is unavailable.
- Maintain operational flexibility, minimize leverage, and prefer variable costs over fixed commitments.
- Explore the development and use of insurance products tailored to cover unknown risks.

#### 13.2.4.2 Managing Unknowable Risks

Unknowable risks, being inherently unpredictable, pose the greatest challenge. Strategies include:

- Diversification across products, markets, and geographies to reduce exposure.
- Exploring future insurance innovations aimed at unknowable risks.

While unknowable risks cannot be fully mitigated, these measures can build resilience and reduce potential losses.

## 13.3 Practice Questions and Problems

### 13.3.1 Financial Innovation

1. Explain the terms disintermediation and reintermediation.
2. What is meant by (a) machine learning and (b) distributed ledger technologies?
3. Why might a central bank switch from fiat money to digital money?
4. Explain how P2P lending and equity crowdfunding work.
5. "When thinking about the impact of FinTech, financial institutions should be thankful for regulation." Explain.
6. "Index funds are a great idea, but if everyone chose them there would be no price discovery." Do you agree with this argument?

### 13.3.2 Risk Management Mistakes to Avoid

1. Discuss and name a few mistakes to avoid.